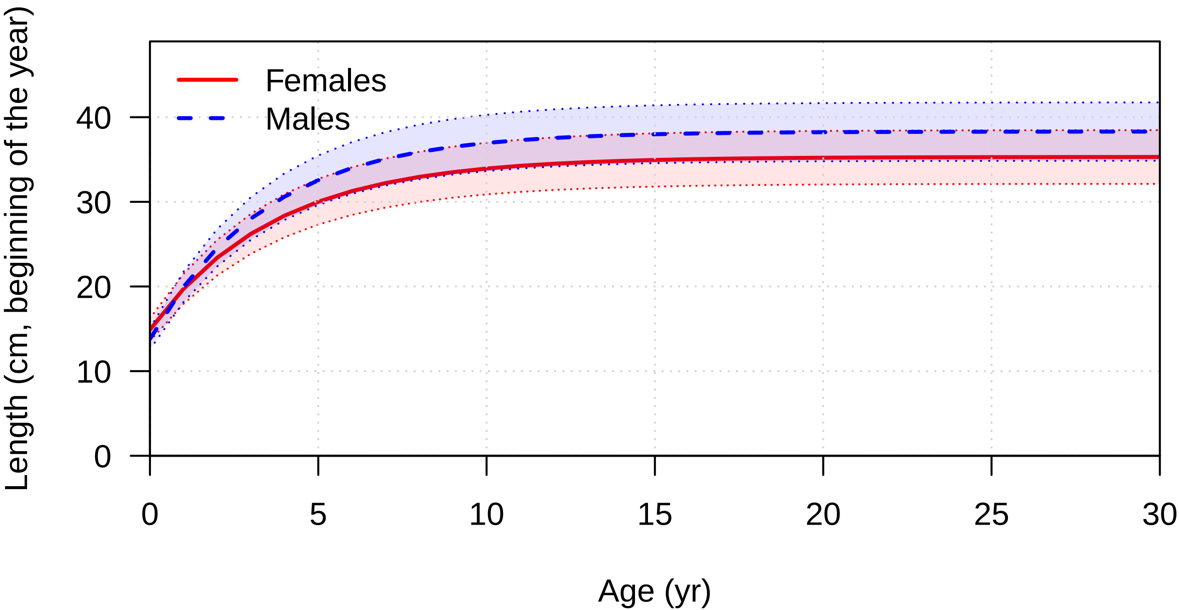
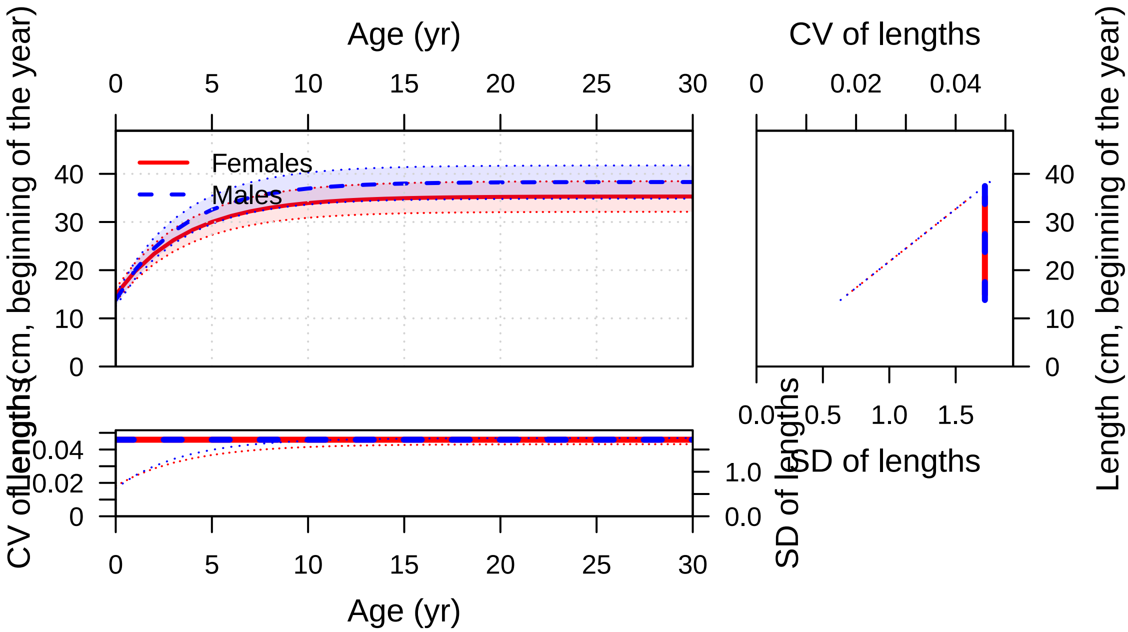
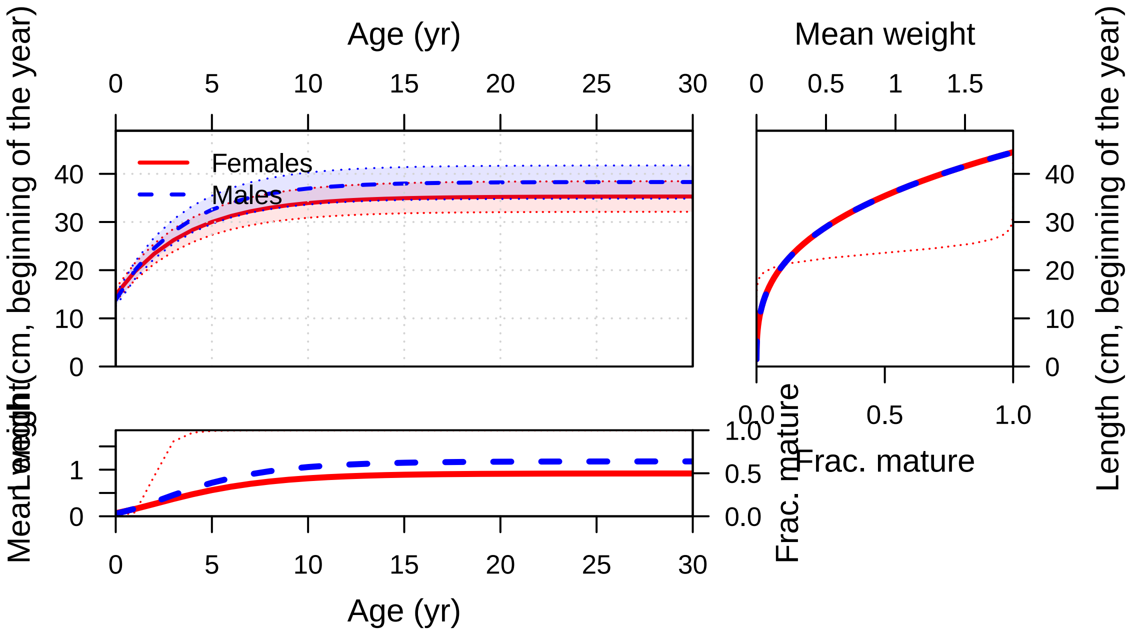
Plots created using the 'r4ss' package in R
Stock Synthesis version: 3.30.19.0
StartTime: Fri Feb 13 12:15:23 2026
Data_File: data.ss
Control_File: control.ss

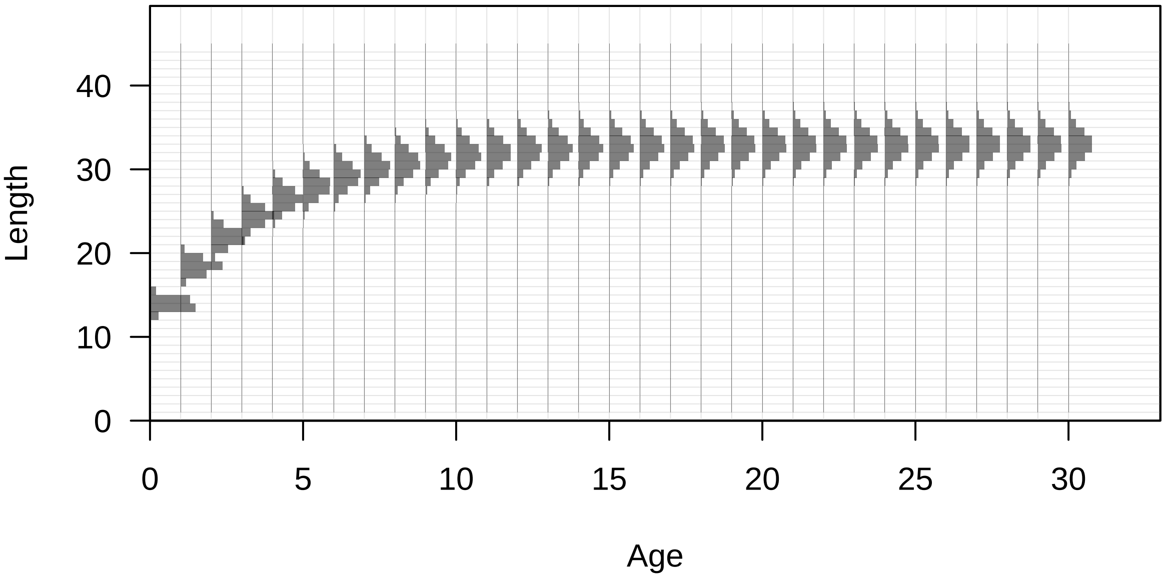


Females
Males
Length (cm, beginning of the year)
0
10
20
30
40
0
5
10
15
20
25
30
Age (yr)

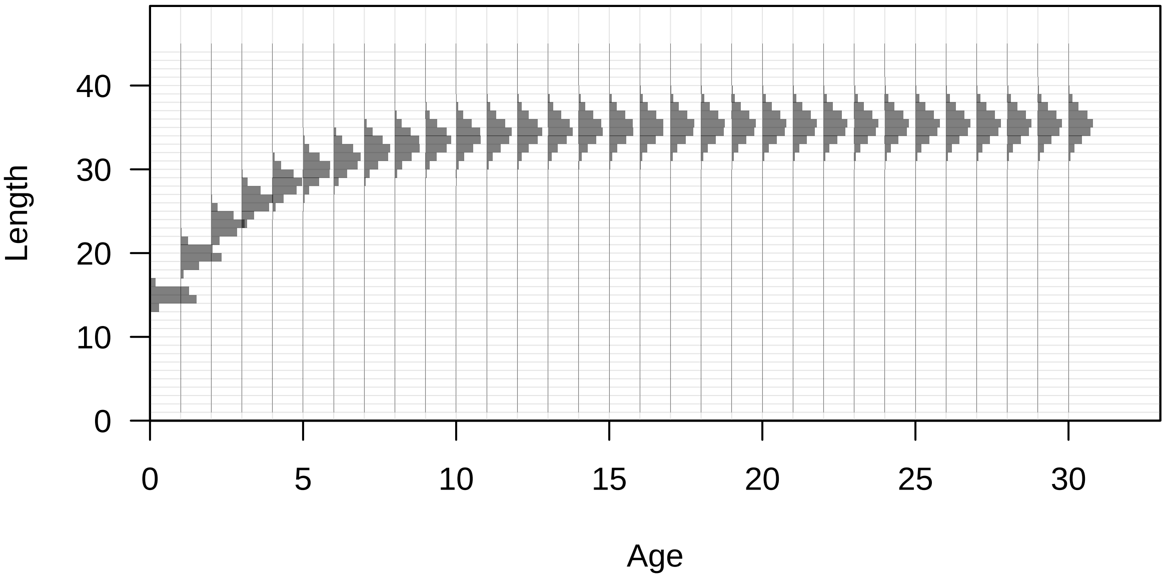


Age (yr)
CV of lengths
Females
Males
Length (cm, beginning of the year)
CV of lengths
SD of lengths
SD of lengths
Age (yr)
Length (cm, beginning of the year)

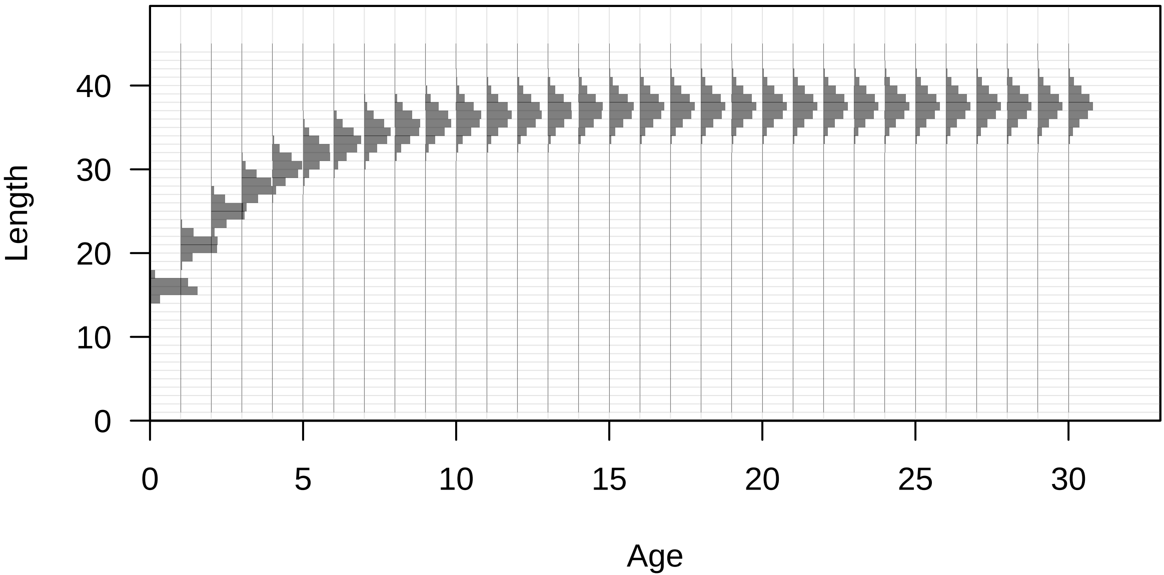

Age (yr)
Mean weight
Females
Males
Length (cm, beginning of the year)
Frac. mature
Mean weight
Age (yr)
Frac. mature
Frac. mature
Length (cm, beginning of the year)



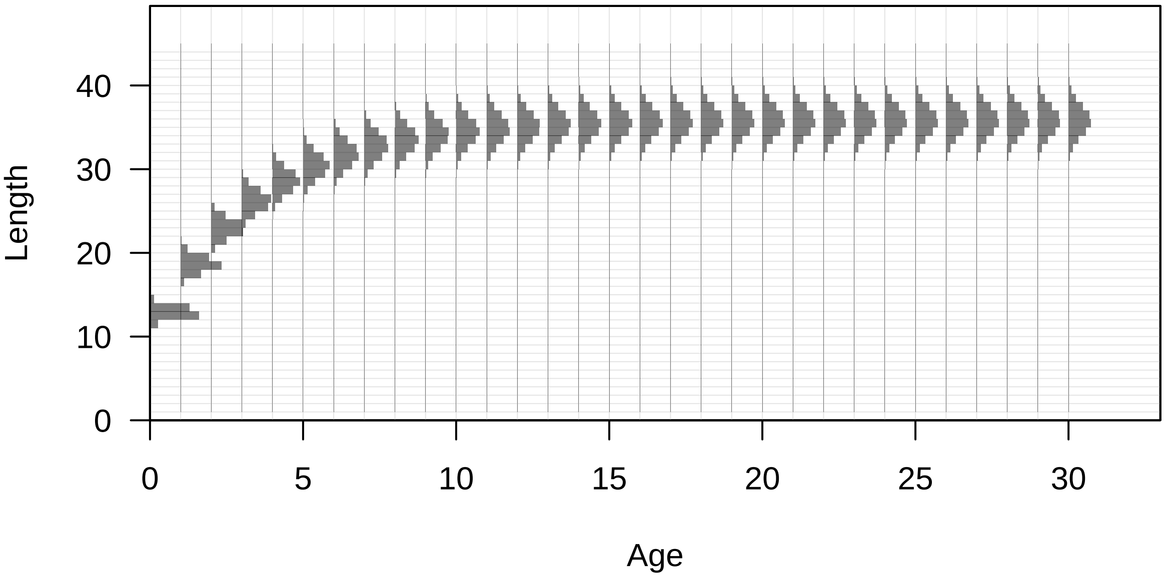
Length
Age
0
10
20
30
40
0
5
10
15
20
25
30



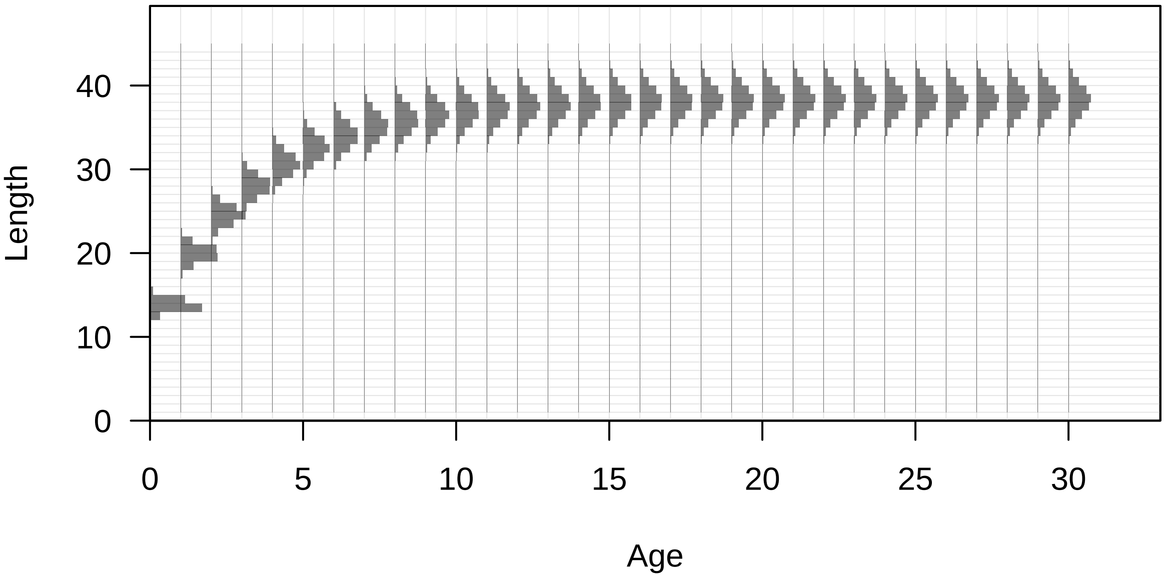
Length
Age
0
10
20
30
40
0
5
10
15
20
25
30

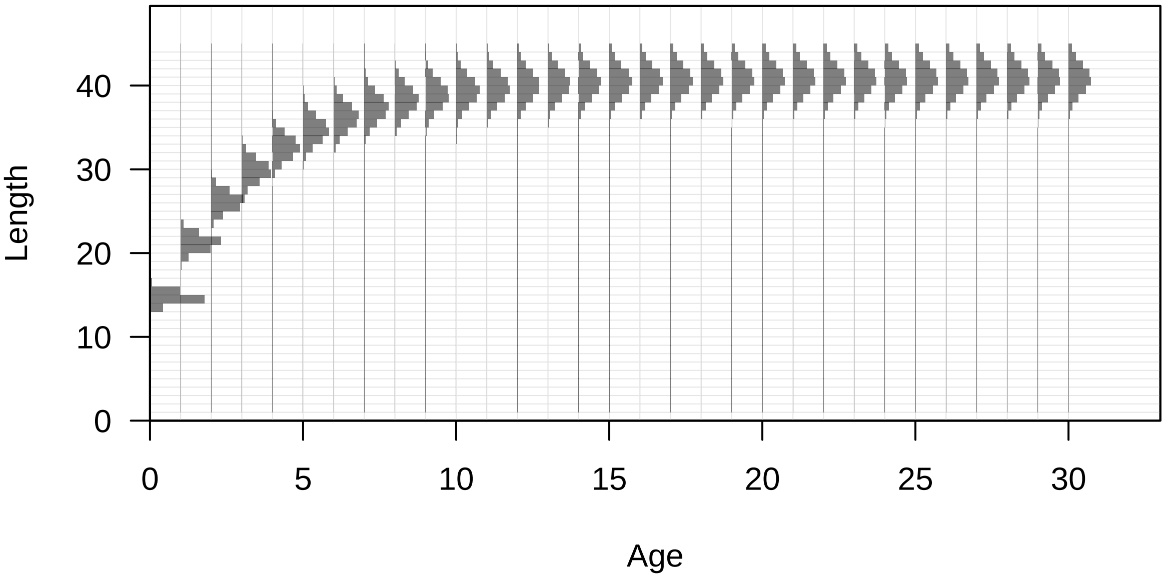


Length
0
10
20
30
40
0
5
10
15
20
25
30
Age

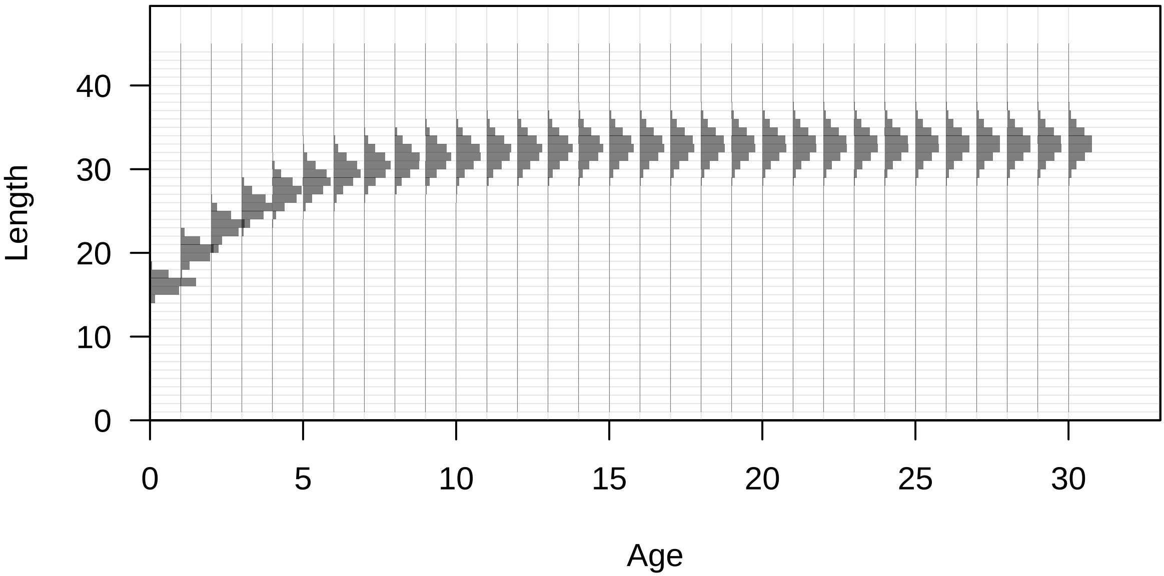


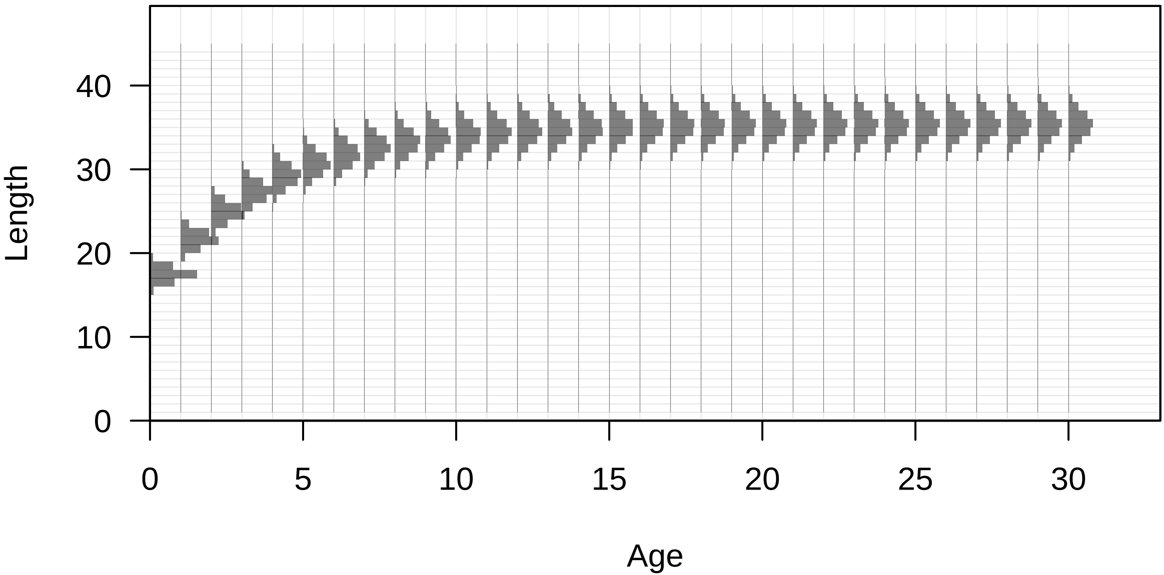
Length
0
10
20
30
40
0
5
10
15
20
25
30
Age


Length
Age
0
10
20
30
40
0
5
10
15
20
25
30

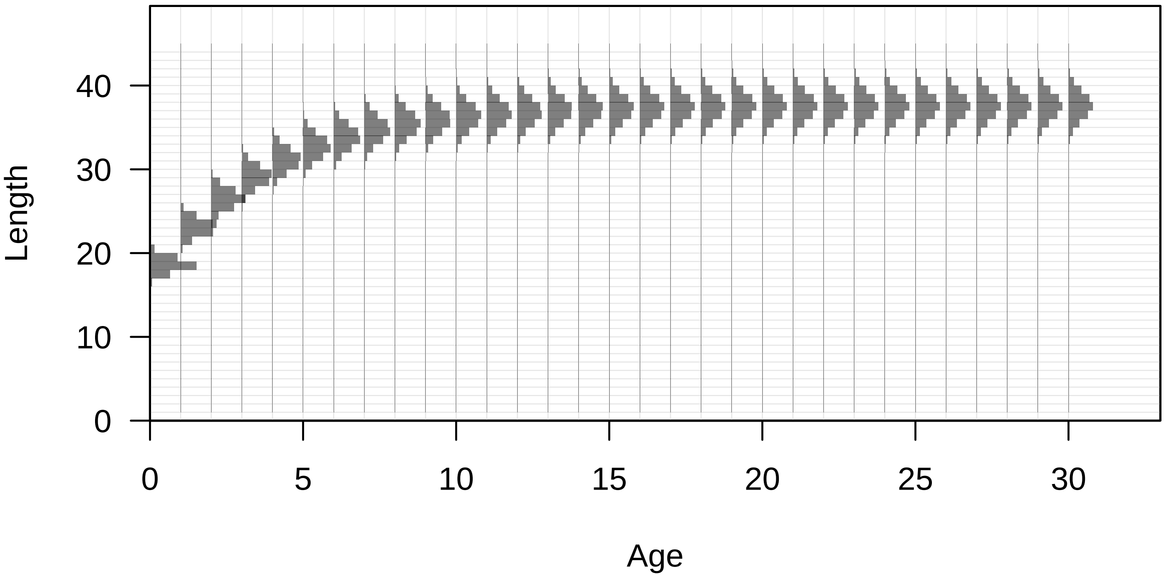


Length
0
10
20
30
40
0
5
10
15
20
25
30
Age

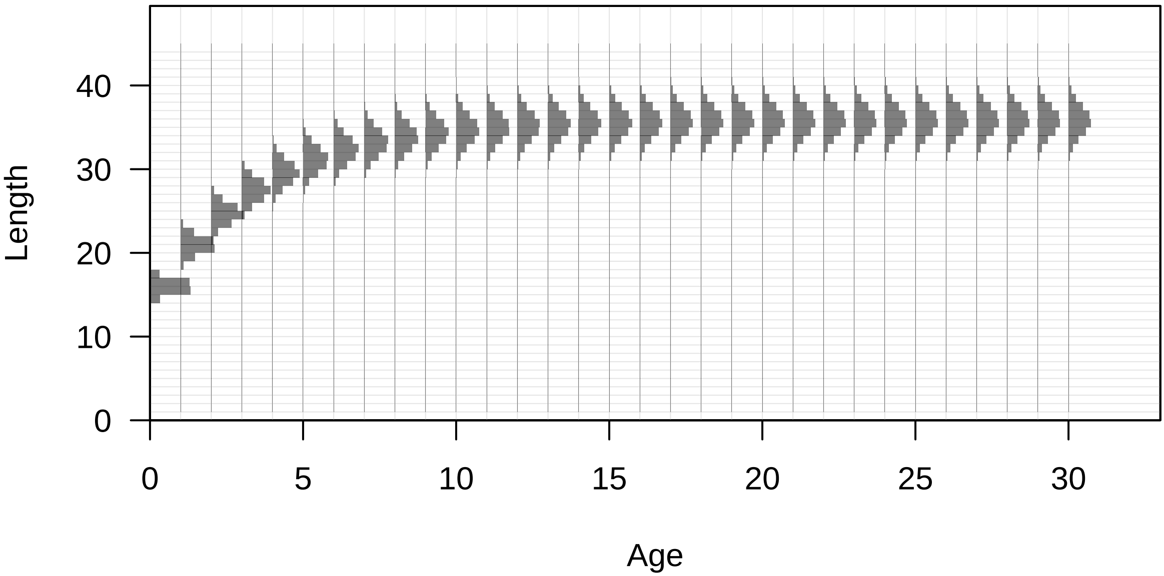

Length
0
10
20
30
40
0
5
10
15
20
25
30
Age



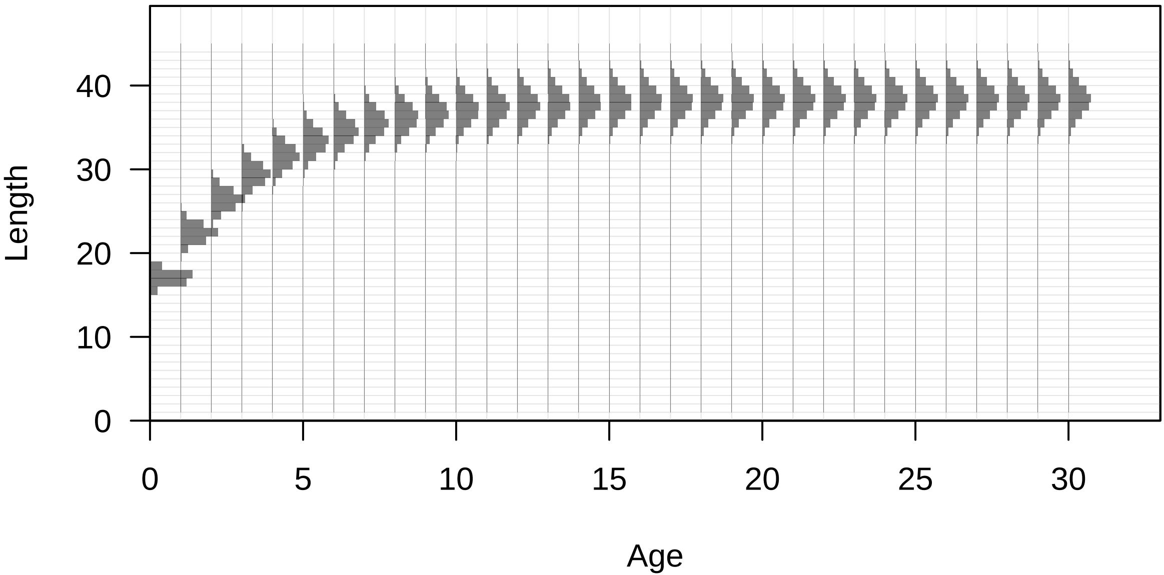
Length
Age
0
10
20
30
40
0
5
10
15
20
25
30



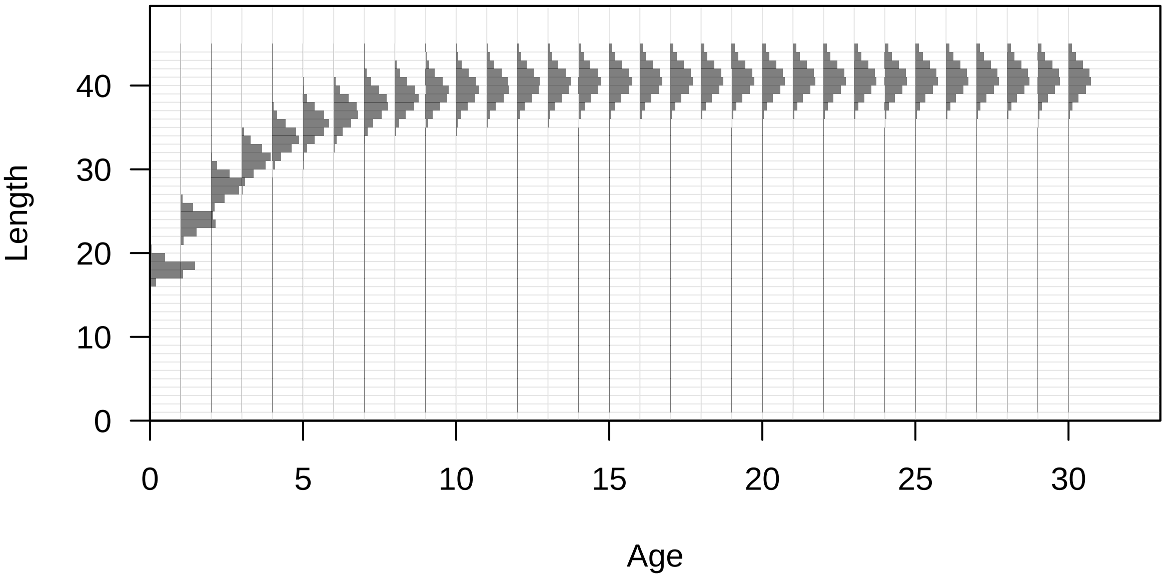
Length
0
10
20
30
40
0
5
10
15
20
25
30
Age



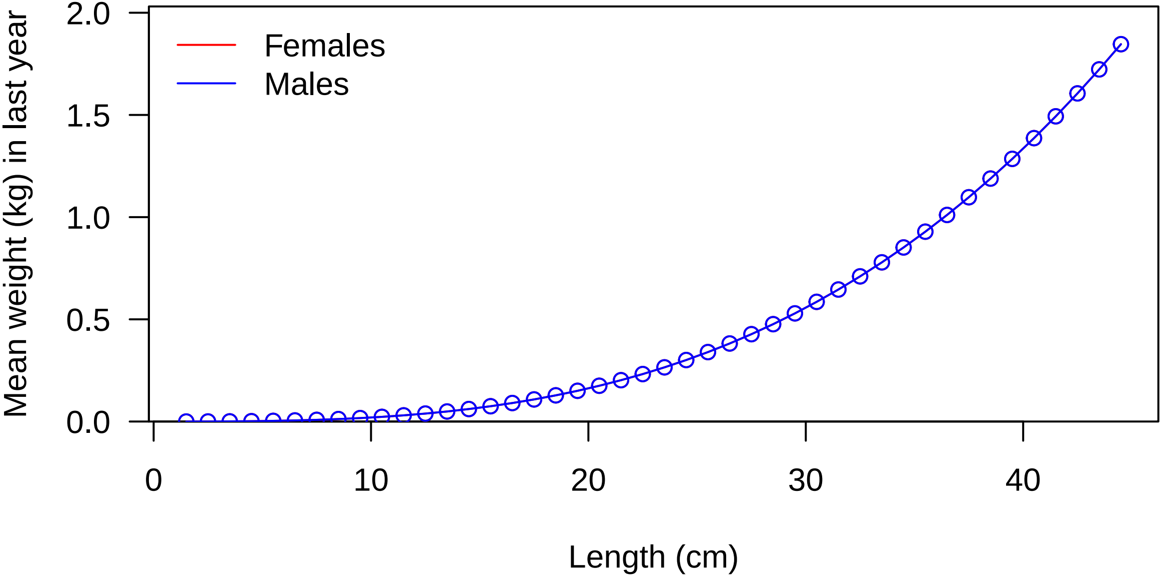
Length
0
10
20
30
40
0
5
10
15
20
25
30
Age

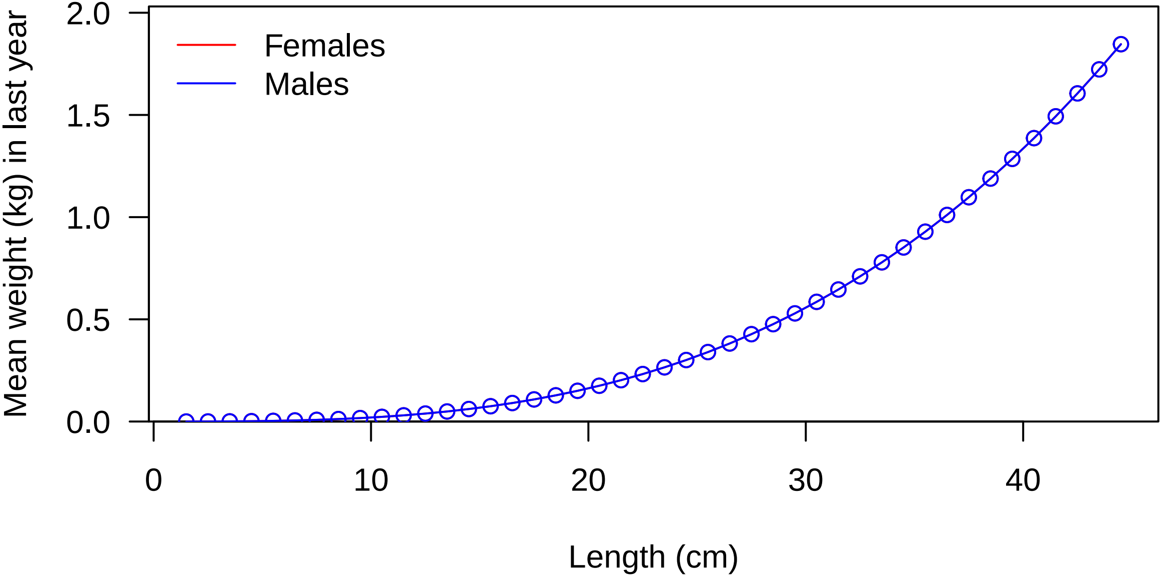


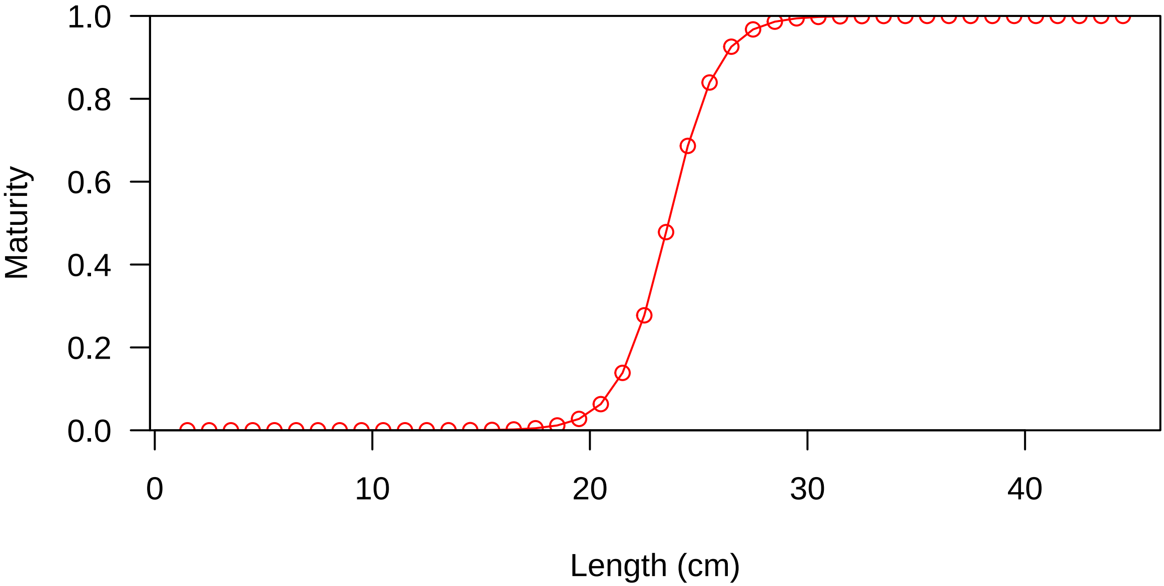
Length
0
10
20
30
40
0
5
10
15
20
25
30
Age



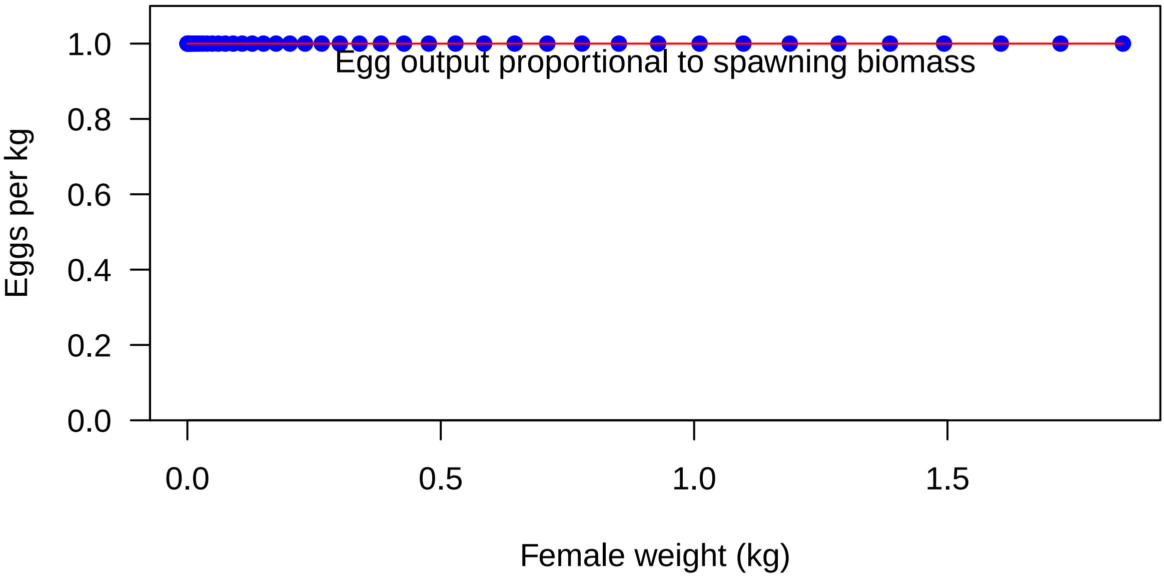
Length
0
10
20
30
40
0
5
10
15
20
25
30
Age


Mean weight (kg) in last year
2.0
1.5
1.0
0.5
0.0
Females
Males
0
10
20
30
40
Length (cm)

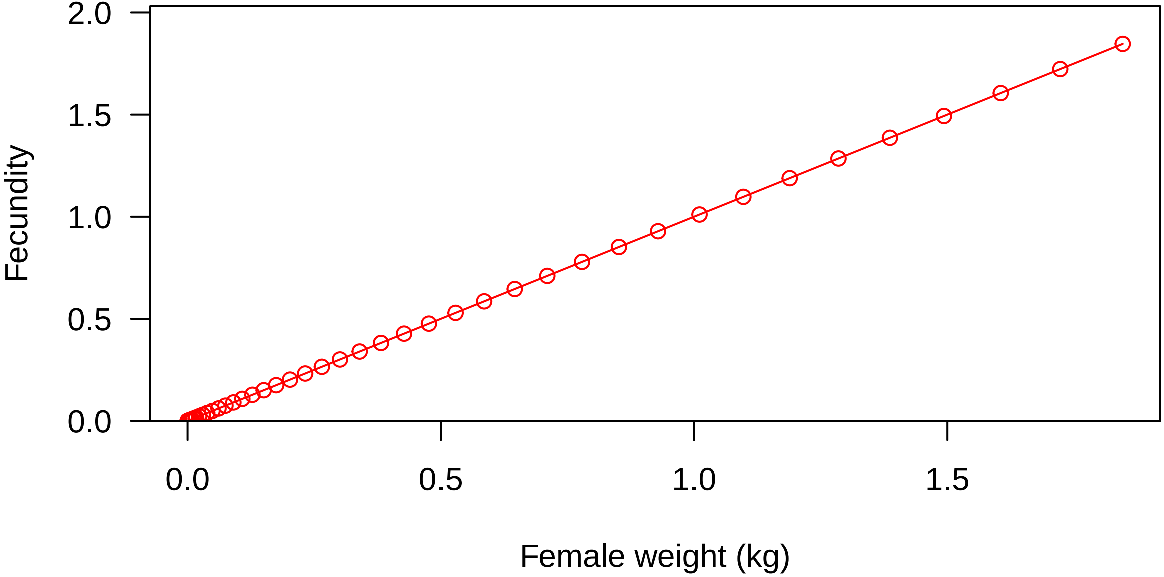

Females
Males
Mean weight (kg) in last year
2.0
1.5
1.0
0.5
0.0
0
10
20
30
40
Length (cm)



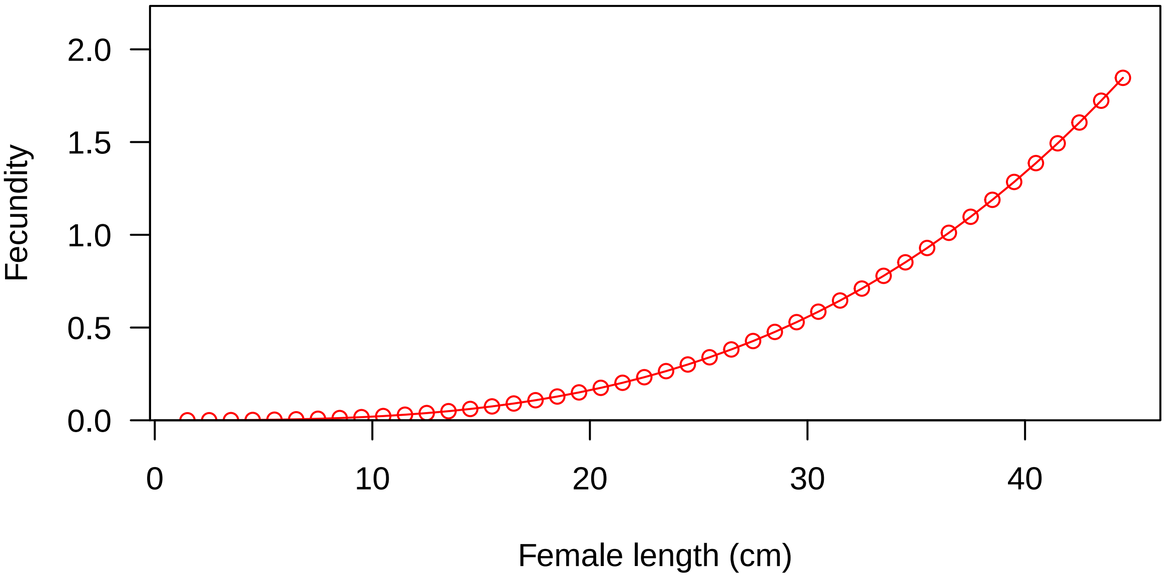
1.0
0.8
0.6
0.4
0.2
0.0
Maturity
0
10
20
30
40
Length (cm)

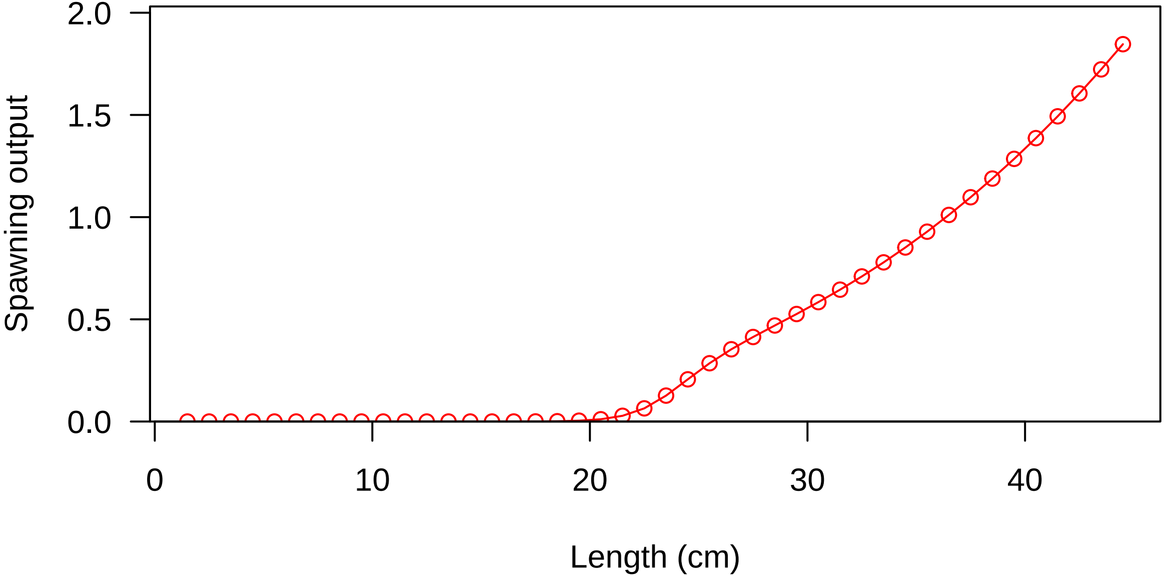


Egg output proportional to spawning biomass
Eggs per kg
1.0
0.8
0.6
0.4
0.2
0.0
0.0
0.5
1.0
1.5
Female weight (kg)

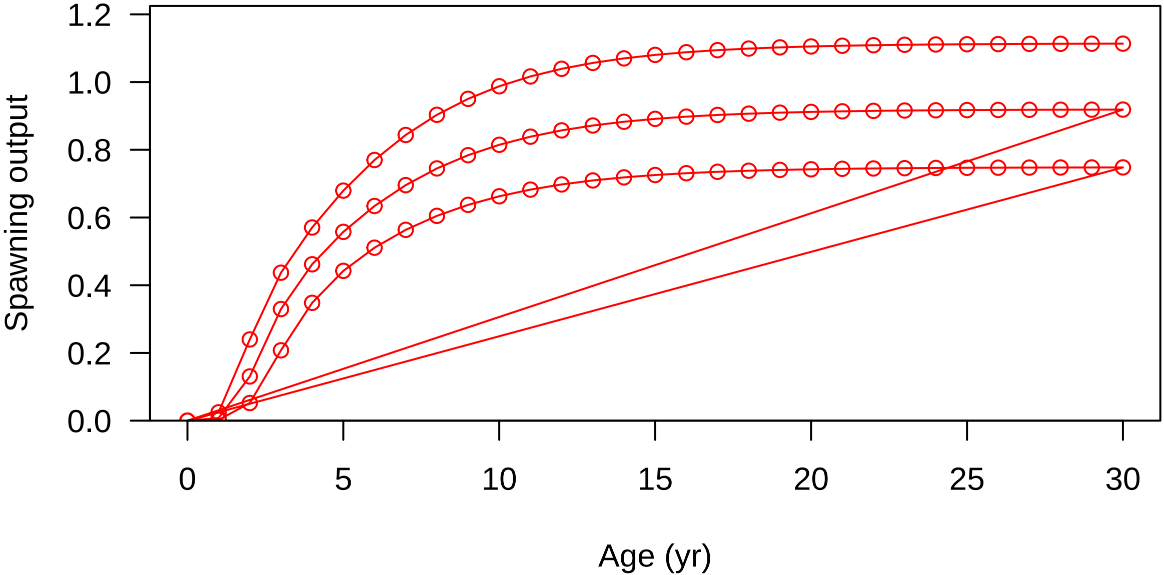


Fecundity
2.0
1.5
1.0
0.5
0.0
0.0
0.5
1.0
1.5
Female weight (kg)

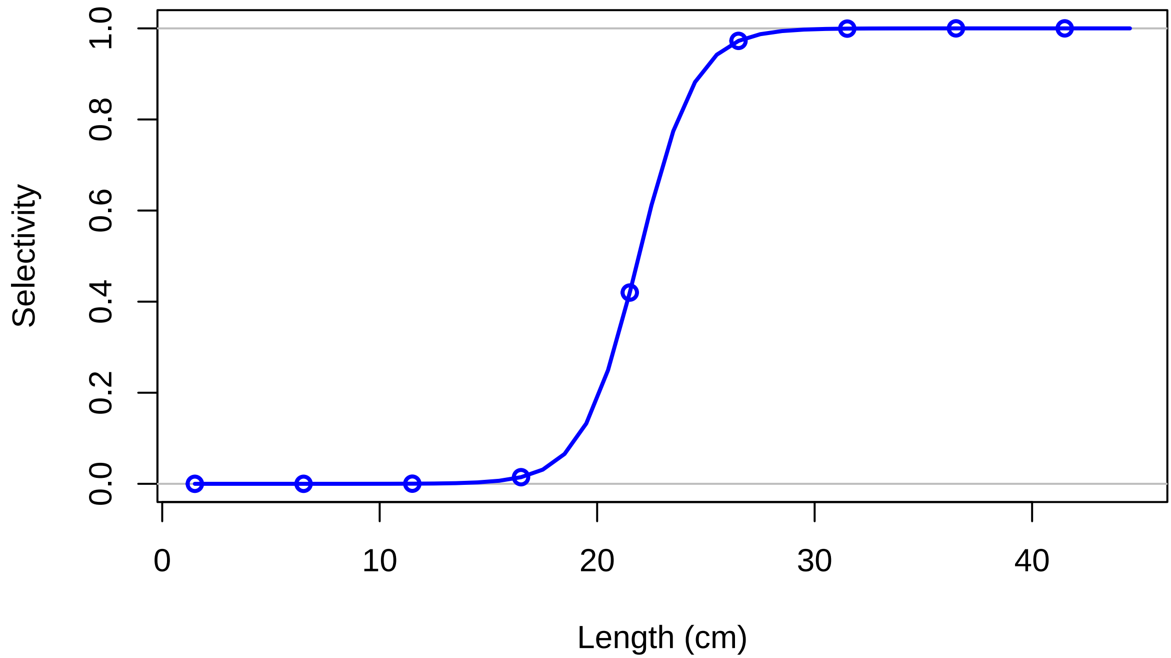

Fecundity
0.0
0.5
1.0
1.5
2.0
0
10
20
30
40
Female length (cm)



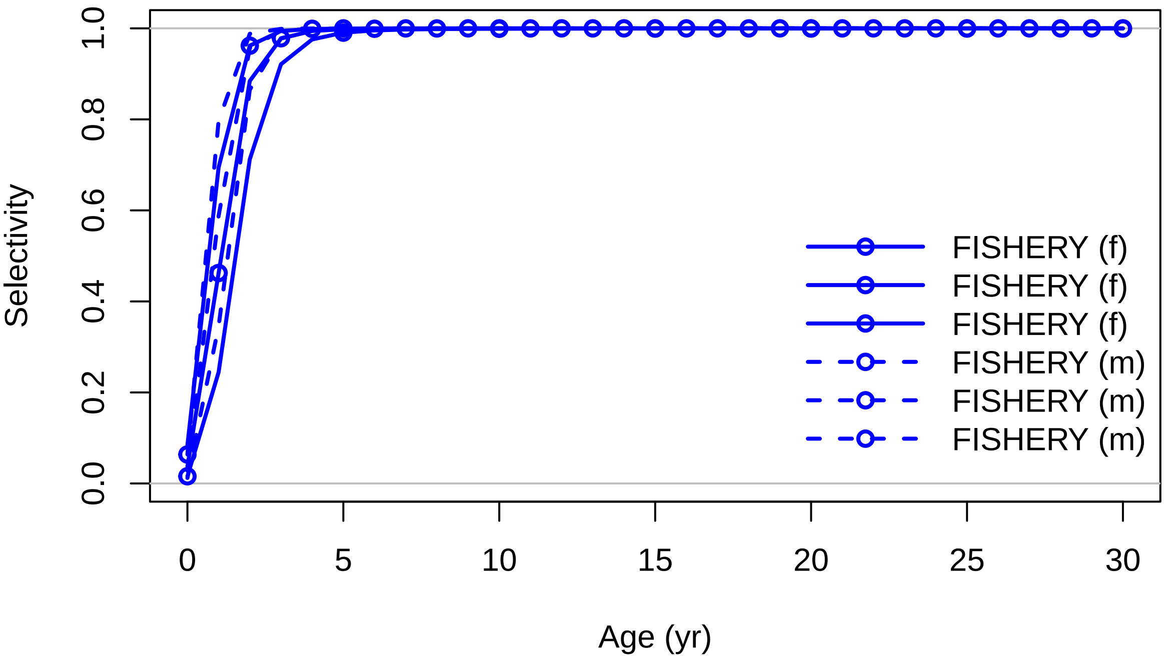
Spawning output
2.0
1.5
1.0
0.5
0.0
0
10
20
30
40
Length (cm)



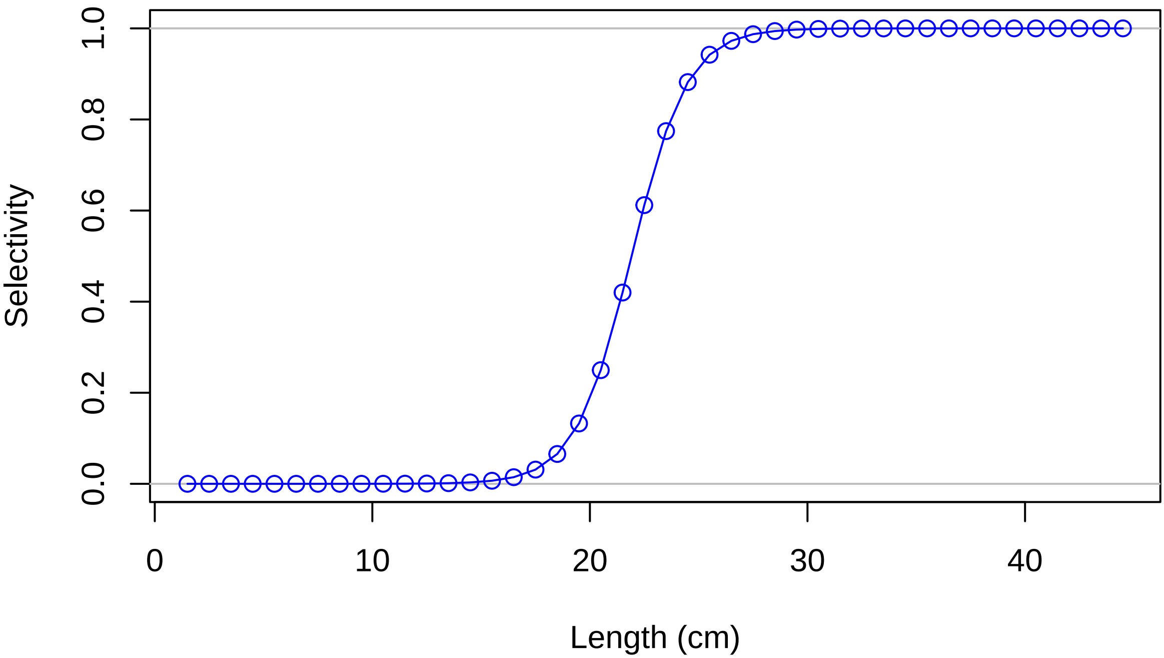
Spawning output
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0
5
10
15
20
25
30
Age (yr)



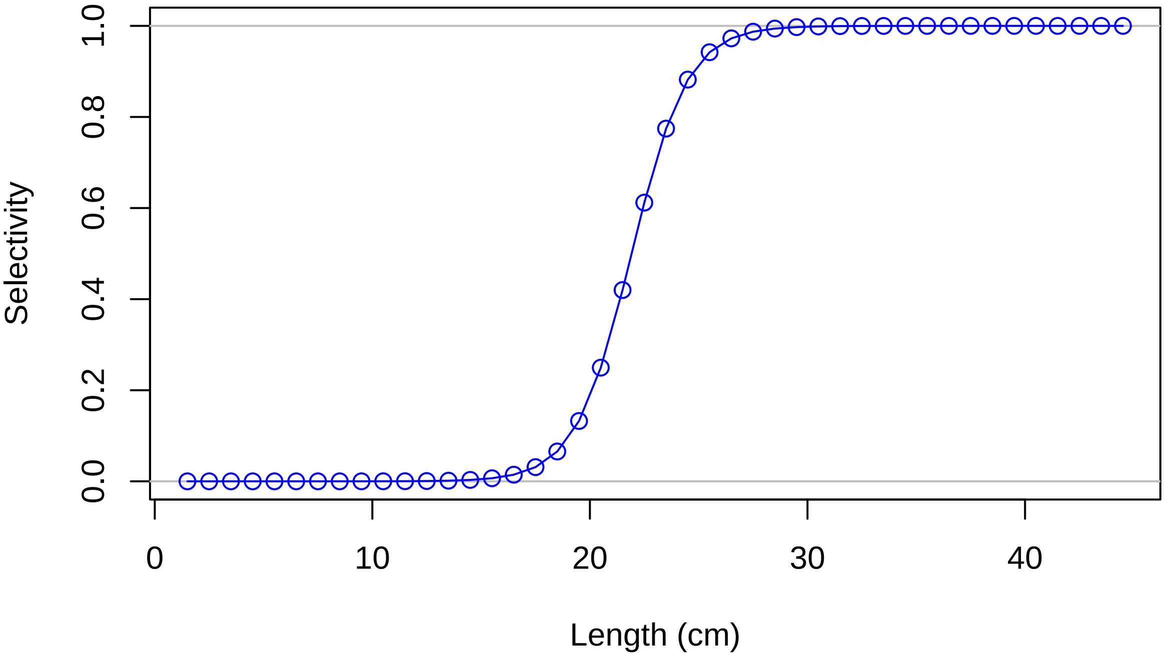
Selectivity
0.0
0.2
0.4
0.6
0.8
1.0
0
10
20
30
40
Length (cm)



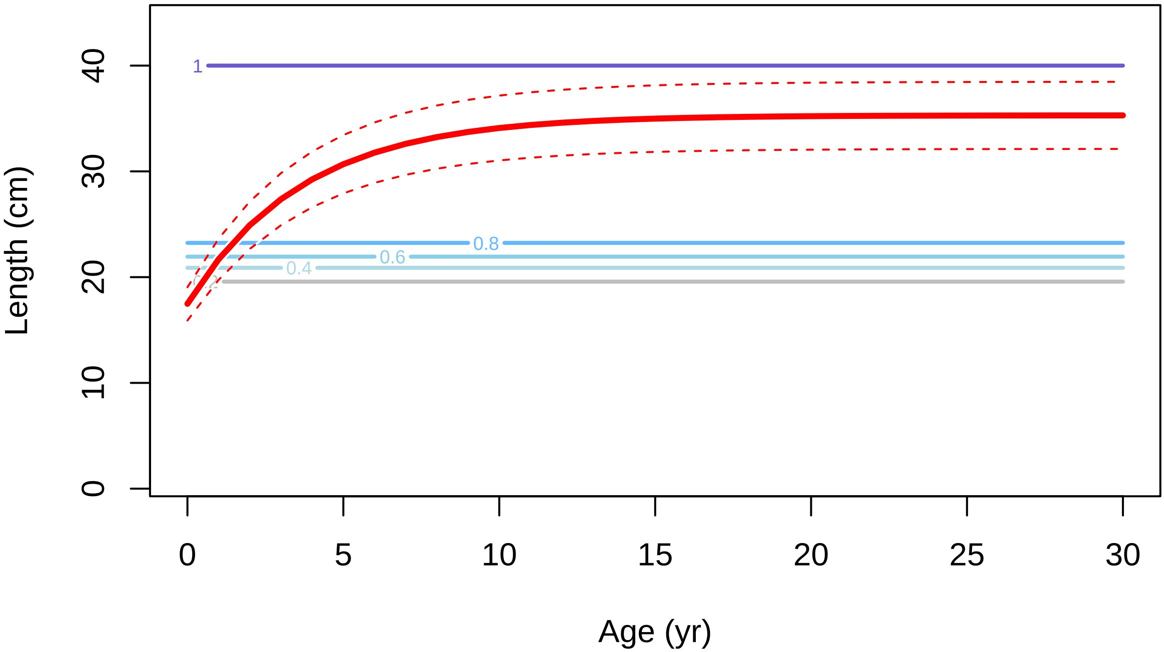
Selectivity
0.0
0.2
0.4
0.6
0.8
1.0
0
5
10
15
20
25
30
Age (yr)
FISHERY (f)
FISHERY (f)
FISHERY (f)
FISHERY (m)
FISHERY (m)
FISHERY (m)

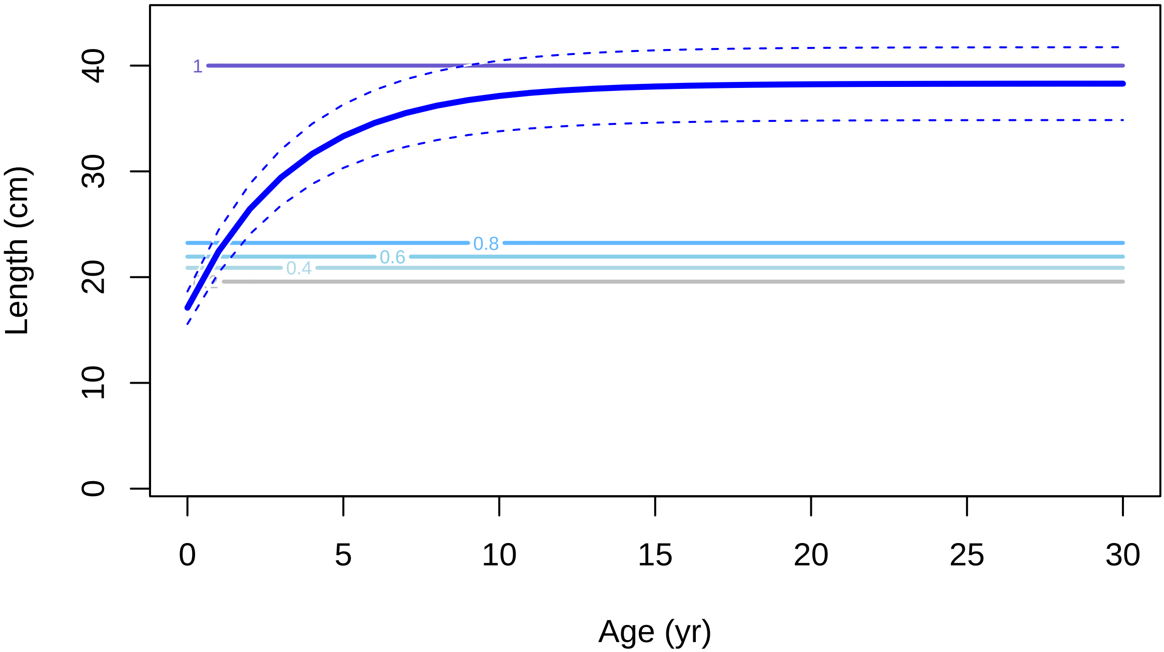


Selectivity
0.0
0.2
0.4
0.6
0.8
1.0
0
10
20
30
40
Length (cm)

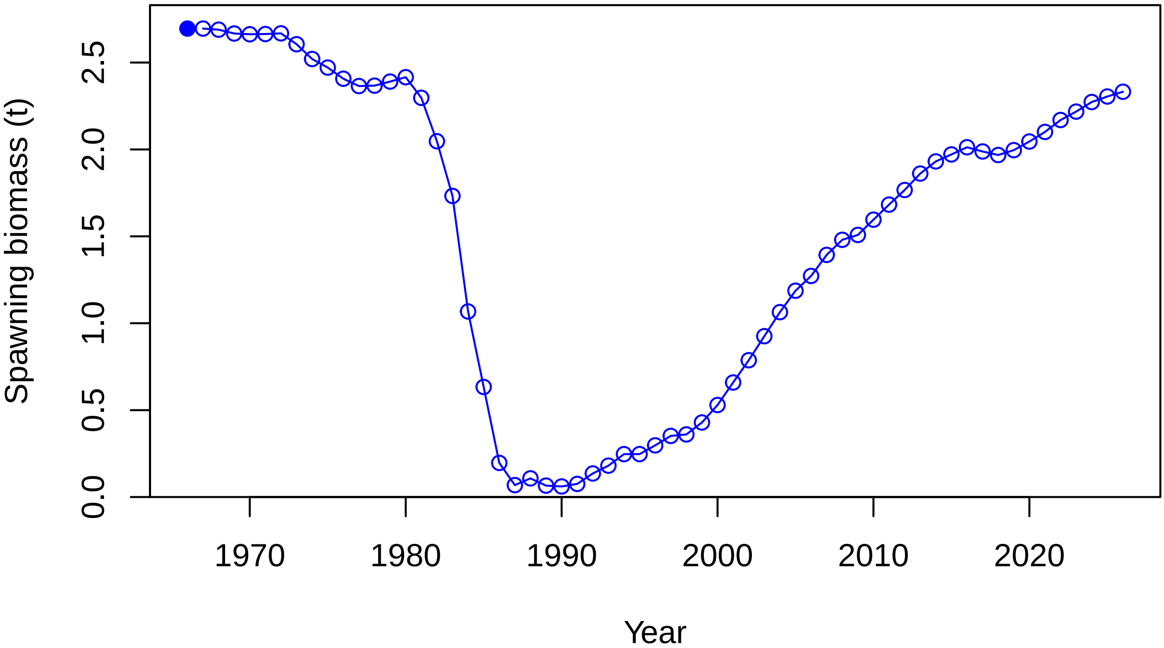


Selectivity
1.0
0.8
0.6
0.4
0.2
0.0
0
10
20
30
40
Length (cm)

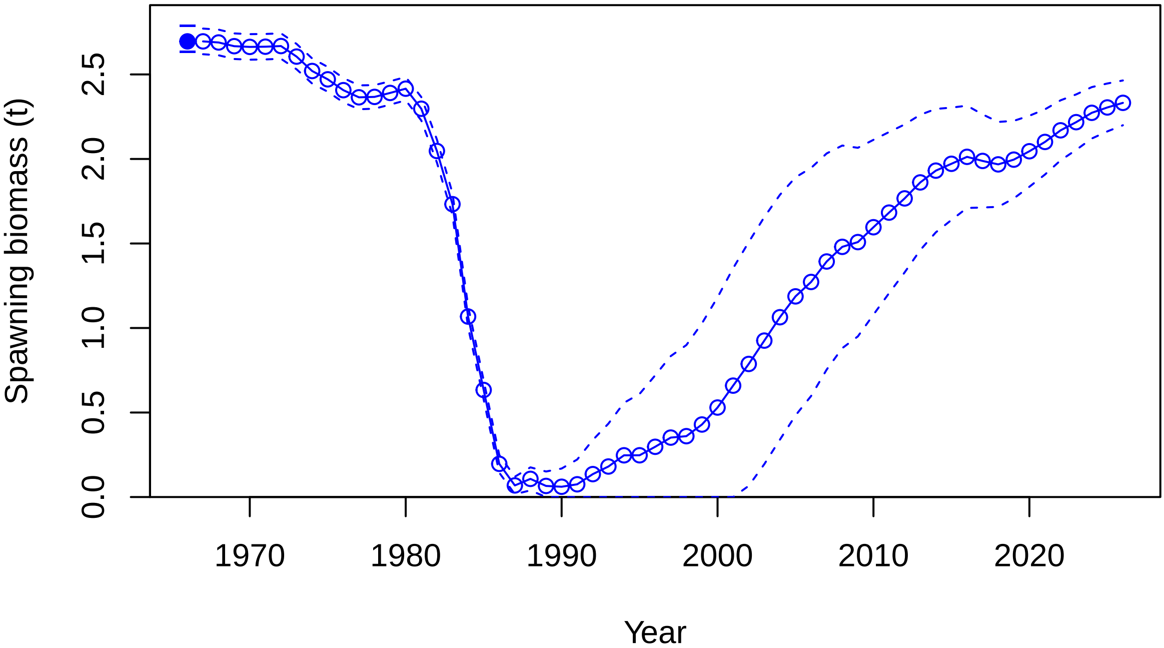

Length (cm)
0
10
20
30
40
1
0.8
0.6
0.4
0.2
0
5
10
15
20
25
30
Age (yr)



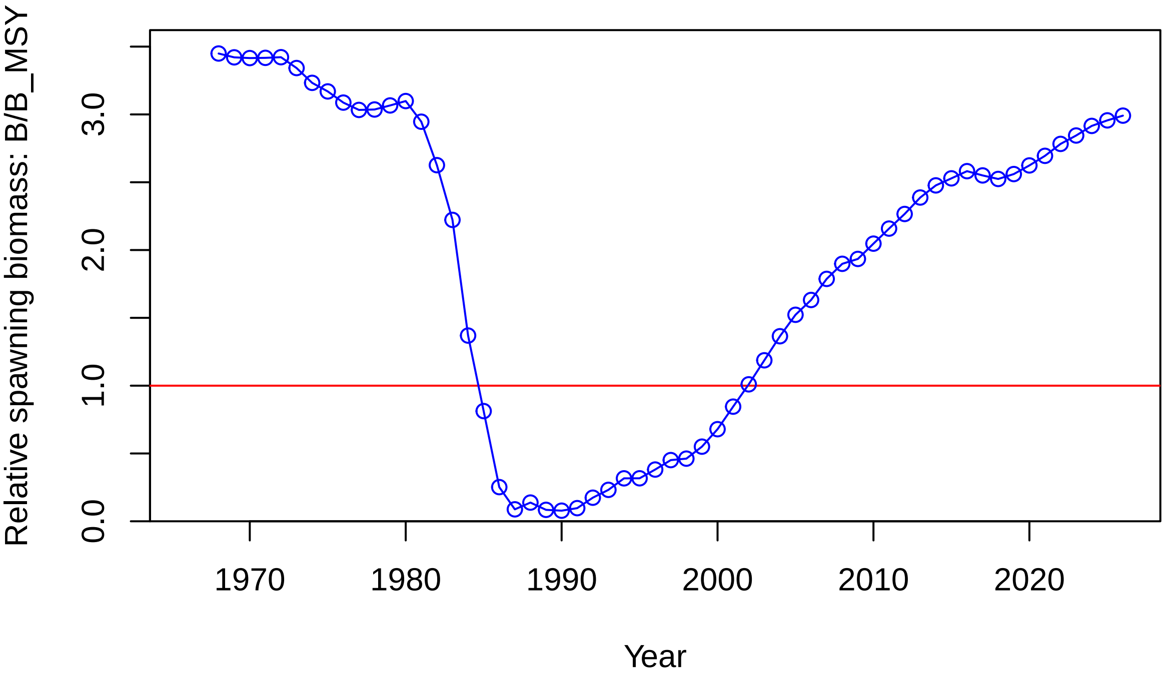
Length (cm)
0
10
20
30
40
Age (yr)
0
5
10
15
20
25
30
1
0.8
0.6
0.4



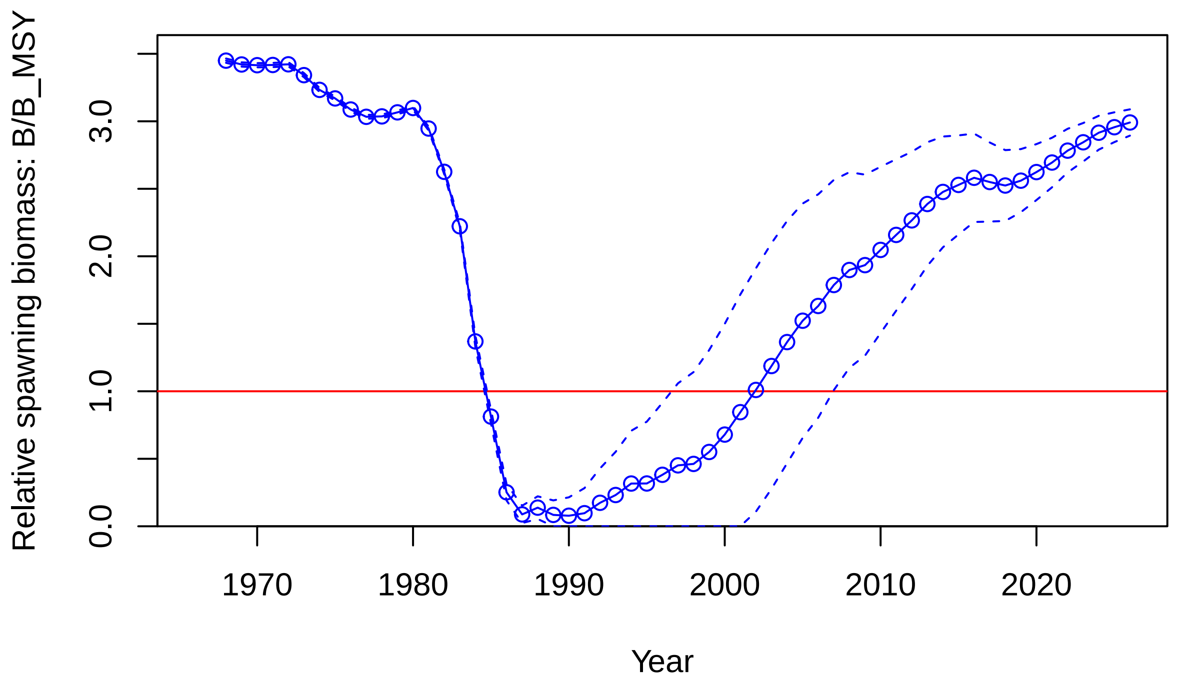
Spawning biomass (t)
0.0
0.5
1.0
1.5
2.0
2.5
1970
1980
1990
2000
2010
2020
Year



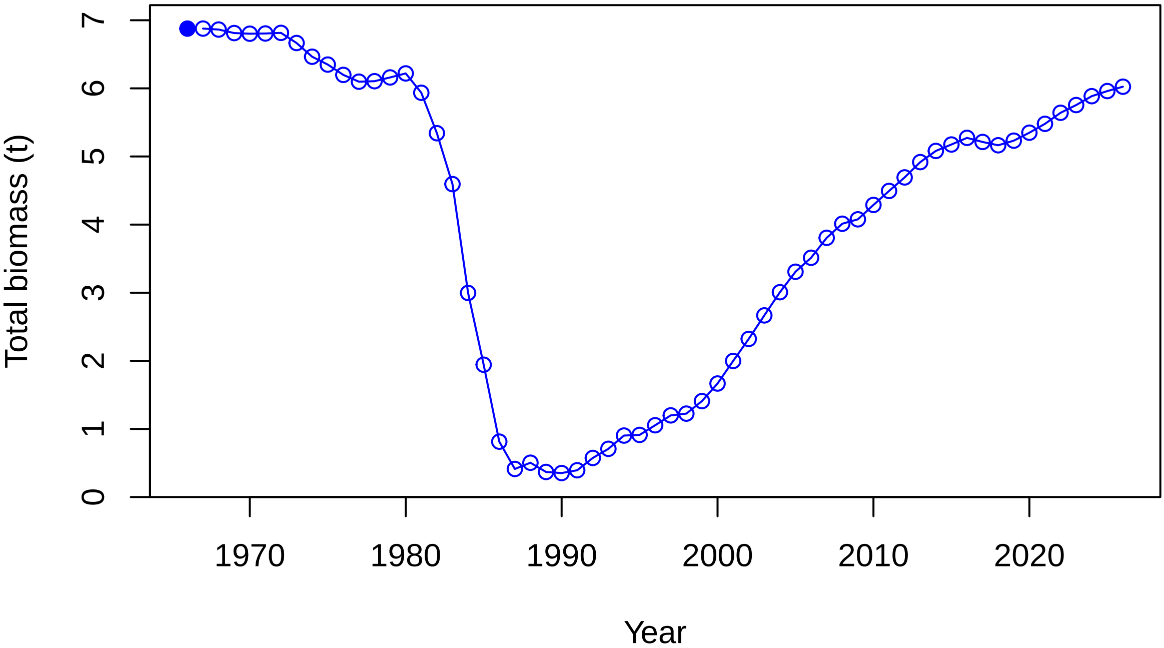
Spawning biomass (t)
0.0
0.5
1.0
1.5
2.0
2.5
1970
1980
1990
2000
2010
2020
Year

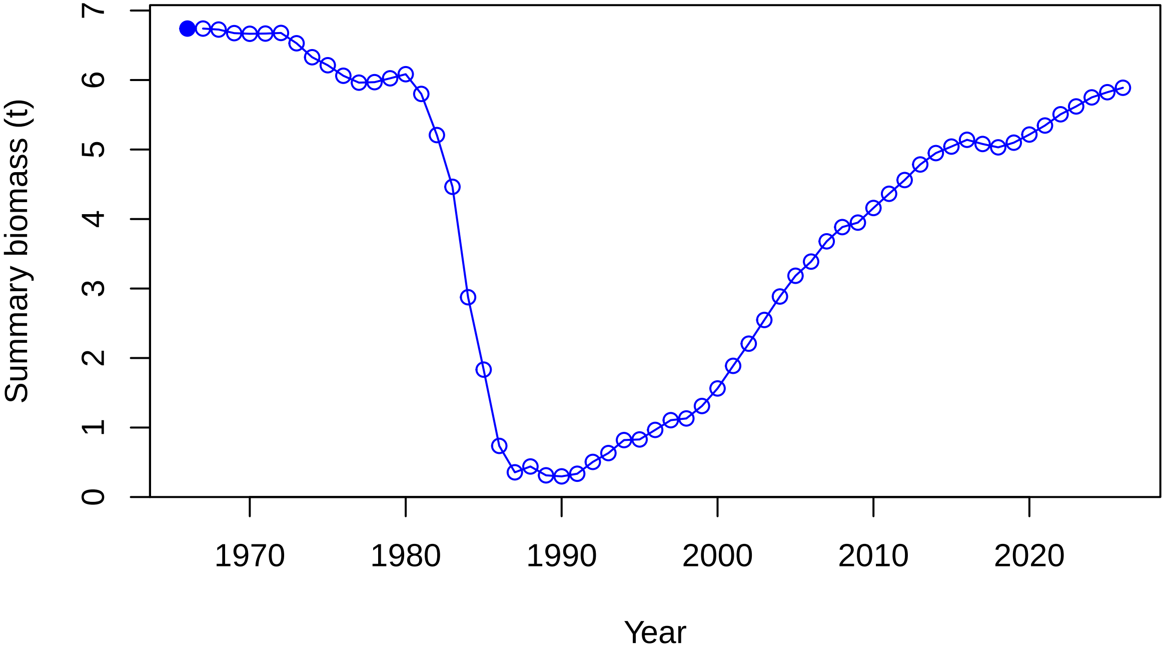


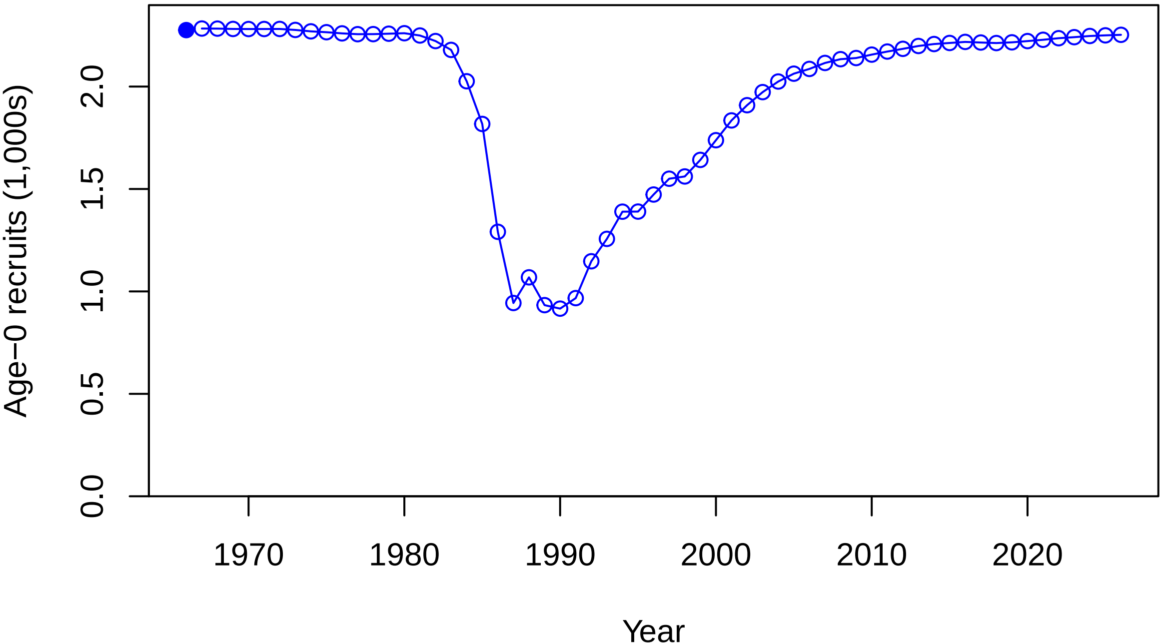
Relative spawning biomass: B/B_MSY
0.0
1.0
2.0
3.0
1970
1980
1990
2000
2010
2020
Year



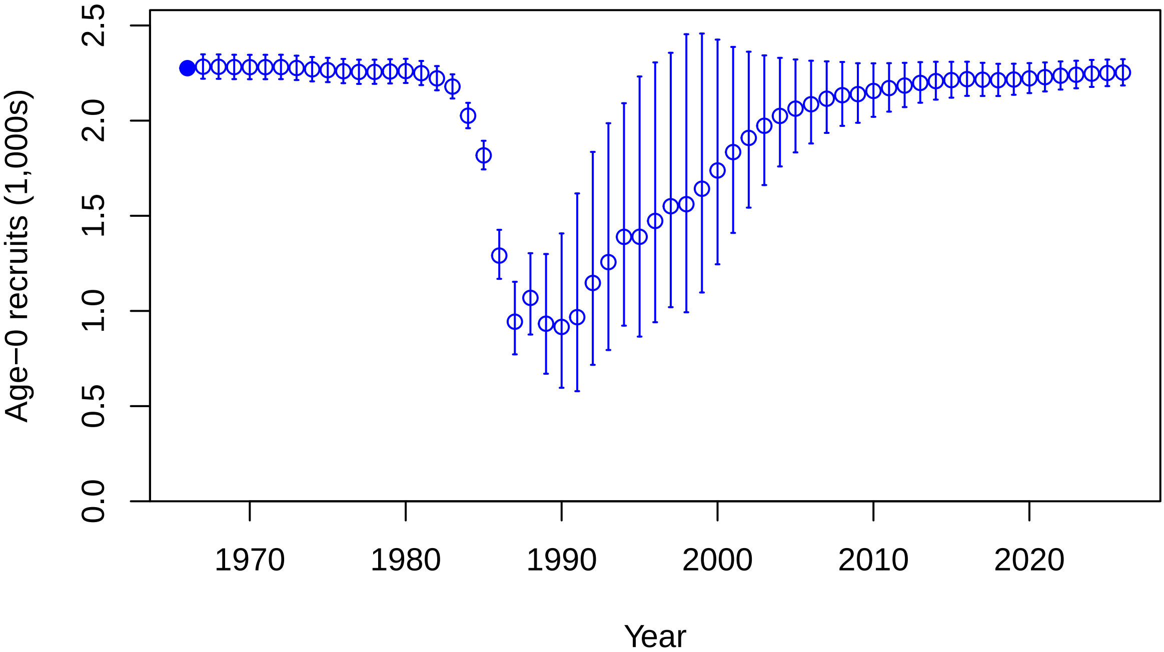
Relative spawning biomass: B/B_MSY
0.0
1.0
2.0
3.0
1970
1980
1990
2000
2010
2020
Year


Total biomass (t)
0
1
2
3
4
5
6
7
1970
1980
1990
2000
2010
2020
Year

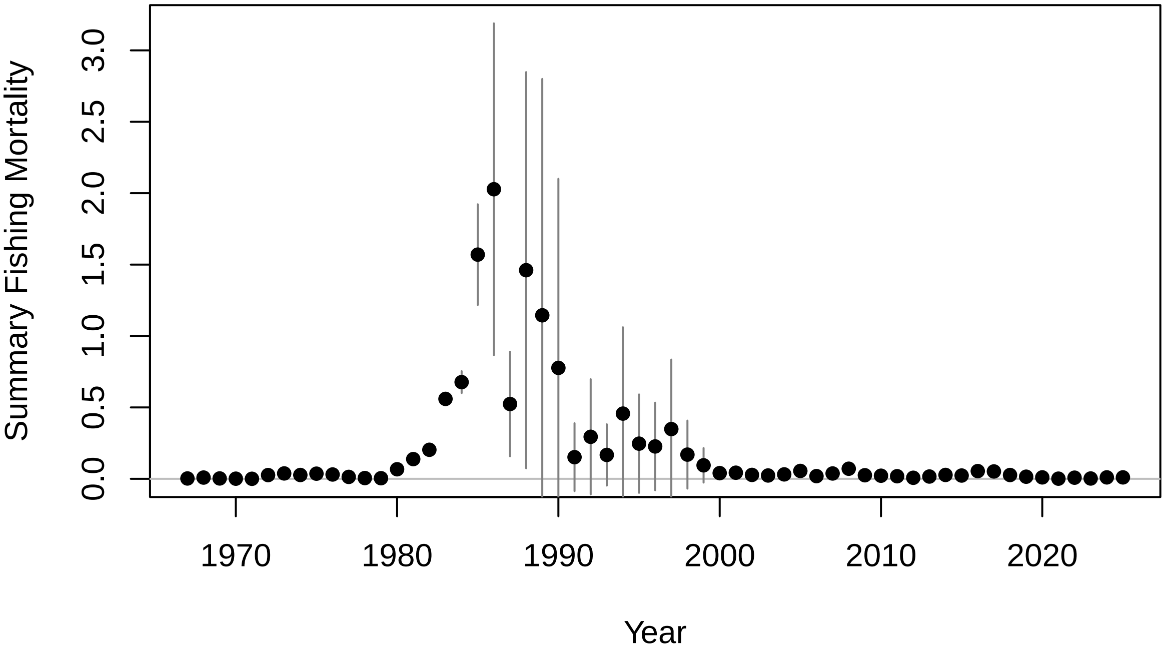

Summary biomass (t)
0
1
2
3
4
5
6
7
1970
1980
1990
2000
2010
2020
Year



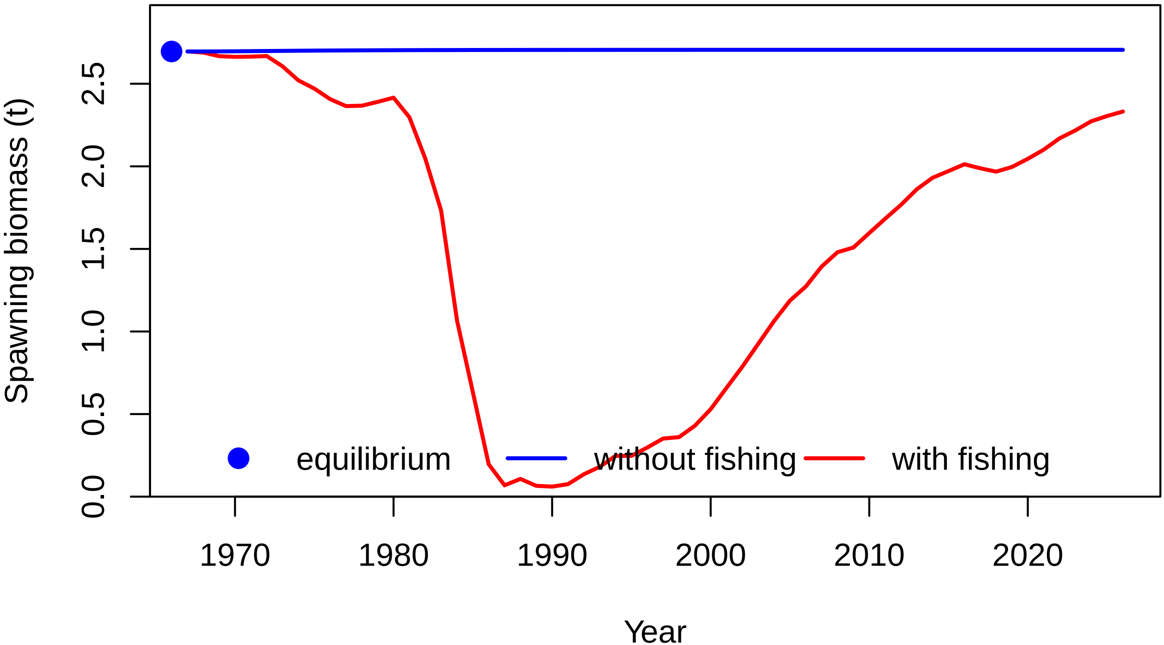
Age−0 recruits (1,000s)
0.0
0.5
1.0
1.5
2.0
1970
1980
1990
2000
2010
2020
Year

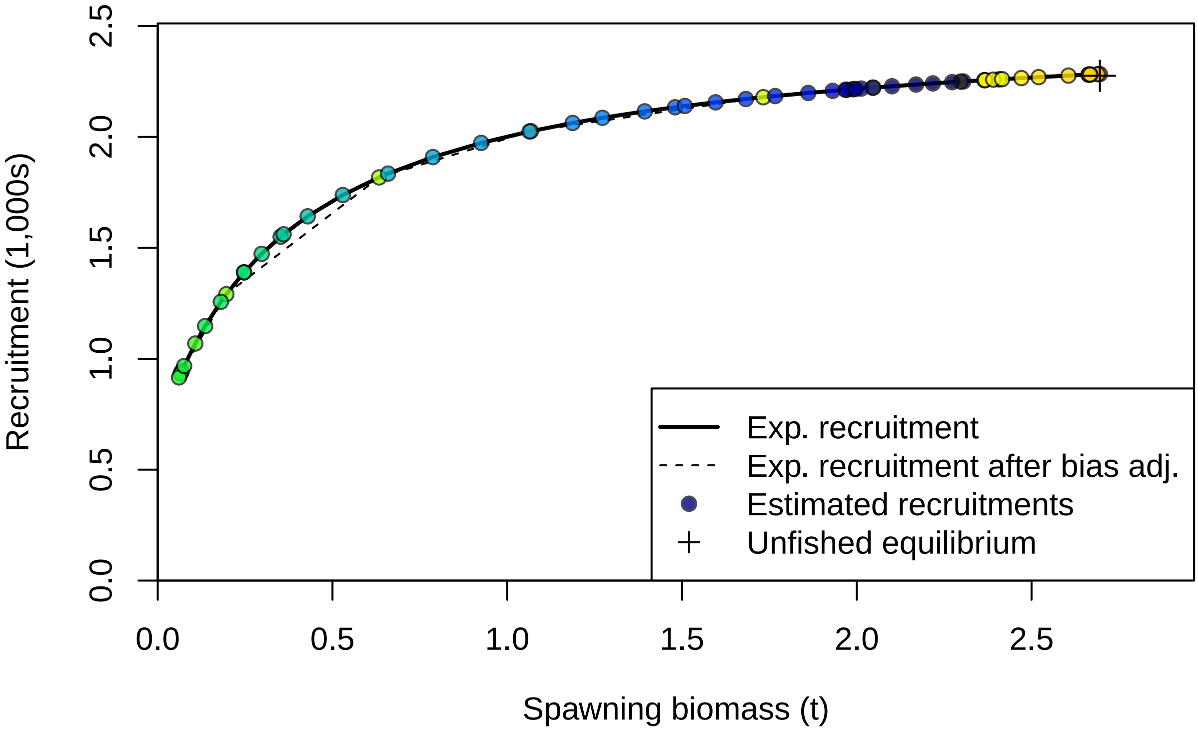


Age−0 recruits (1,000s)
0.0
0.5
1.0
1.5
2.0
2.5
1970
1980
1990
2000
2010
2020
Year

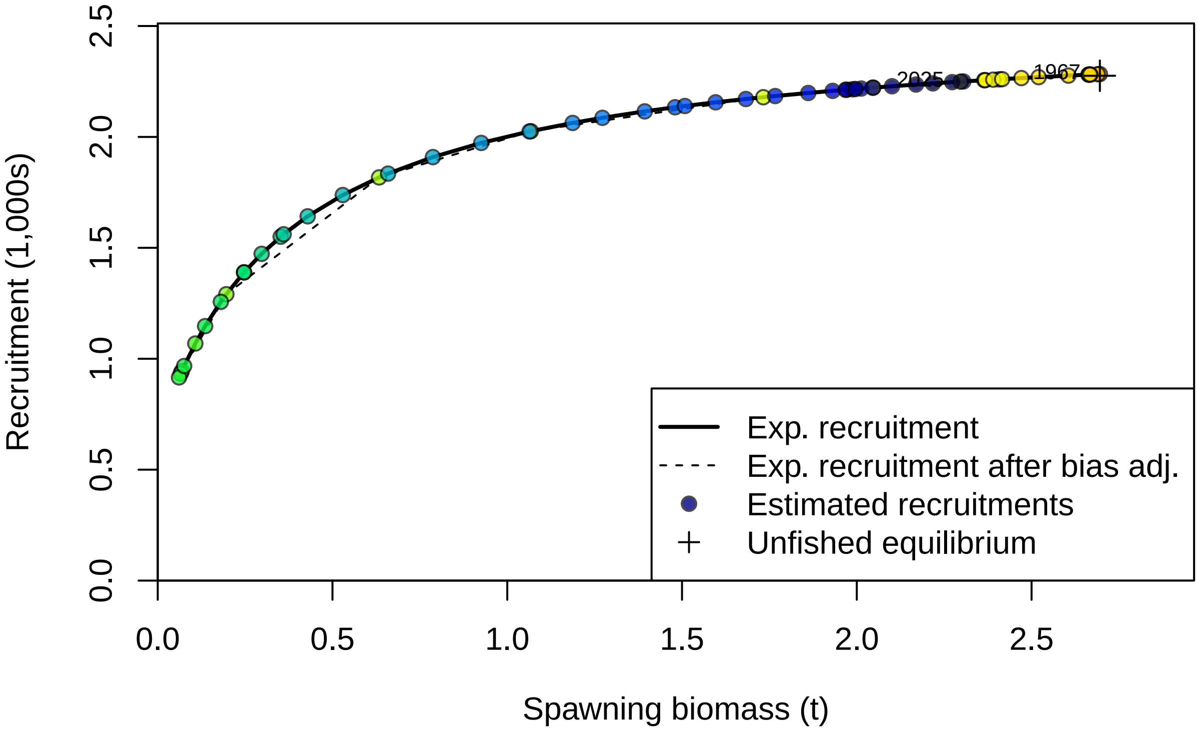


Summary Fishing Mortality
0.0
0.5
1.0
1.5
2.0
2.5
3.0
1970
1980
1990
2000
2010
2020
Year

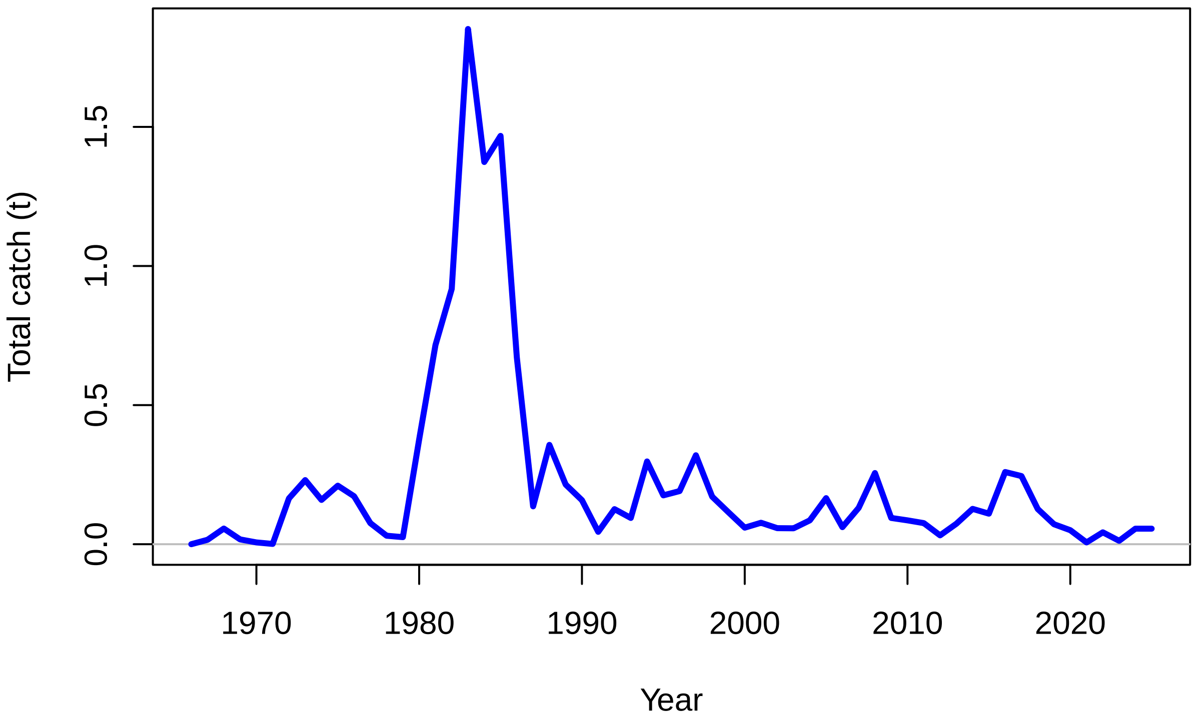

Spawning biomass (t)
0.0
0.5
1.0
1.5
2.0
2.5
1970
1980
1990
2000
2010
2020
Year
equilibrium
without fishing
with fishing



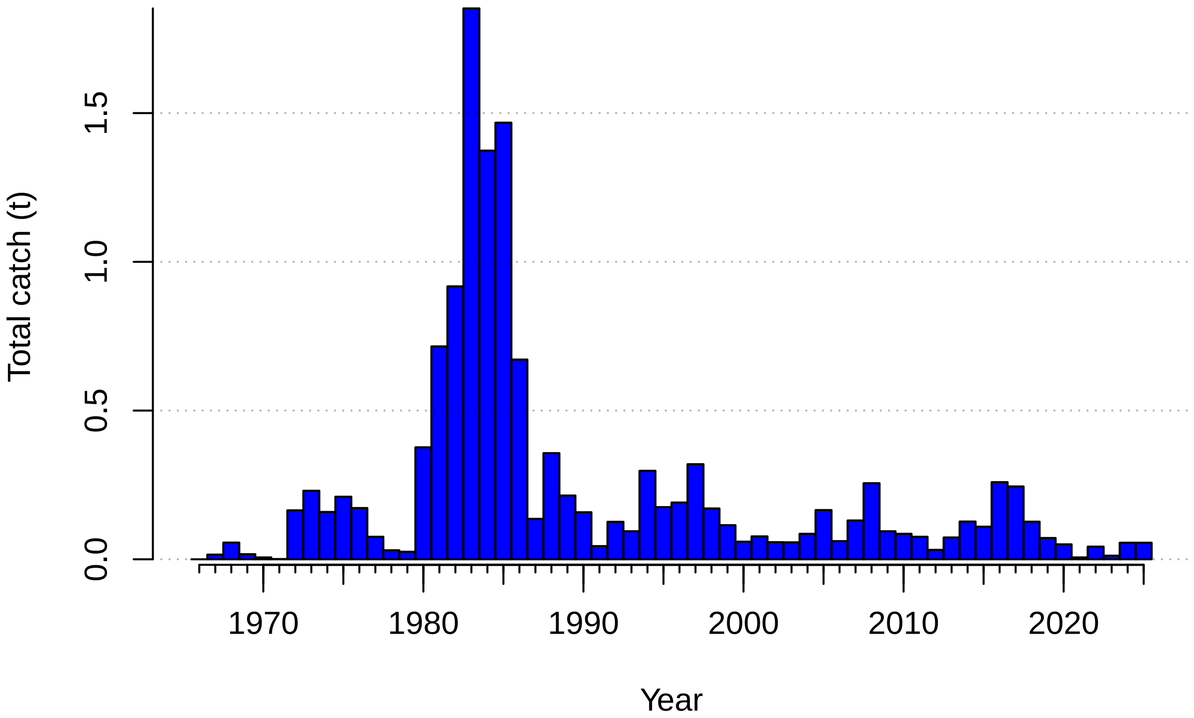
Recruitment (1,000s)
0.0
0.5
1.0
1.5
2.0
2.5
Spawning biomass (t)
Exp. recruitment
Exp. recruitment after bias adj.
Estimated recruitments
Unfished equilibrium



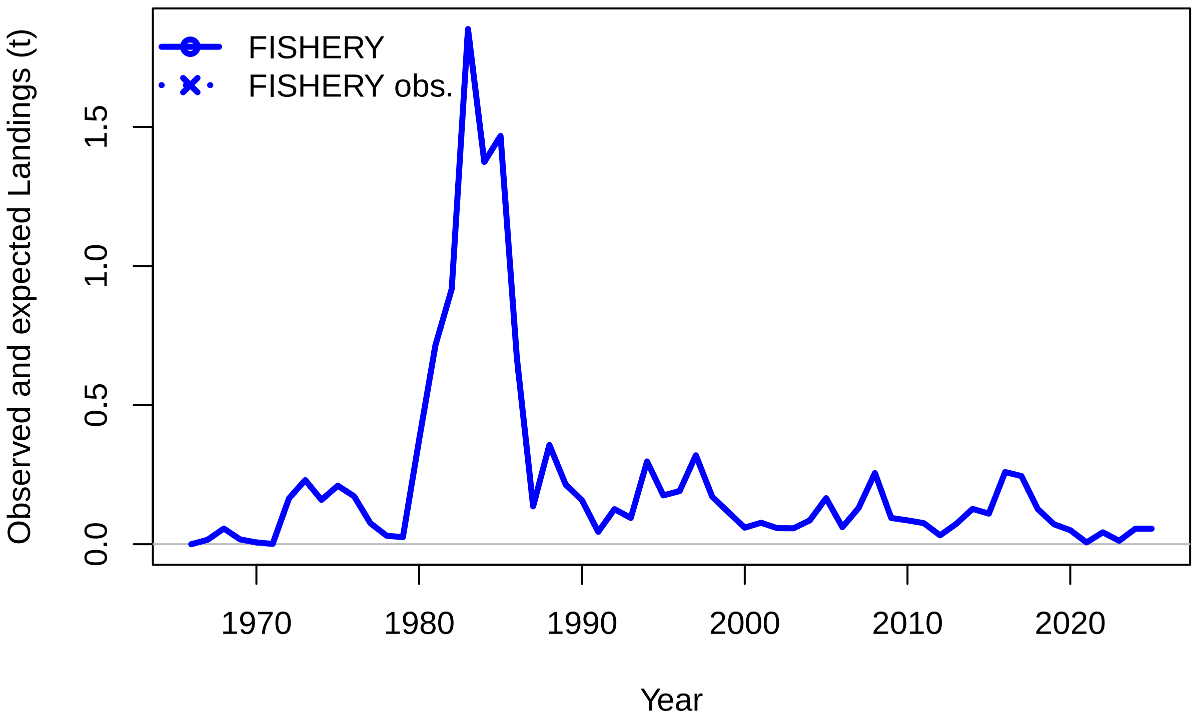
Recruitment (1,000s)
0.0
0.5
1.0
1.5
2.0
2.5
Spawning biomass (t)
0.0
0.5
1.0
1.5
2.0
2.5
2025
1967
Exp. recruitment
Exp. recruitment after bias adj.
Estimated recruitments
Unfished equilibrium



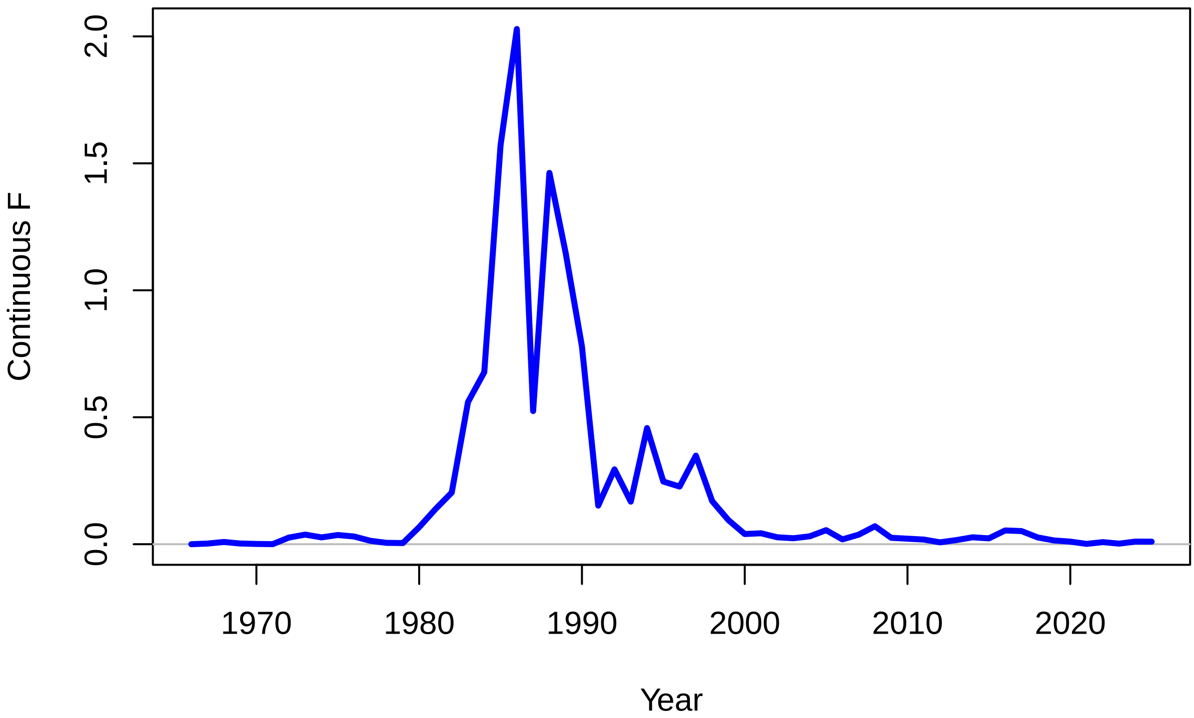
Total catch (t)
0.0
0.5
1.0
1.5
1970
1980
1990
2000
2010
2020
Year



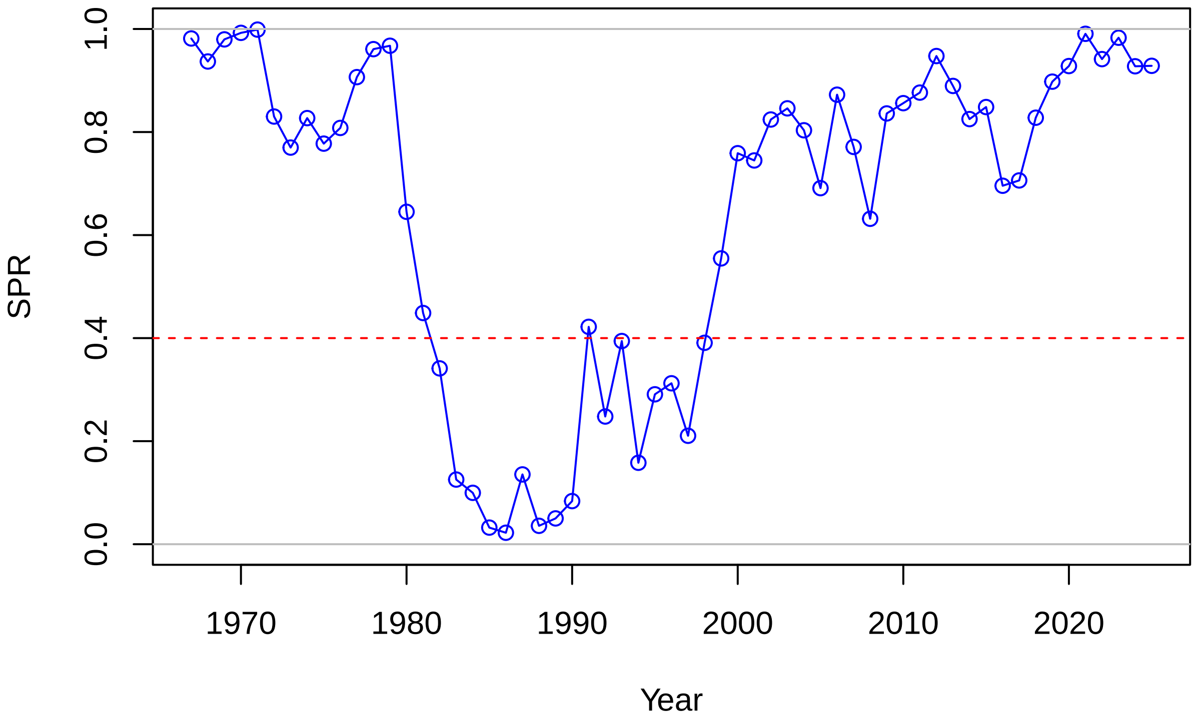
Total catch (t)
0.0
0.5
1.0
1.5
1970
1980
1990
2000
2010
2020
Year

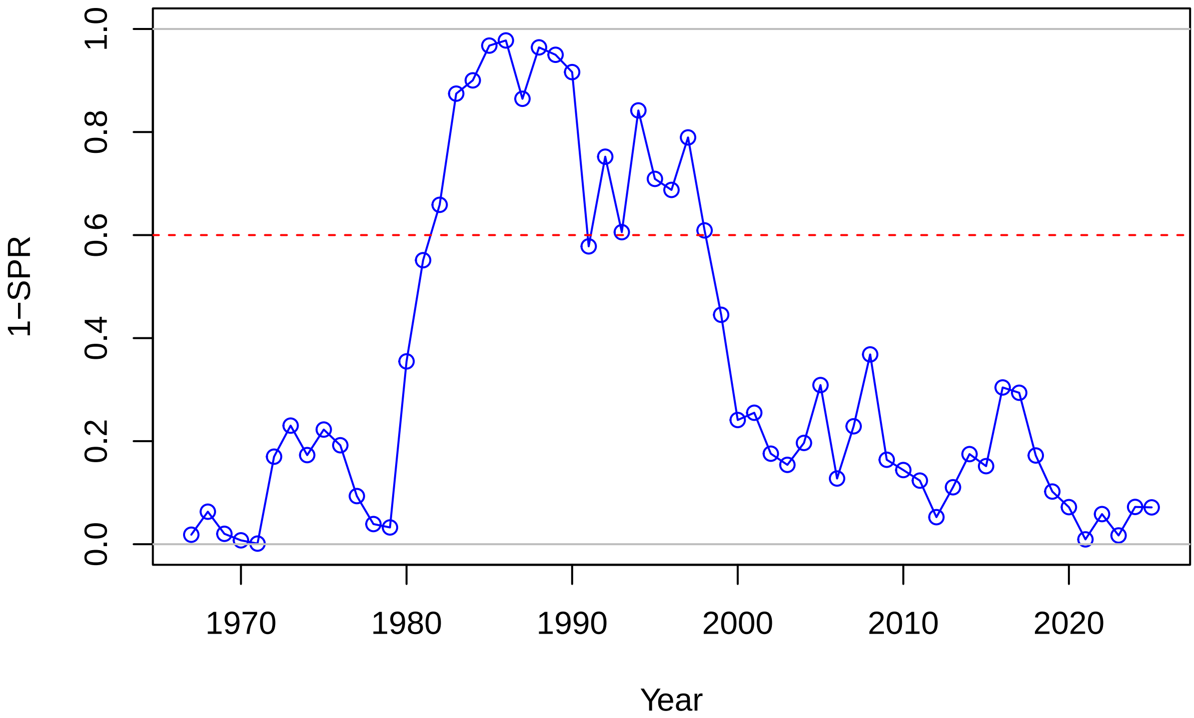


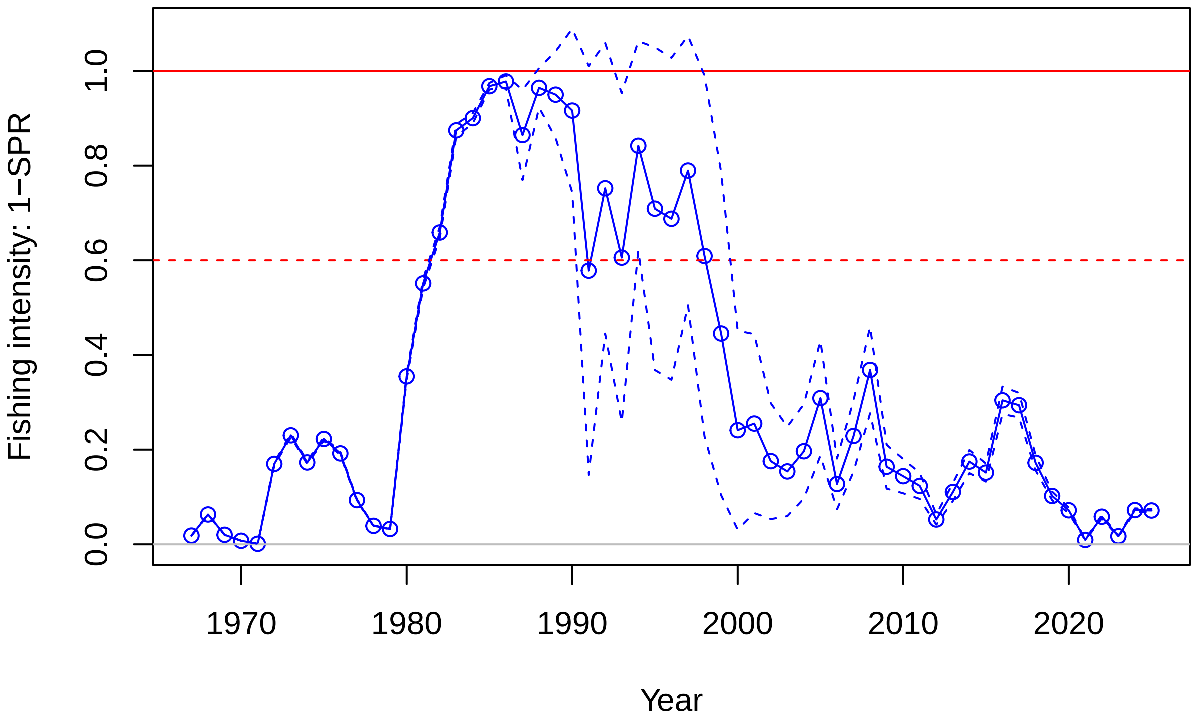
FISHERY
FISHERY obs.
Observed and expected Landings (t)
0.0
0.5
1.0
1.5
1970
1980
1990
2000
2010
2020
Year



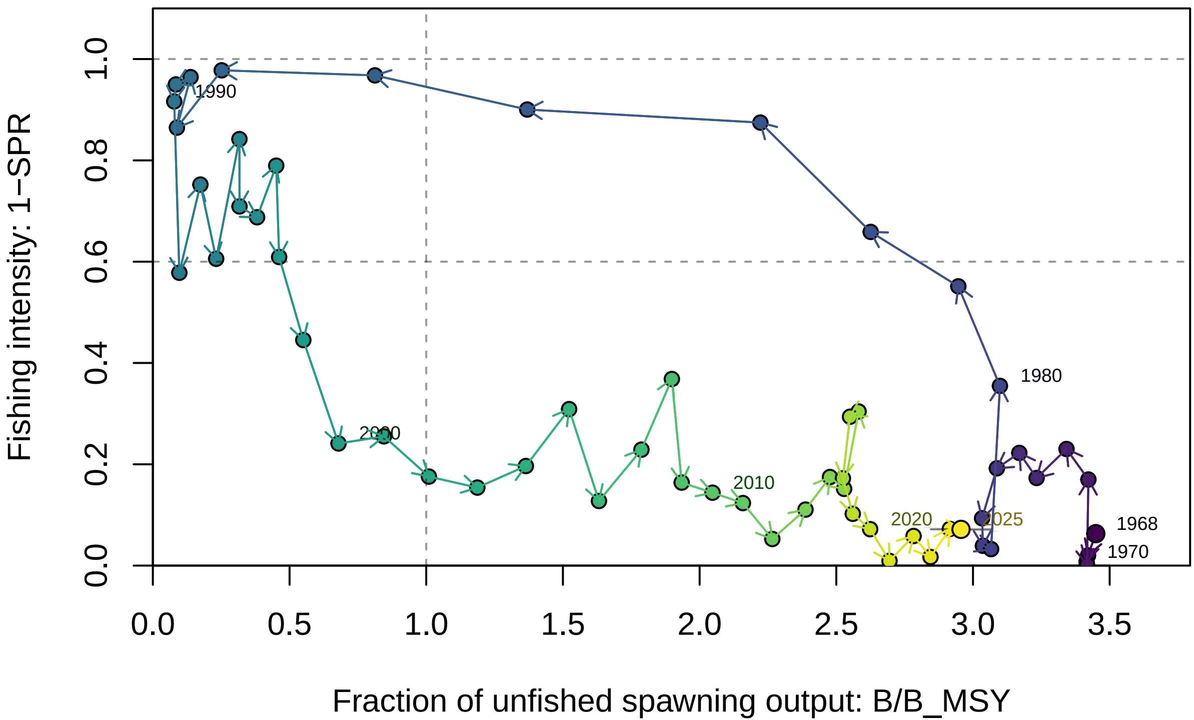
Continuous F
0.0
0.5
1.0
1.5
2.0
1970
1980
1990
2000
2010
2020
Year


SPR
0.0
0.2
0.4
0.6
0.8
1.0
1970
1980
1990
2000
2010
2020
Year

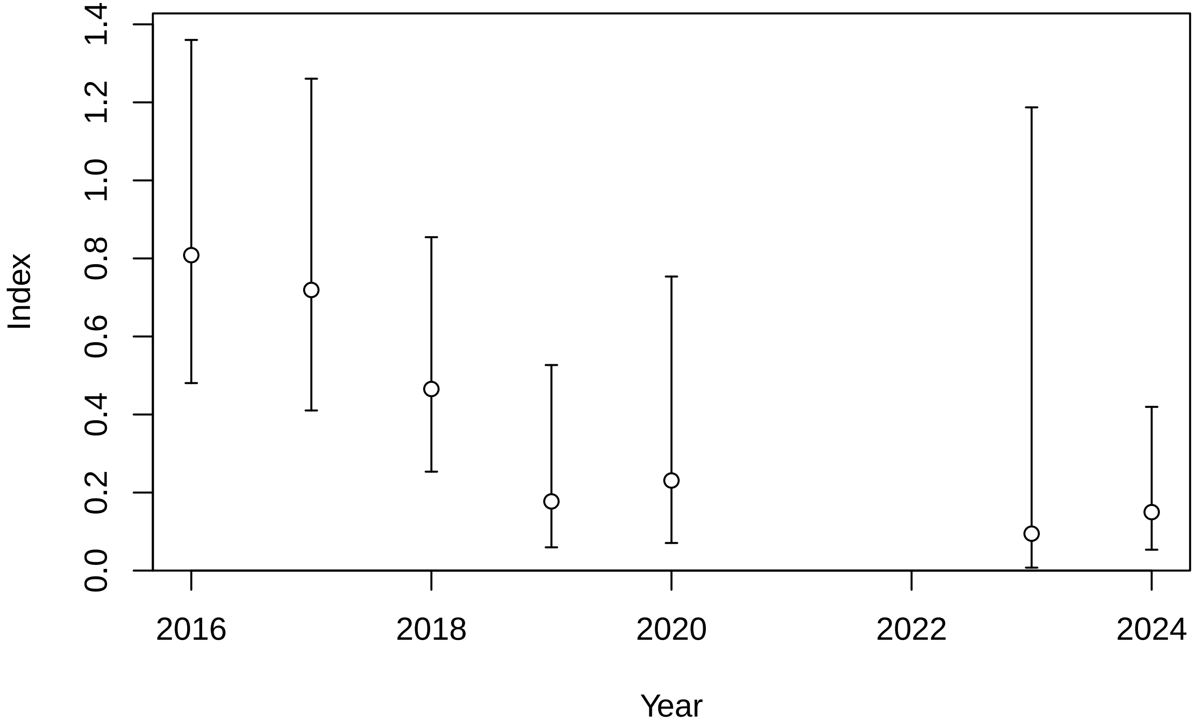

1−SPR
0.0
0.2
0.4
0.6
0.8
1.0
1970
1980
1990
2000
2010
2020
Year

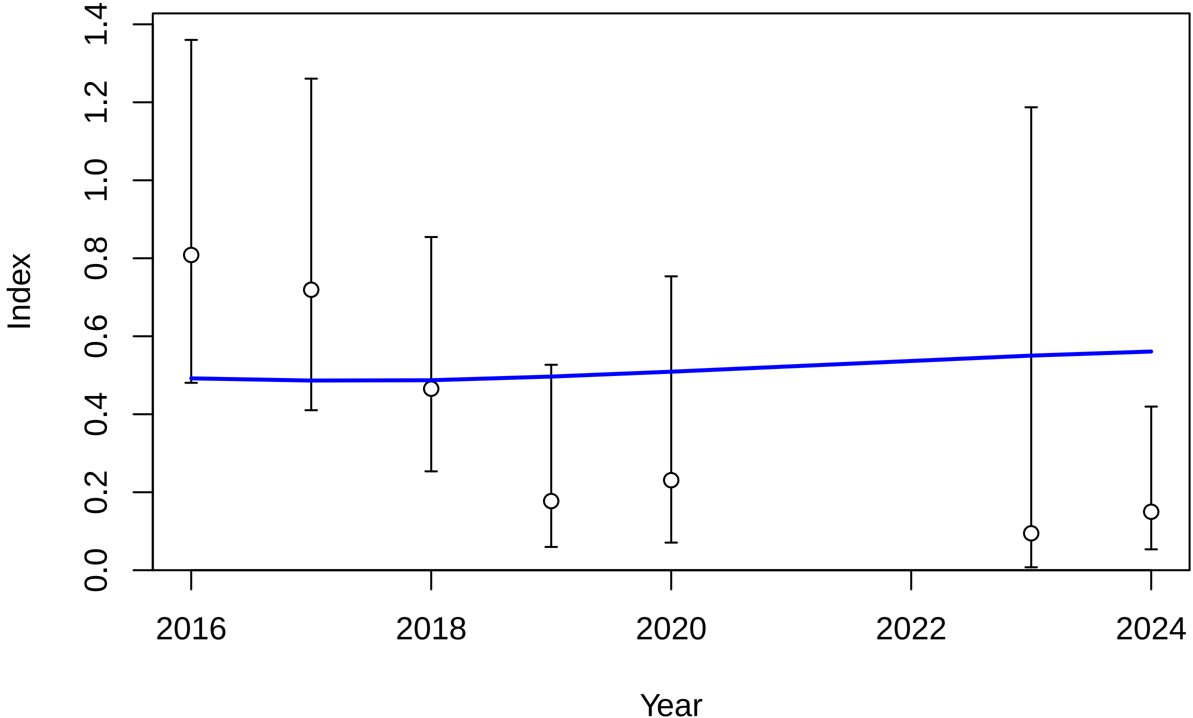


Fishing intensity: 1−SPR
0.0
0.2
0.4
0.6
0.8
1.0
1970
1980
1990
2000
2010
2020
Year

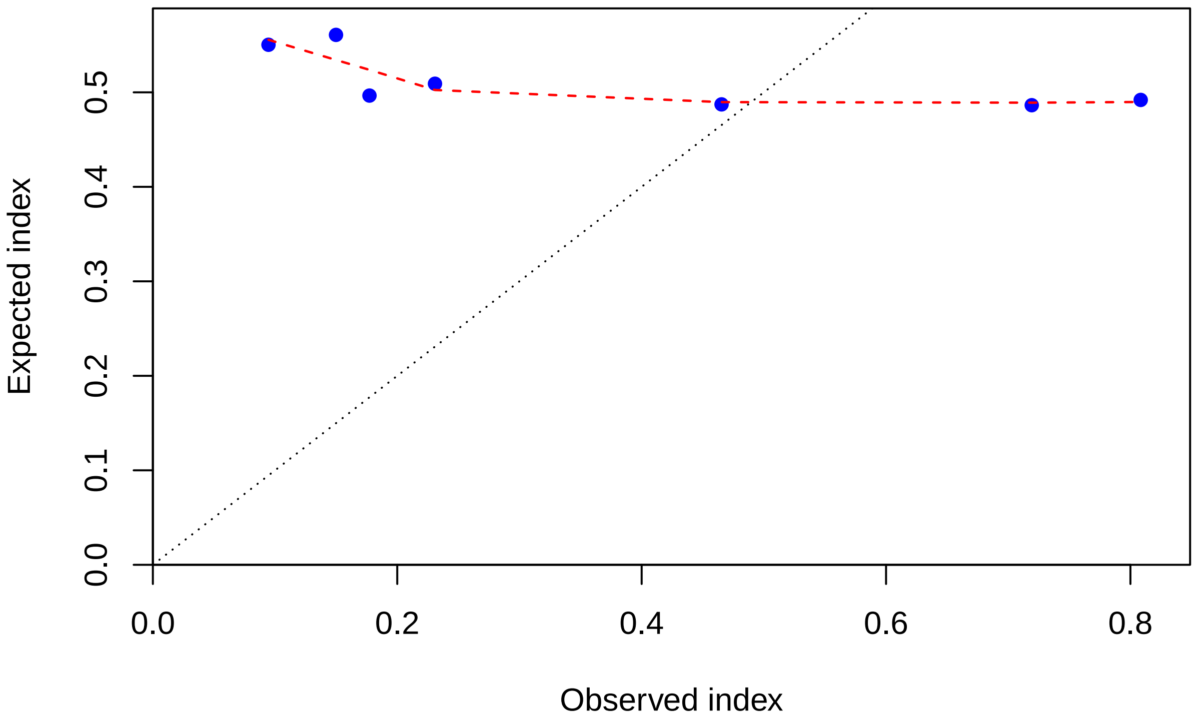


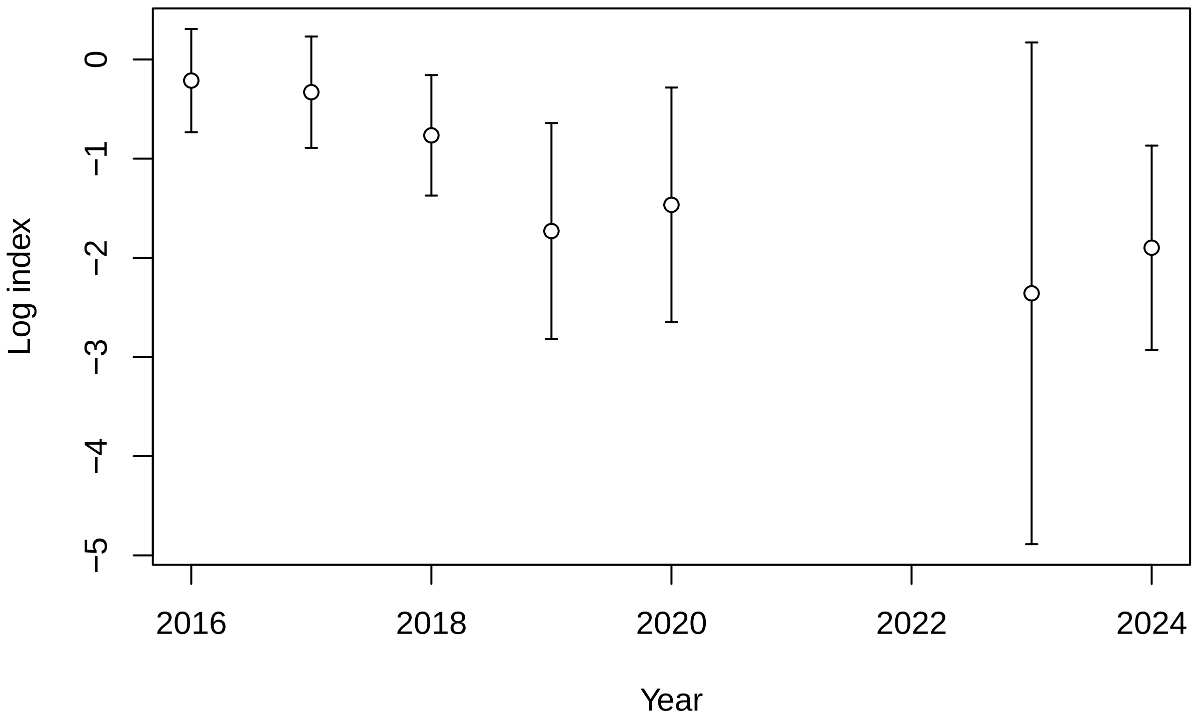
Fishing intensity: 1−SPR
Fraction of unfished spawning output: B/B_MSY
0.0
0.2
0.4
0.6
0.8
1.0
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
1968
1970
1980
1990
2000
2010
2020
2025

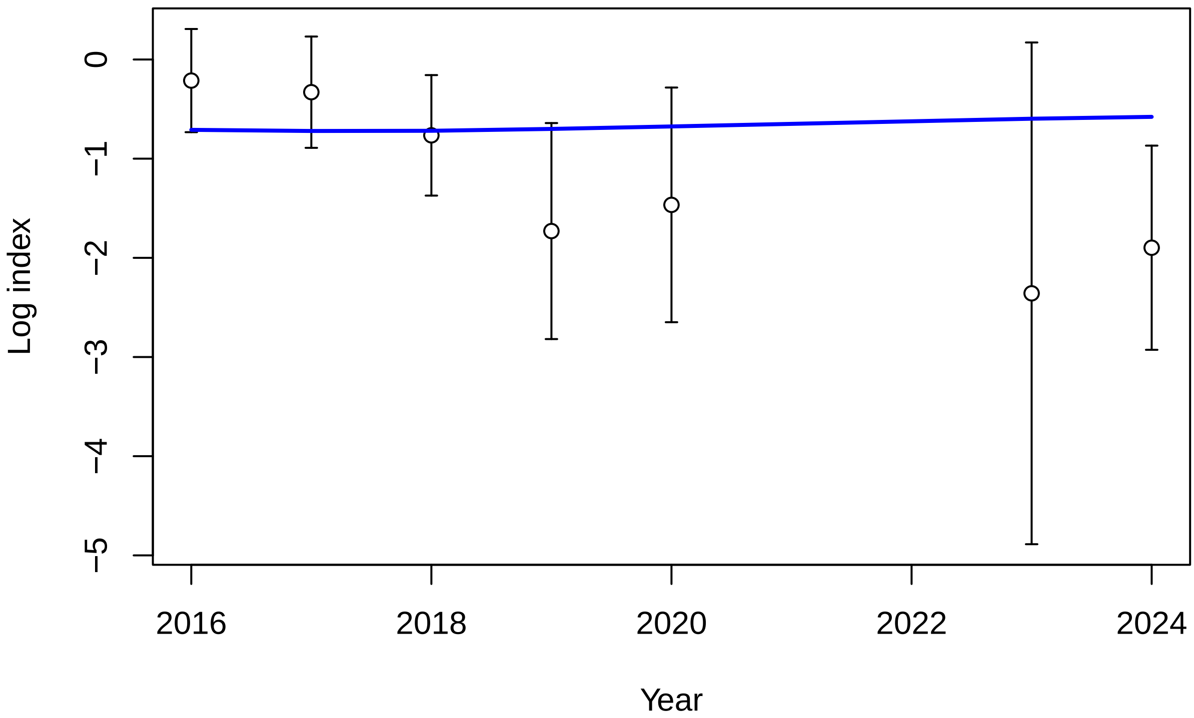

Index
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
2016
2018
2020
2022
2024
Year



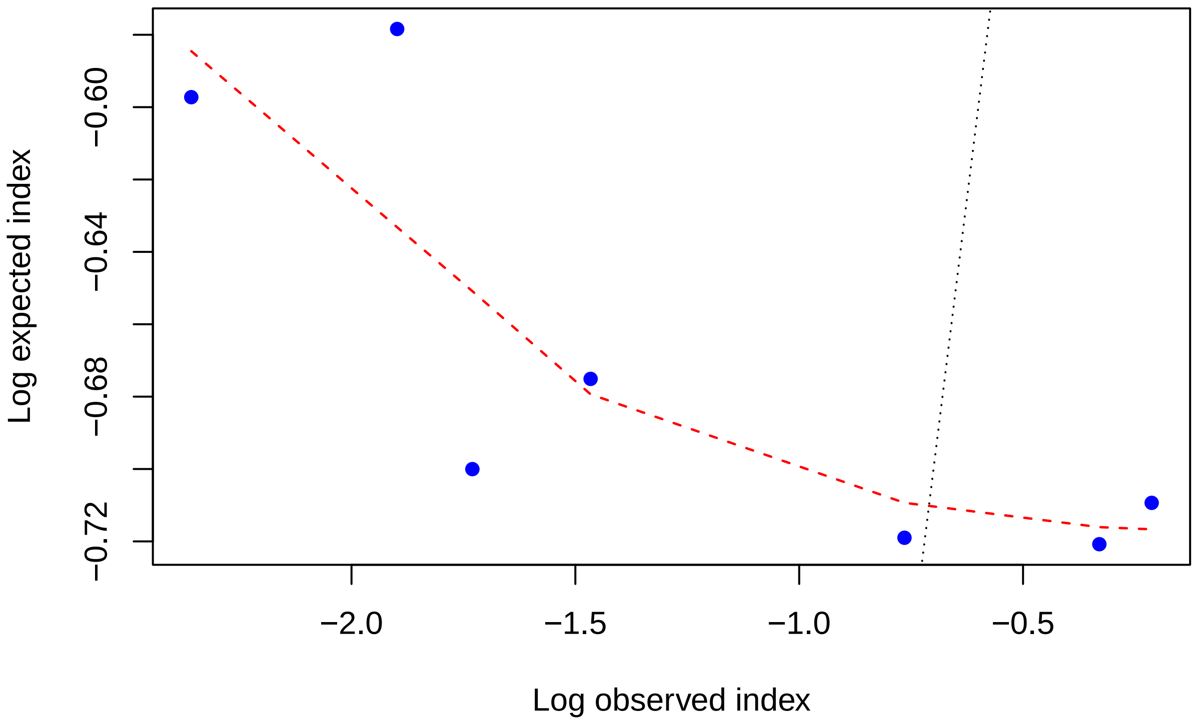
Index
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
2016
2018
2020
2022
2024
Year


Expected index
0.0
0.1
0.2
0.3
0.4
0.5
0.0
0.2
0.4
0.6
0.8
Observed index

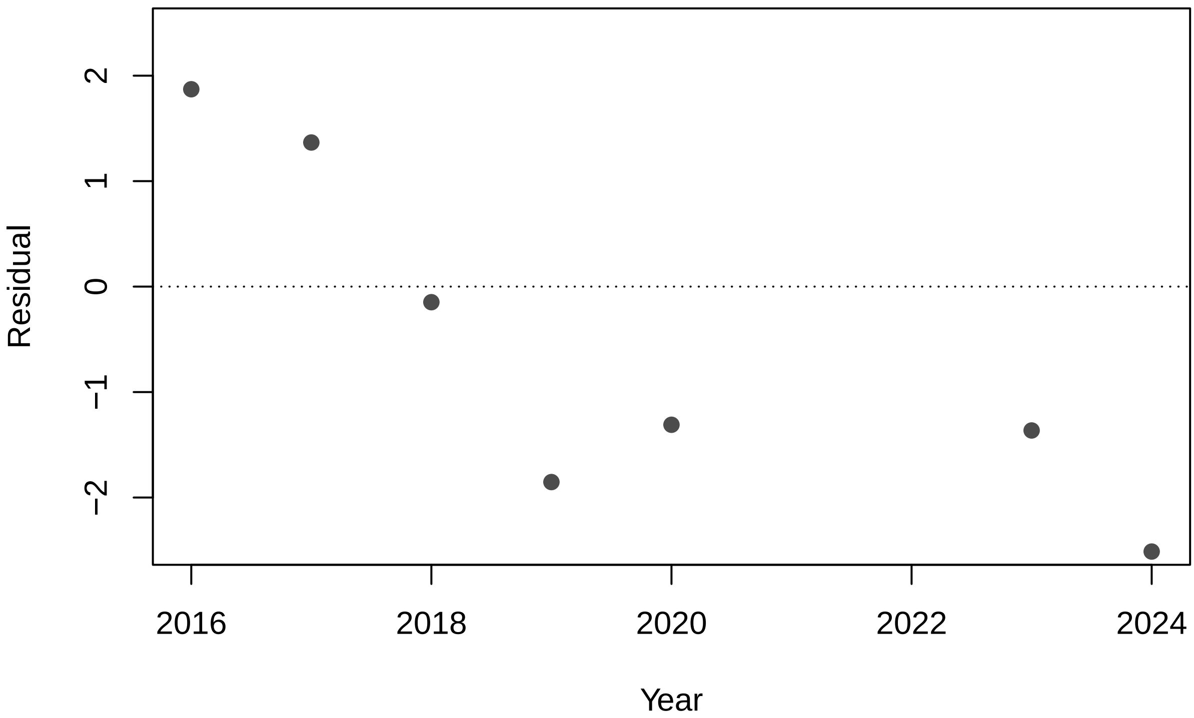

0
−1
−2
−3
−4
−5
Log index
2016
2018
2020
2022
2024
Year



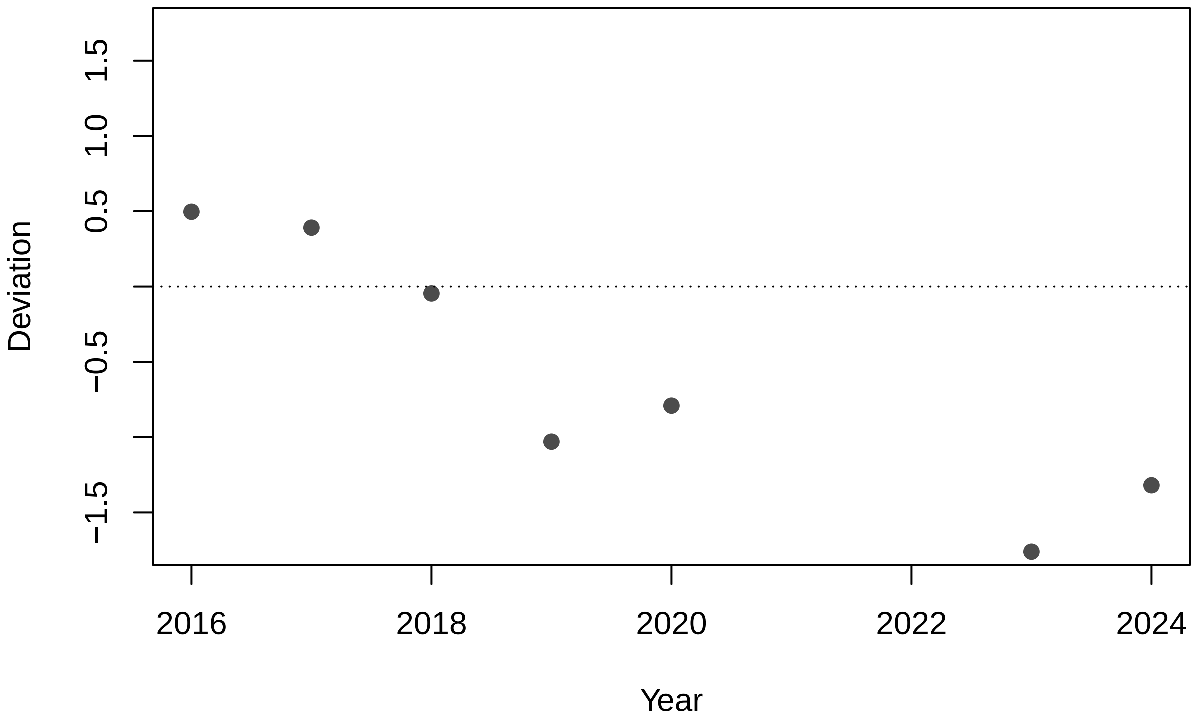
Log index
0
−1
−2
−3
−4
−5
2016
2018
2020
2022
2024
Year



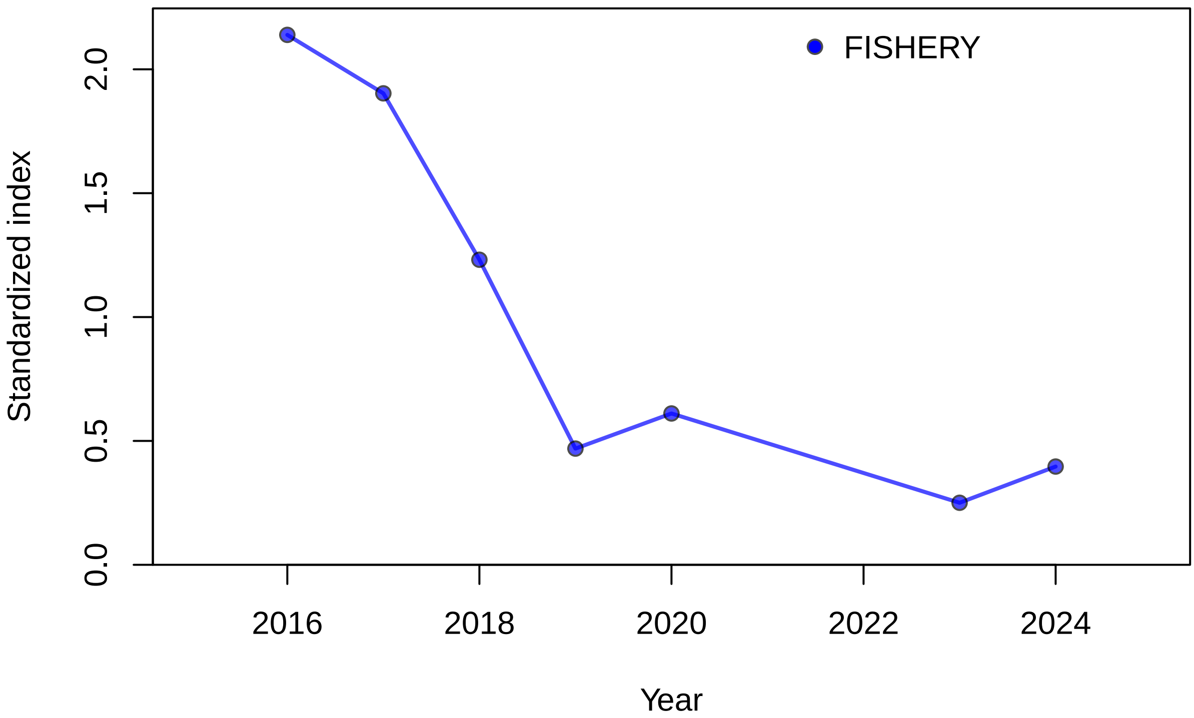
Log expected index
−0.60
−0.64
−0.68
−0.72
−2.0
−1.5
−1.0
−0.5
Log observed index

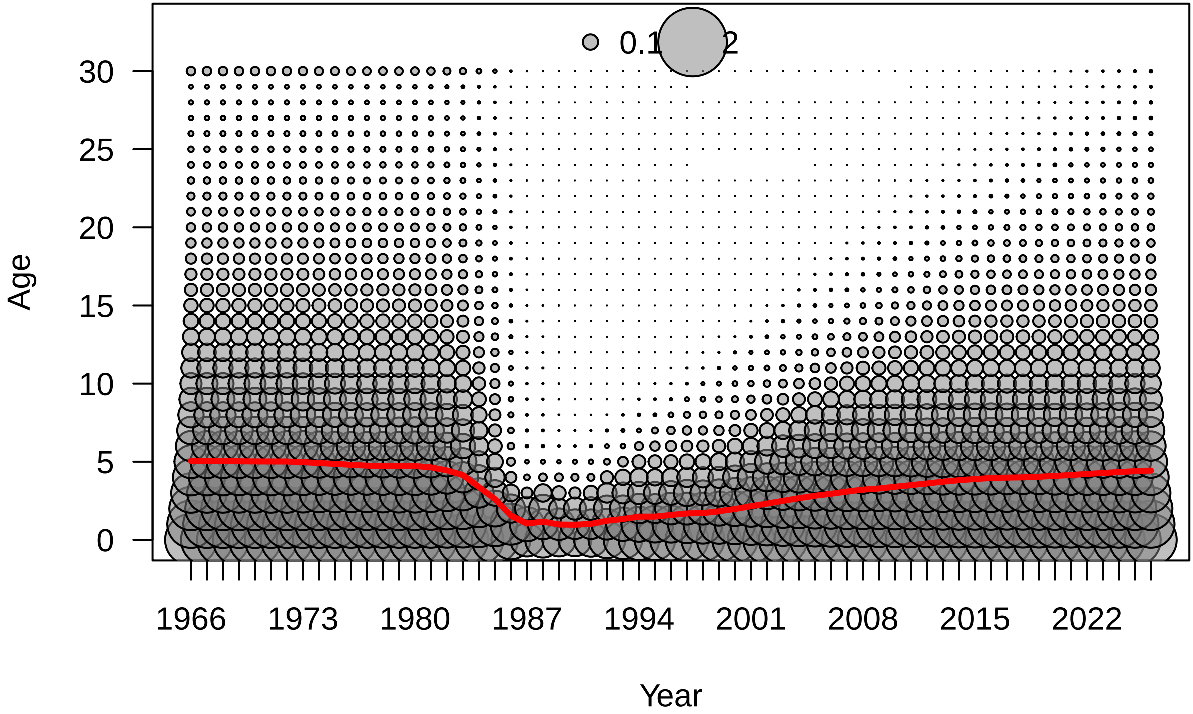


Residual
2
1
0
−1
−2
2016
2018
2020
2022
2024
Year

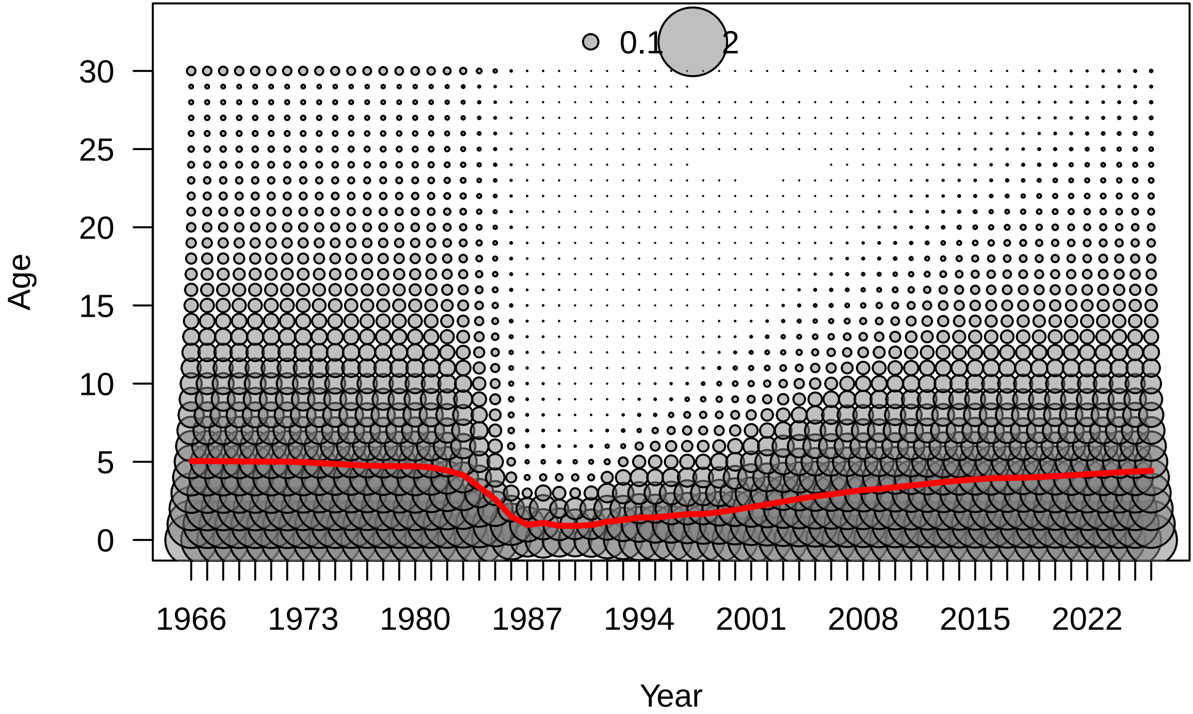


Deviation
1.5
1.0
0.5
−0.5
−1.5
2016
2018
2020
2022
2024
Year

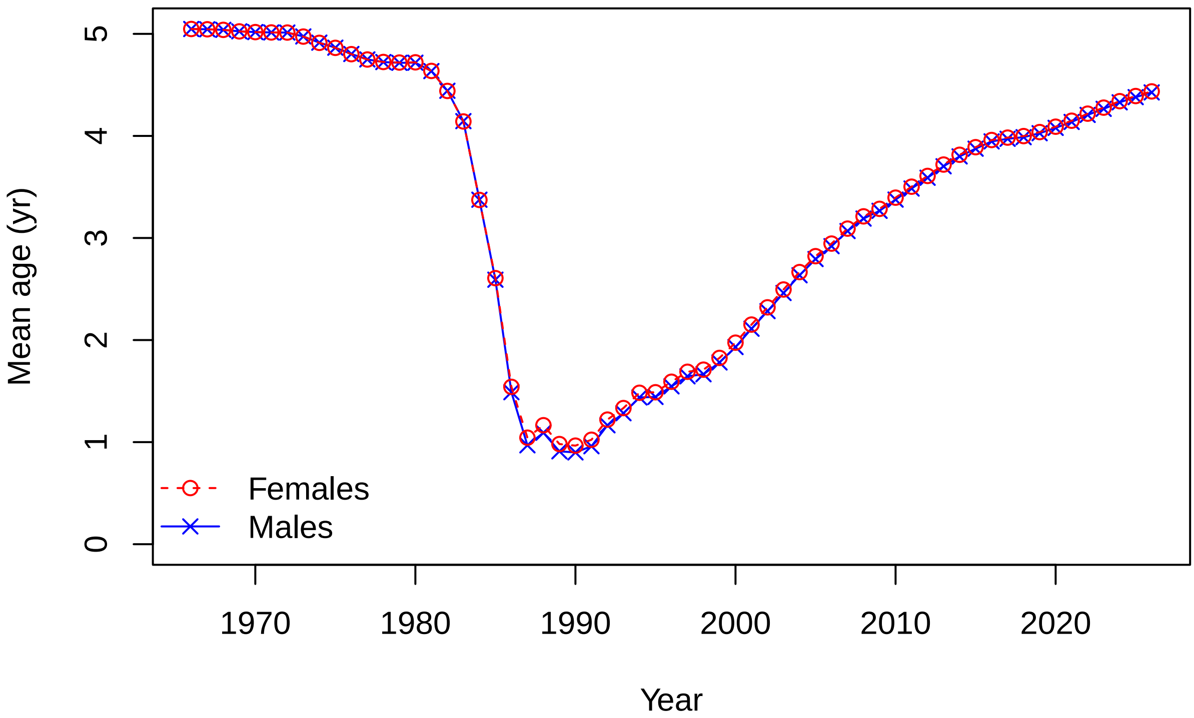

FISHERY
Standardized index
0.0
0.5
1.0
1.5
2.0
2016
2018
2020
2022
2024
Year



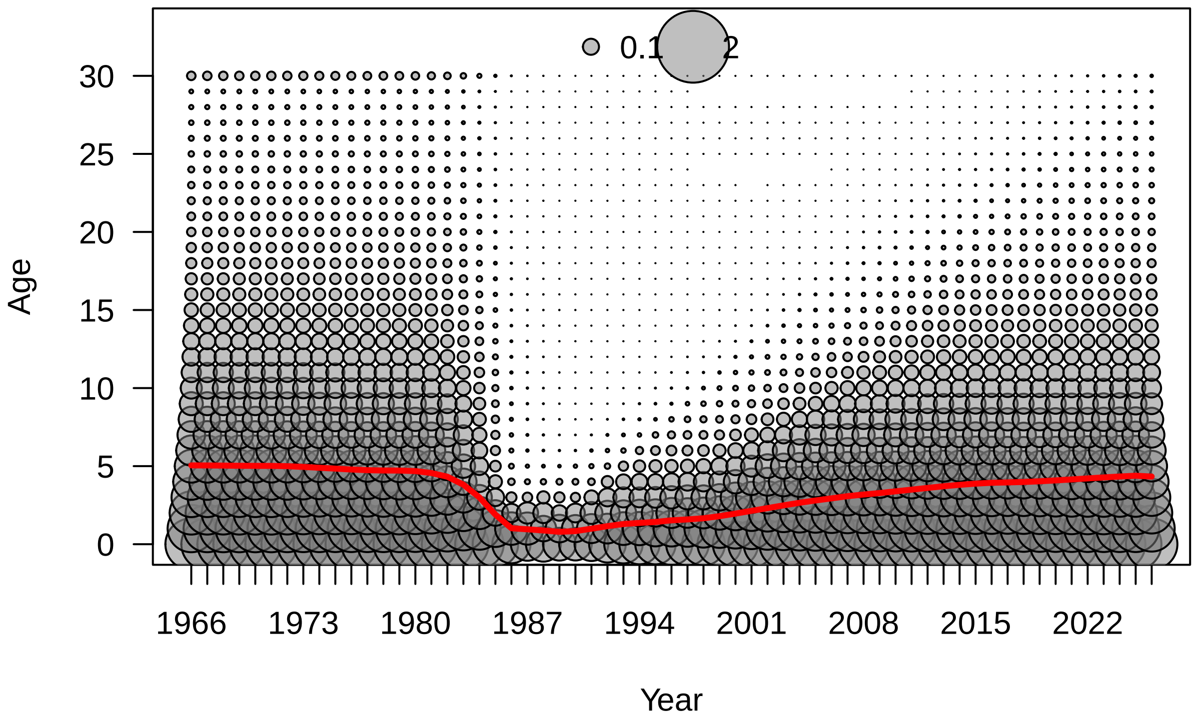
0.1
2
30
25
20
15
10
5
0
Age
1966
1973
1980
1987
1994
2001
2008
2015
2022
Year

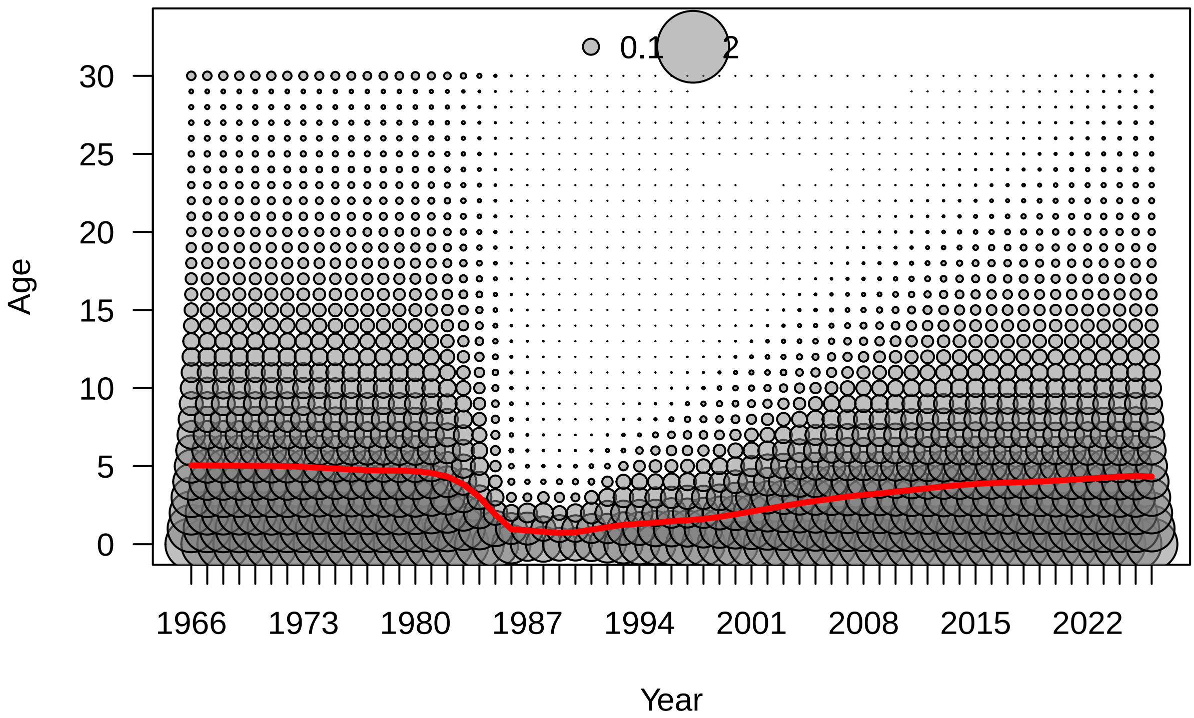


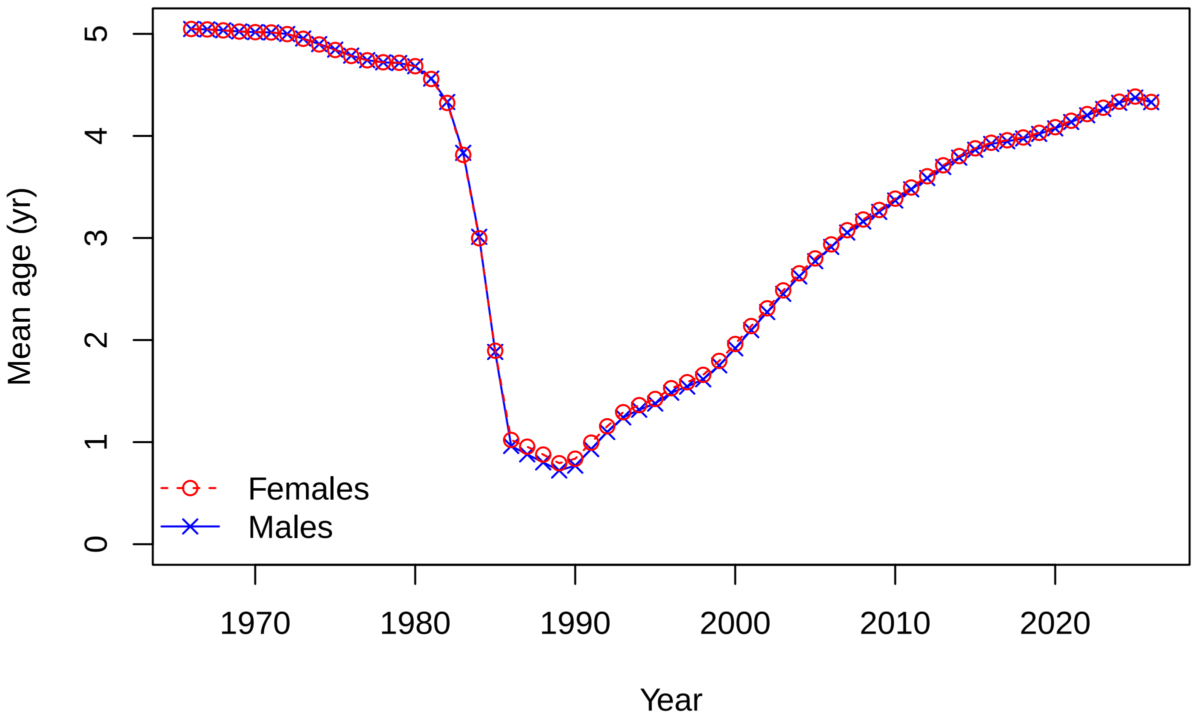
0.1
2
Age
30
25
20
15
10
5
0
Year
1966
1973
1980
1987
1994
2001
2008
2015
2022



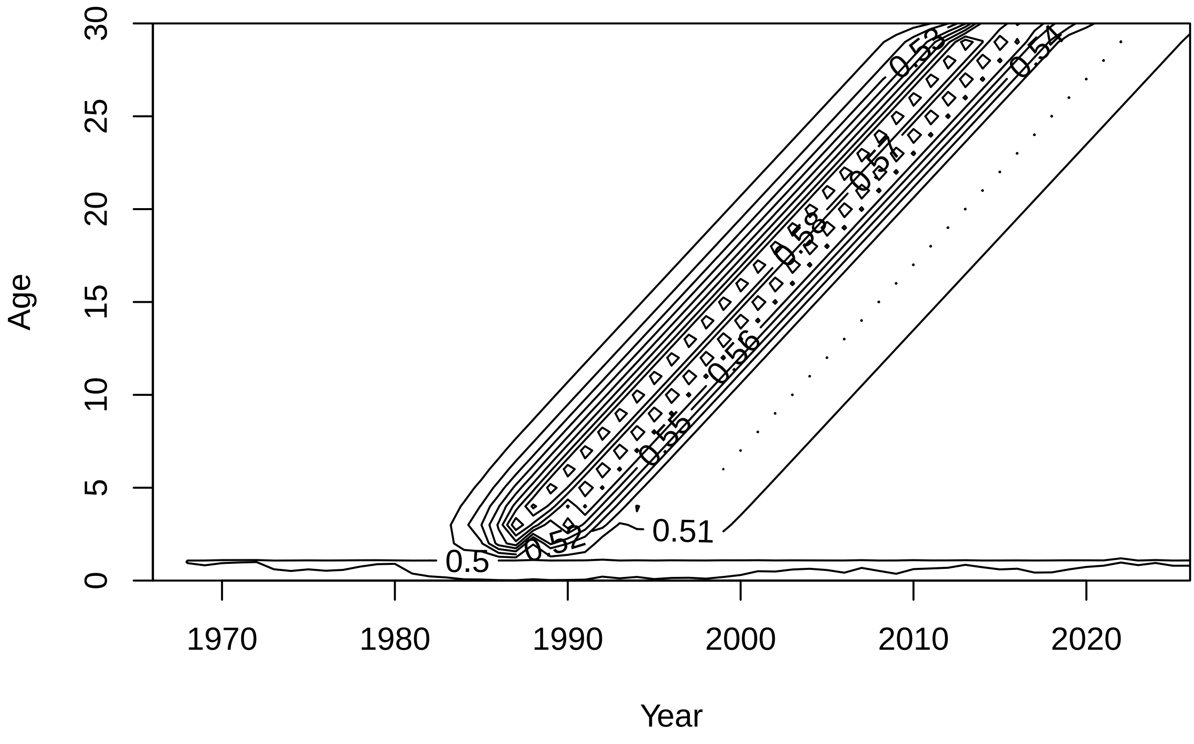
Mean age (yr)
0
1
2
3
4
5
1970
1980
1990
2000
2010
2020
Year
Females
Males


0.1
2
30
25
20
15
10
5
0
Age
1966
1973
1980
1987
1994
2001
2008
2015
2022
Year

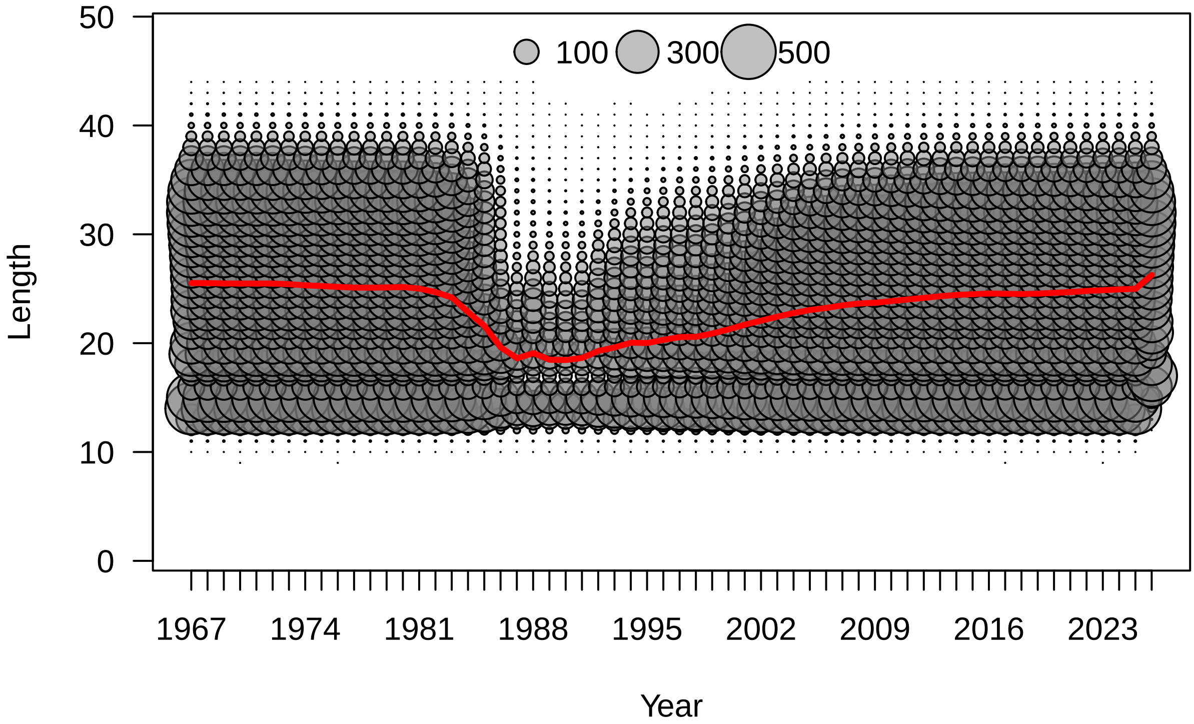

0.1
2
Age
30
25
20
15
10
5
0
1966
1973
1980
1987
1994
2001
2008
2015
2022
Year



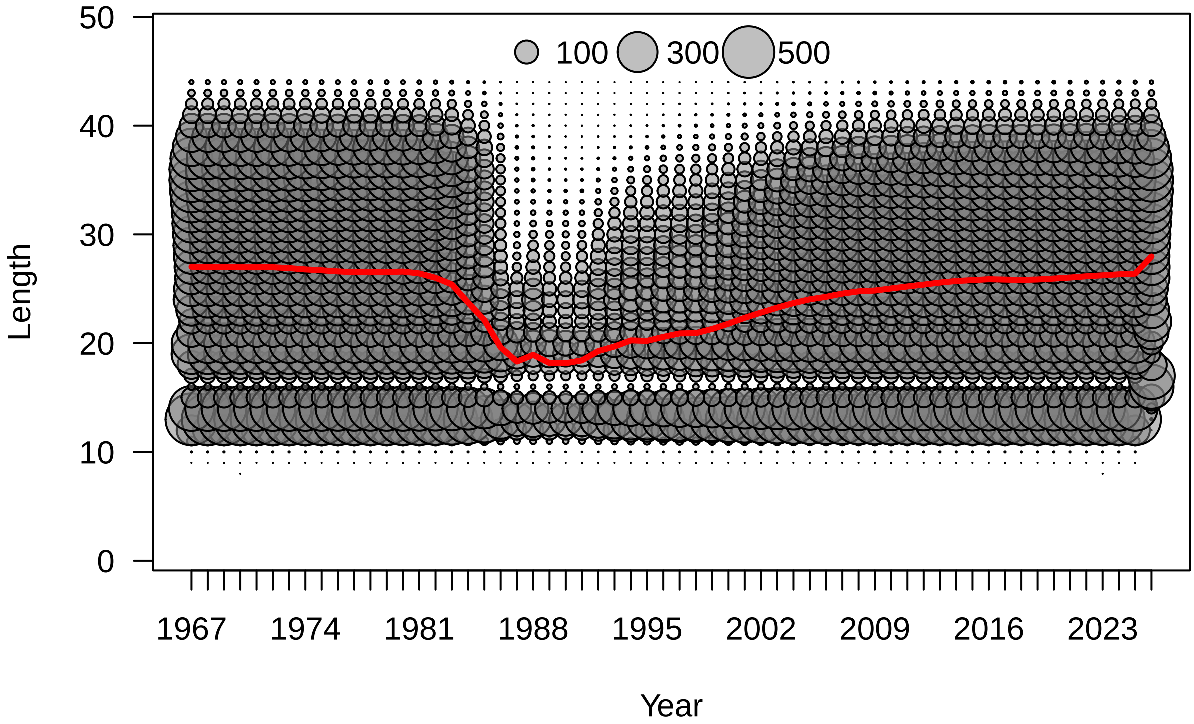
Mean age (yr)
0
1
2
3
4
5
Females
Males
1970
1980
1990
2000
2010
2020
Year



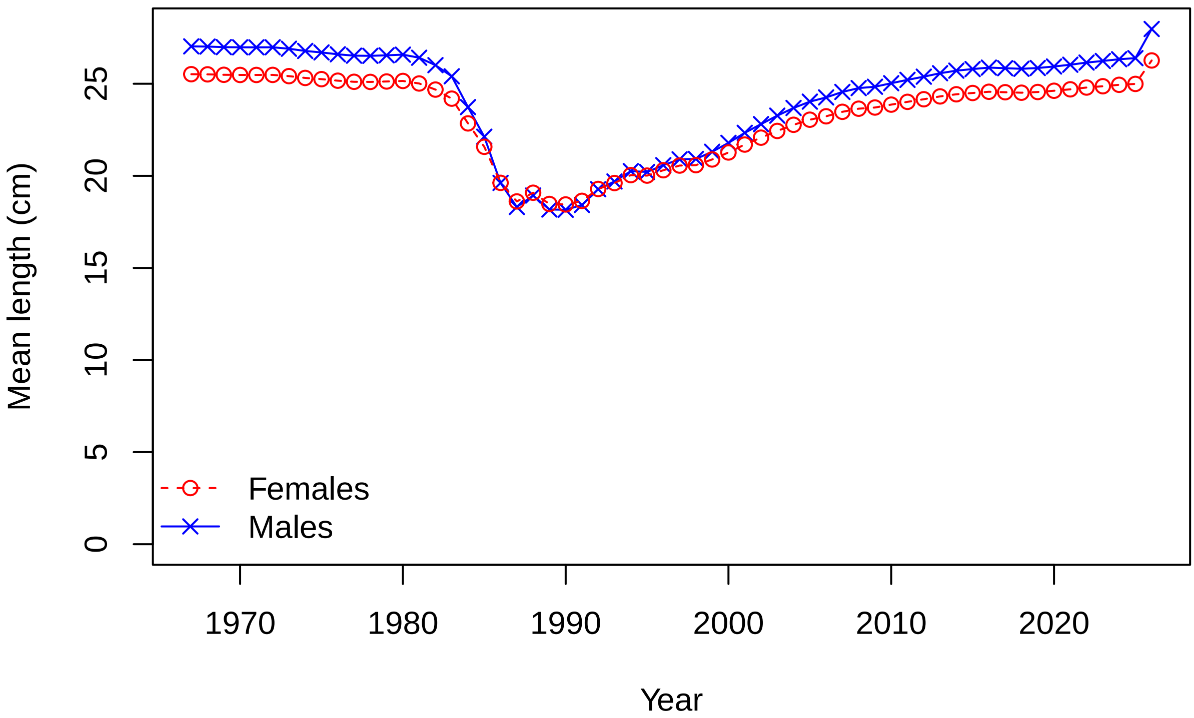
Age
Year
0
5
10
15
20
25
30
1970
1980
1990
2000
2010
2020
0.5
0.51
0.52
0.53
0.54
0.55
0.56
0.57
0.58



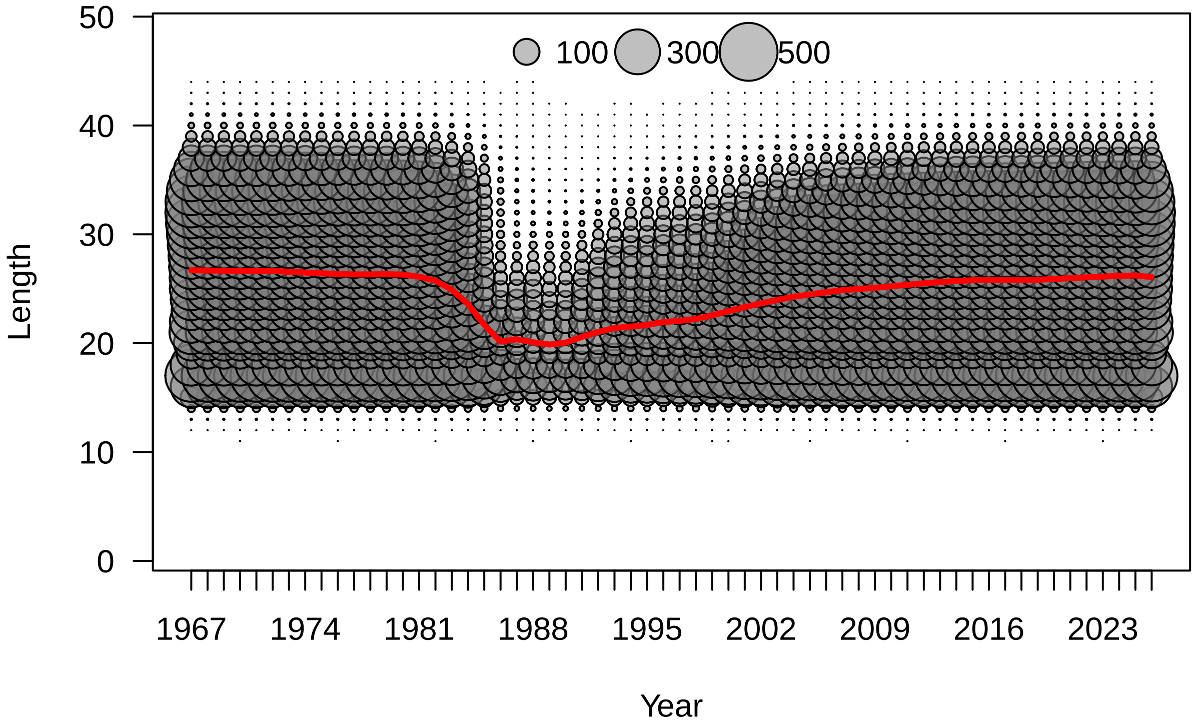
100
300
500
Length
50
40
30
20
10
0
1967
1974
1981
1988
1995
2002
2009
2016
2023
Year



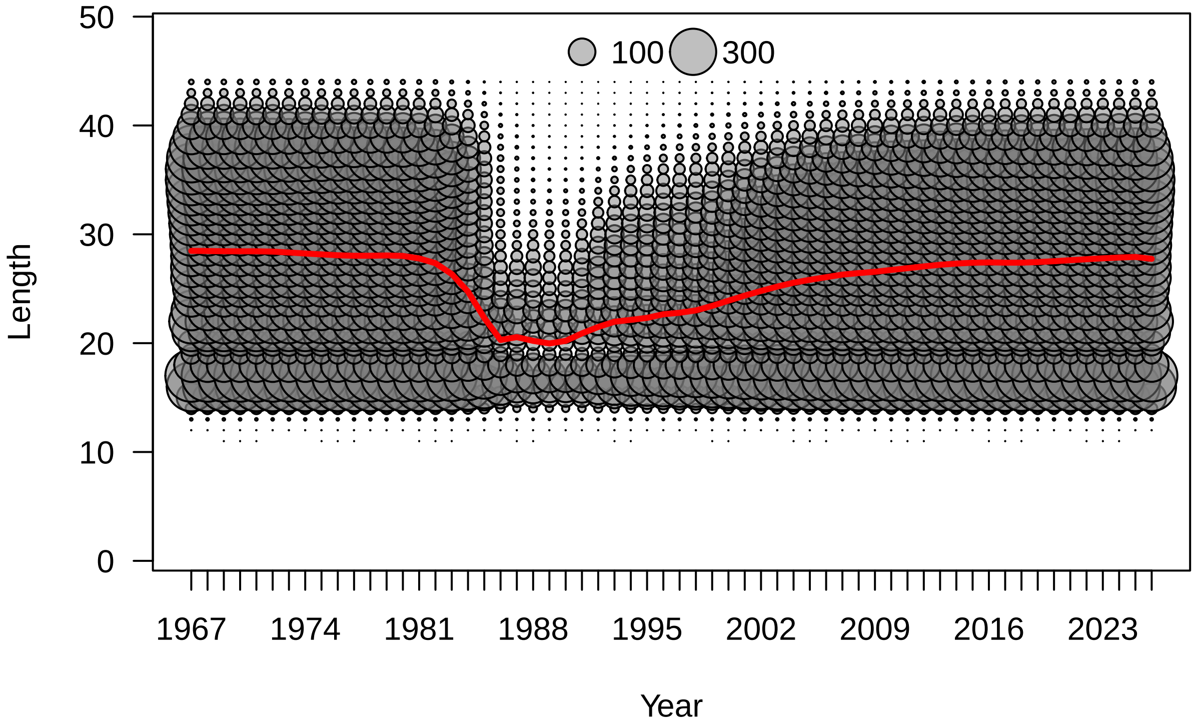
50
40
30
20
10
0
Length
100
300
500
1967
1974
1981
1988
1995
2002
2009
2016
2023
Year



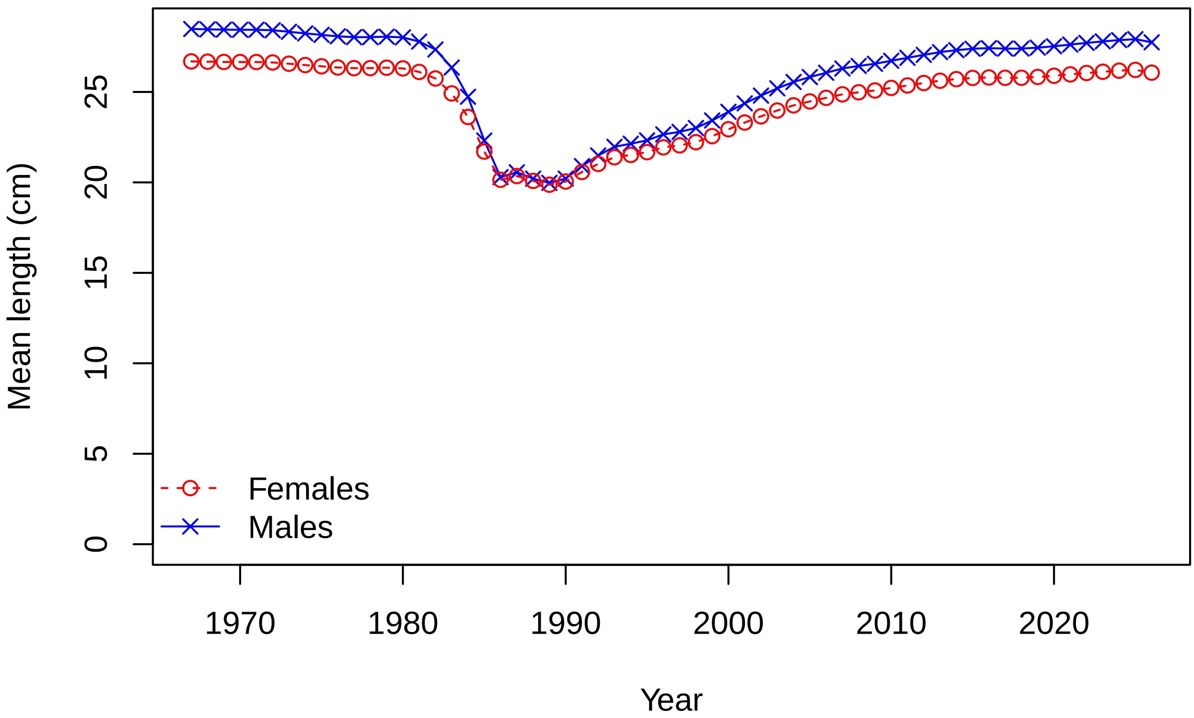
Mean length (cm)
0
5
10
15
20
25
1970
1980
1990
2000
2010
2020
Year
Females
Males



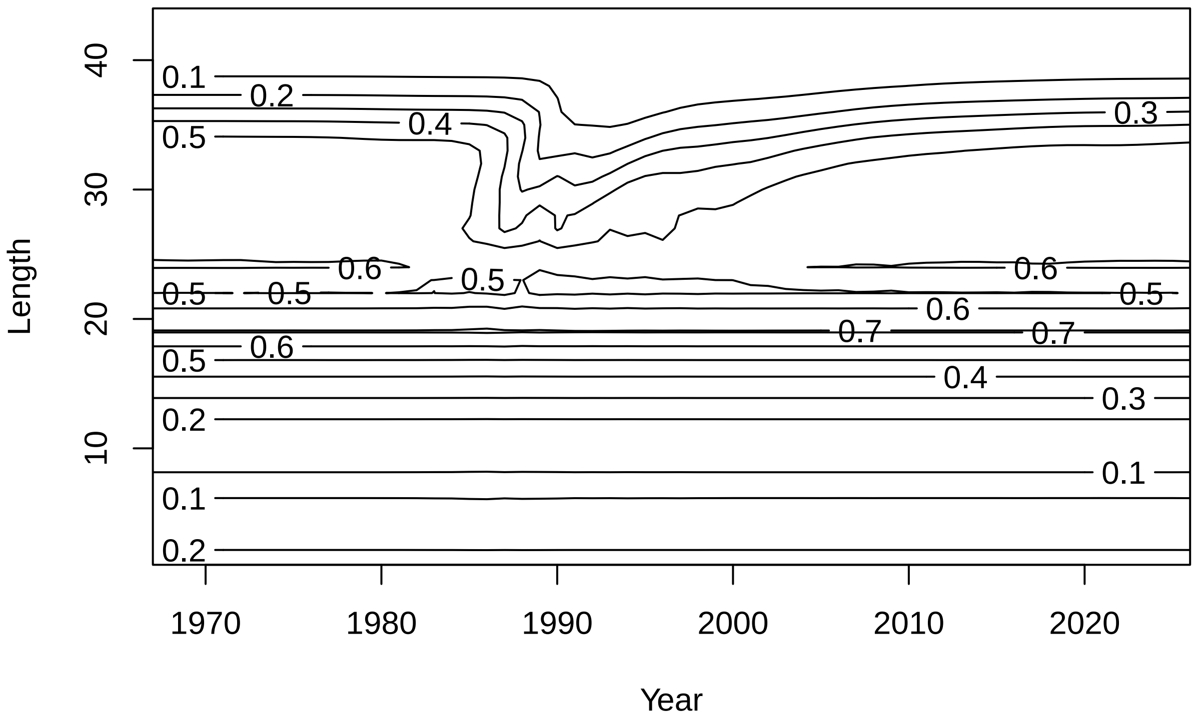
100
300
500
Length
Year
0
10
20
30
40
50
1967
1974
1981
1988
1995
2002
2009
2016
2023



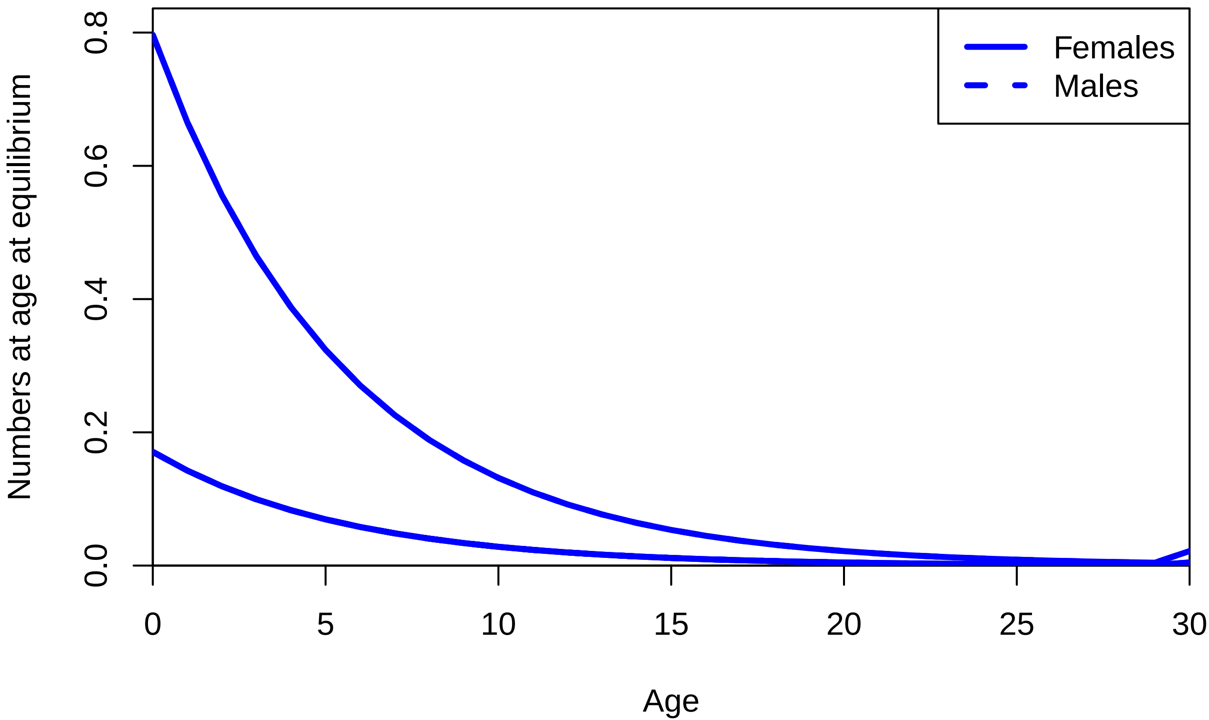
50
40
30
20
10
0
Length
100
300
1967
1974
1981
1988
1995
2002
2009
2016
2023
Year



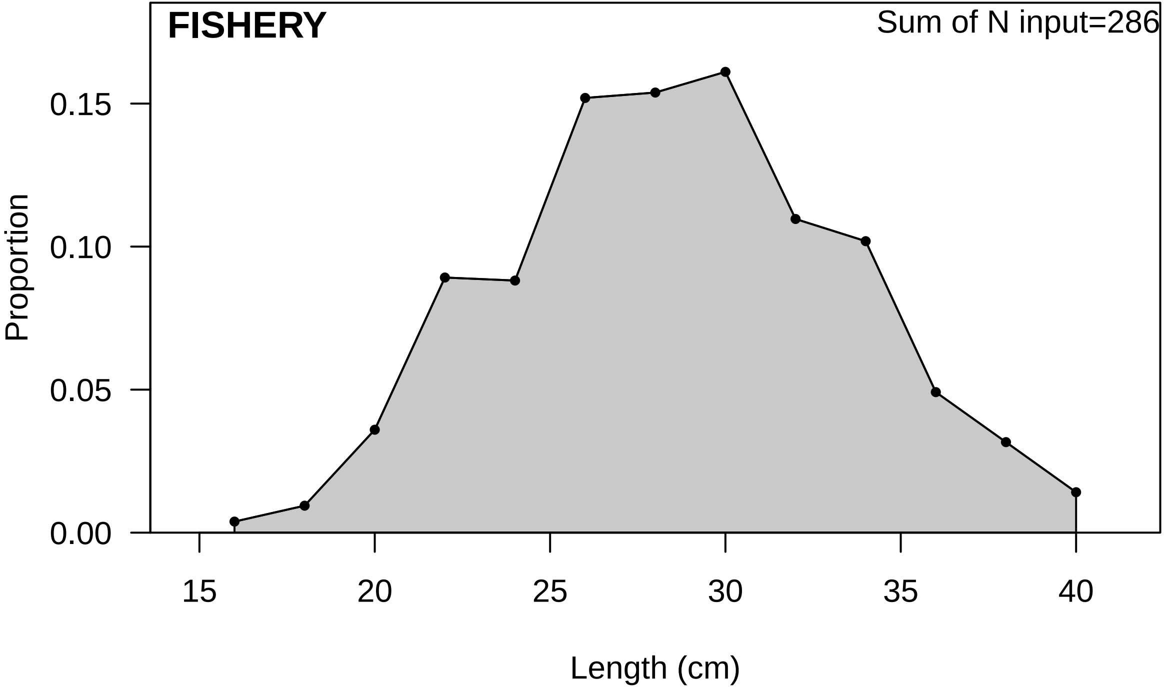
Mean length (cm)
0
5
10
15
20
25
1970
1980
1990
2000
2010
2020
Year
Females
Males

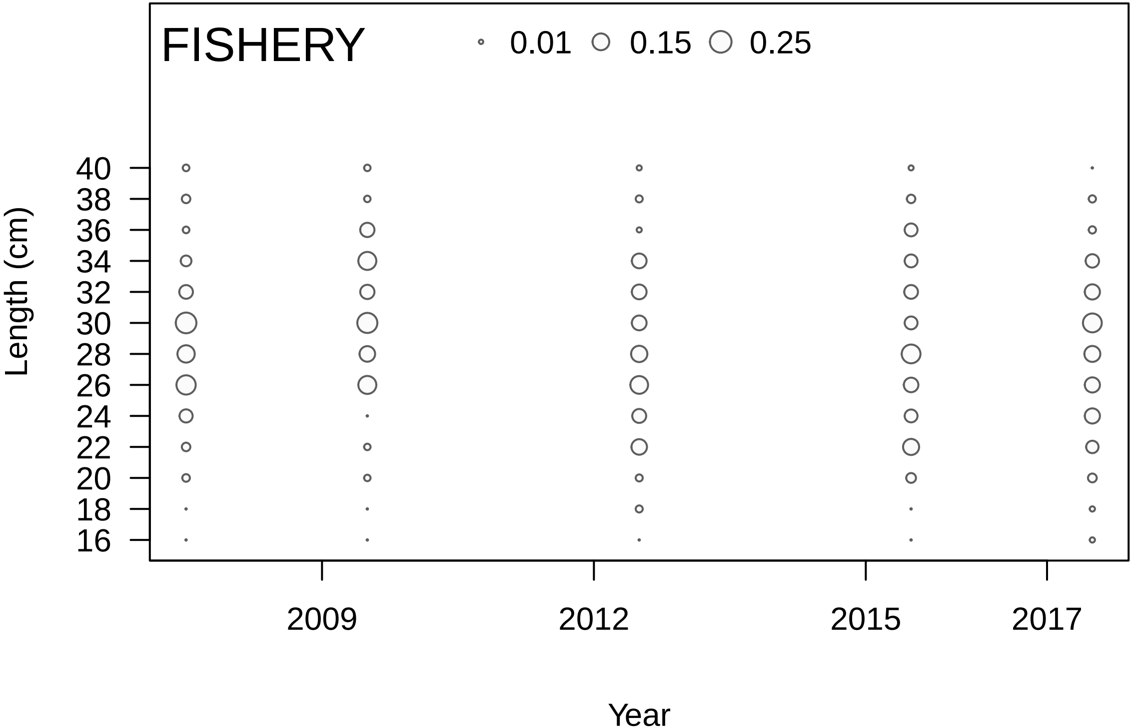


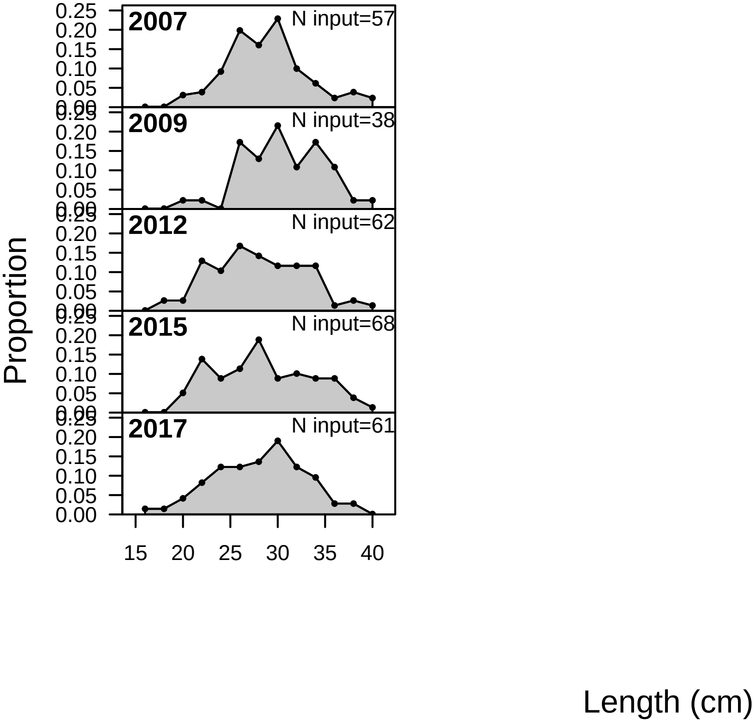
0.1
0.2
0.3
0.4
0.5
0.6
0.5
0.5
0.5
0.6
0.5
0.6
0.7
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.1
0.2
Length
Year
10
20
30
40
1970
1980
1990
2000
2010
2020



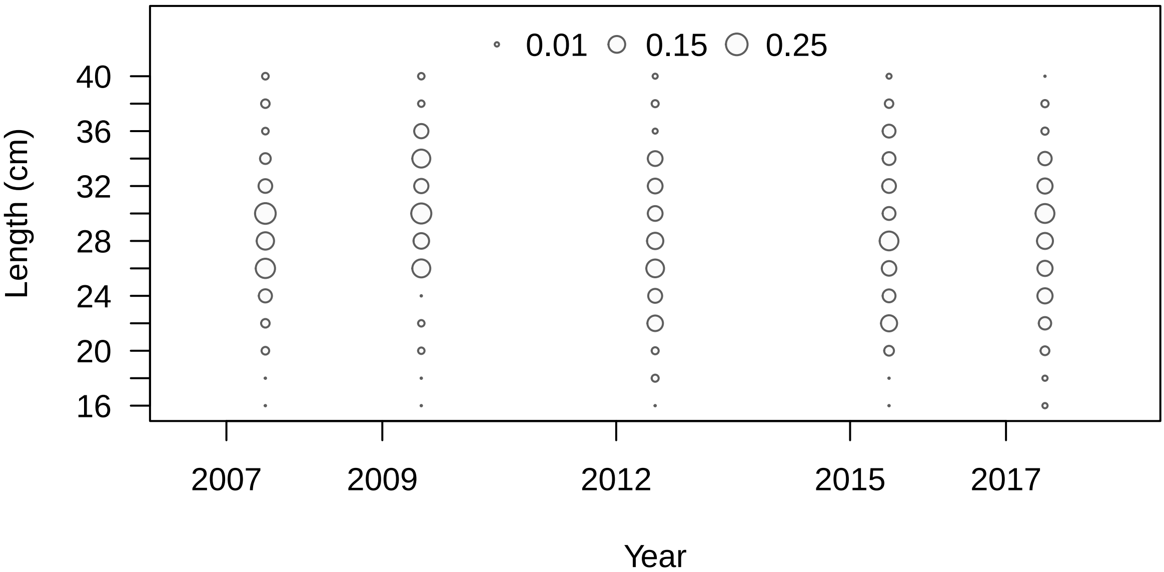
Numbers at age at equilibrium
0.0
0.2
0.4
0.6
0.8
Age
0
5
10
15
20
25
30
Females
Males

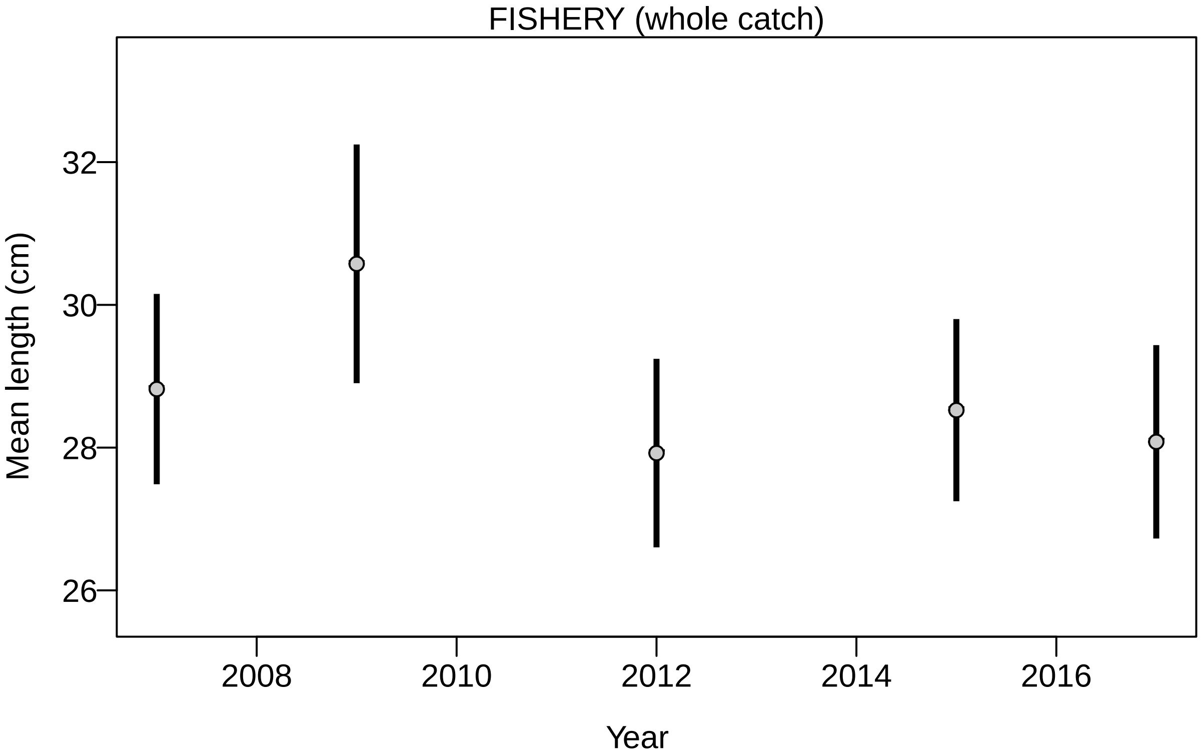

FISHERY
Sum of N input=286
Proportion
0.00
0.05
0.10
0.15
15
20
25
30
35
40
Length (cm)


FISHERY
0.01
0.15
0.25
Length (cm)
40
38
36
34
32
30
28
26
24
22
20
18
16
2009
2012
2015
2017
Year

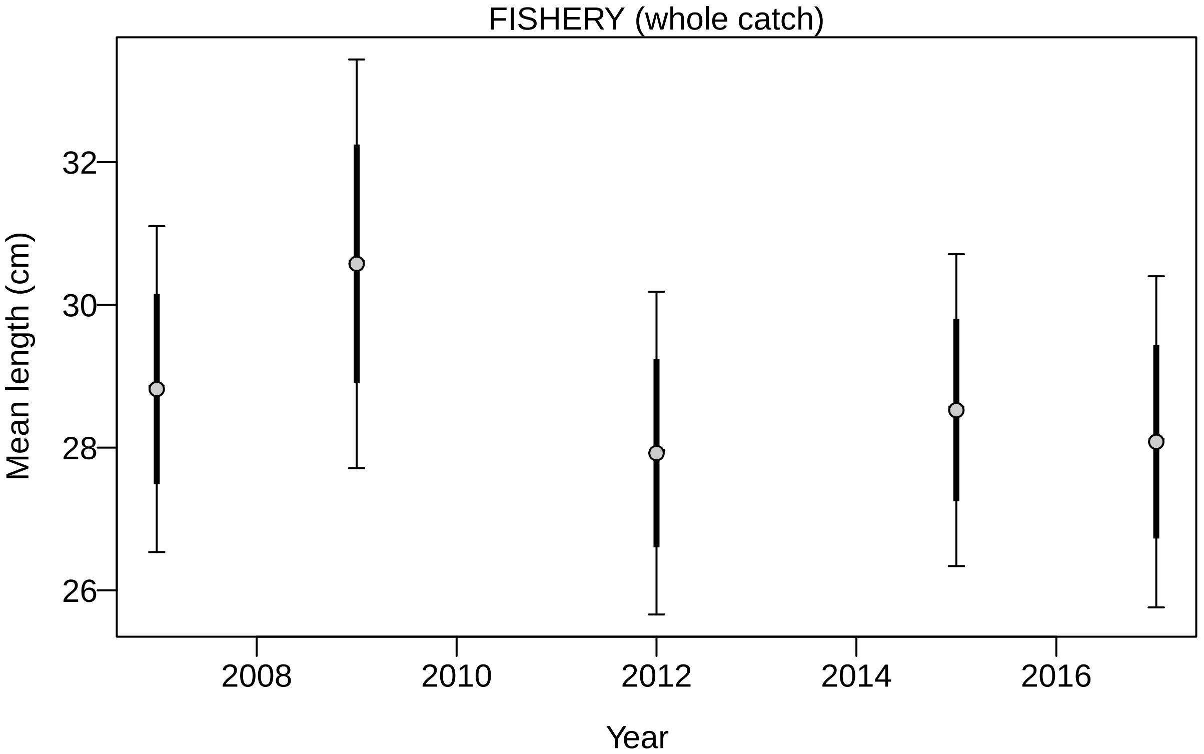


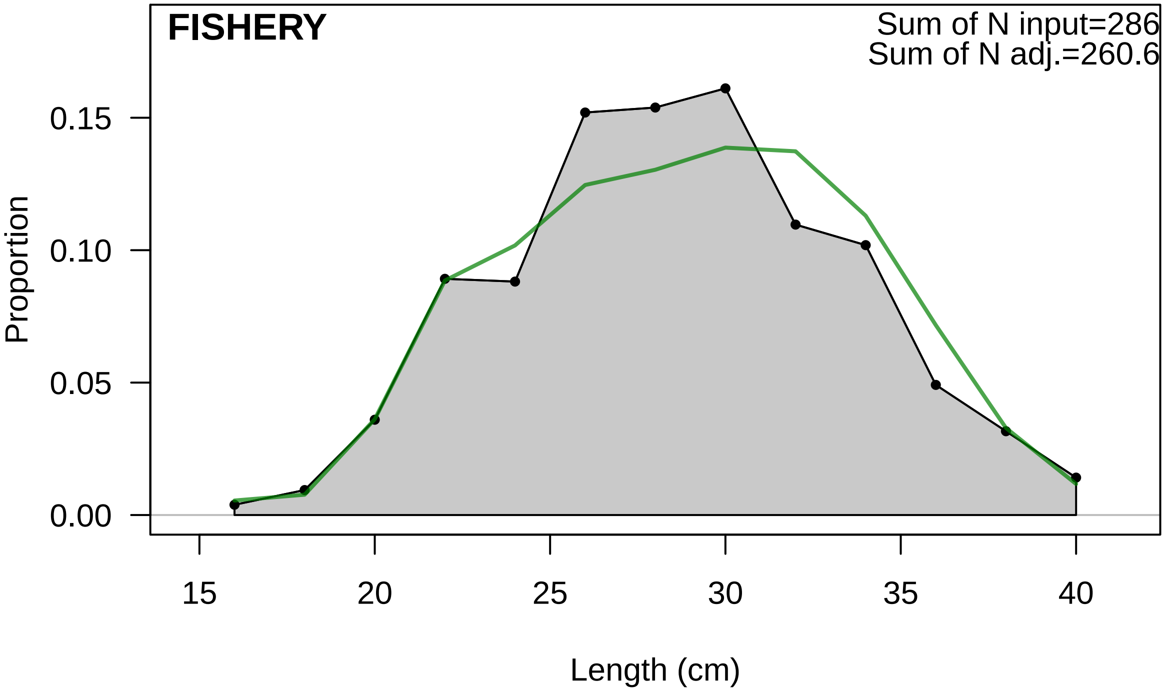
2007
N input=57
2009
N input=38
2012
N input=62
2015
N input=68
2017
N input=61
0.00
0.05
0.10
0.15
0.20
0.25
15
20
25
30
35
40
Proportion
Length (cm)


0.01
0.15
0.25
40
36
32
28
24
20
16
Length (cm)
2007
2009
2012
2015
2017
Year

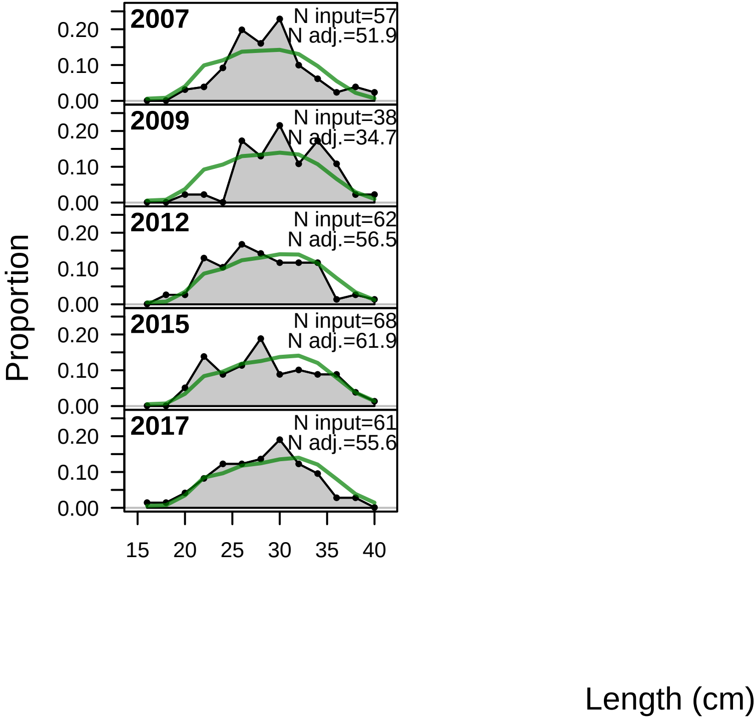


FISHERY (whole catch)
Mean length (cm)
26
28
30
32
2008
2010
2012
2014
2016
Year

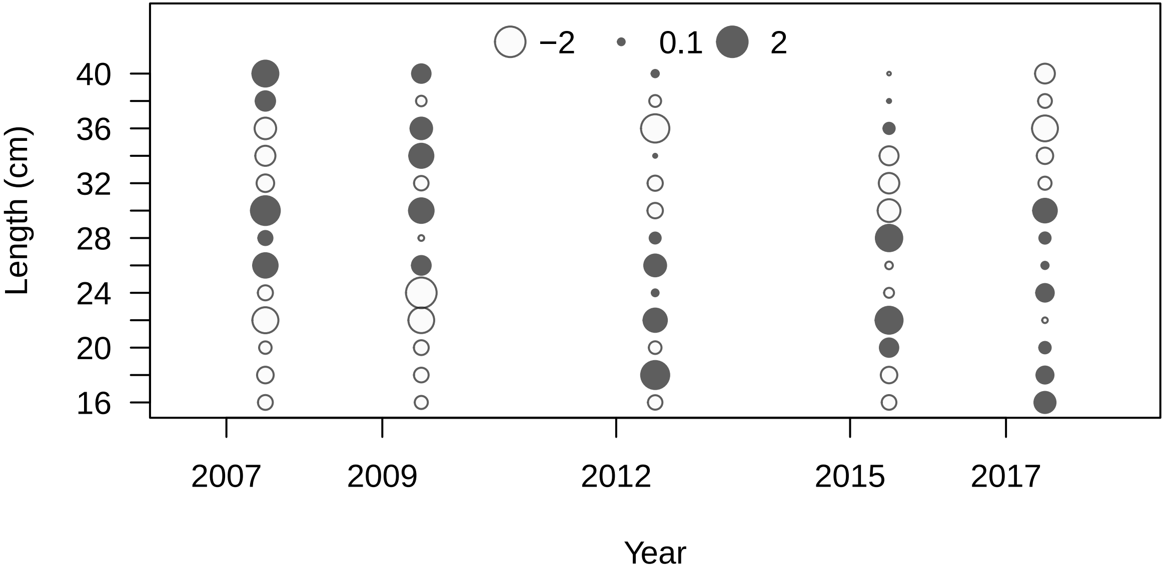

FISHERY (whole catch)
Mean length (cm)
32
30
28
26
2008
2010
2012
2014
2016
Year


FISHERY
Sum of N input=286
Sum of N adj.=260.6
Proportion
0.00
0.05
0.10
0.15
15
20
25
30
35
40
Length (cm)

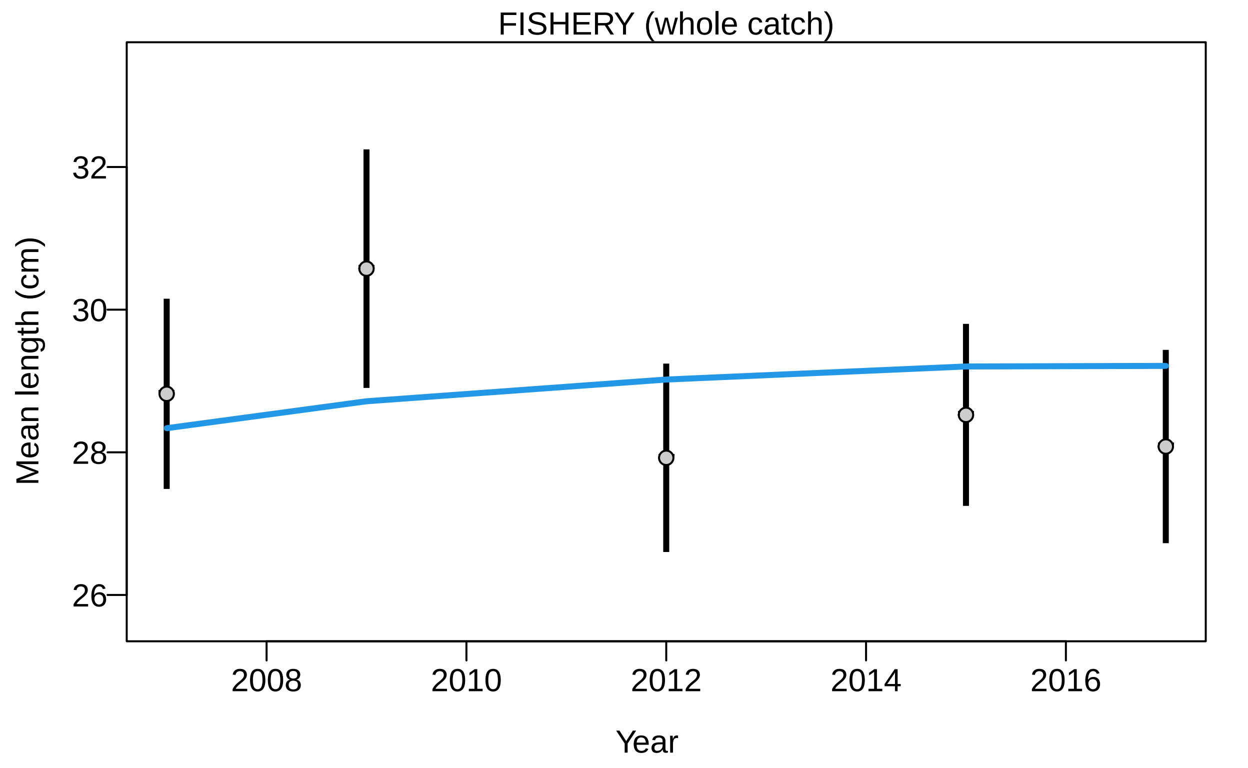

FISHERY
−2
0.1
2
Length (cm)
40
38
36
34
32
30
28
26
24
22
20
18
16
2009
2012
2015
2017
Year

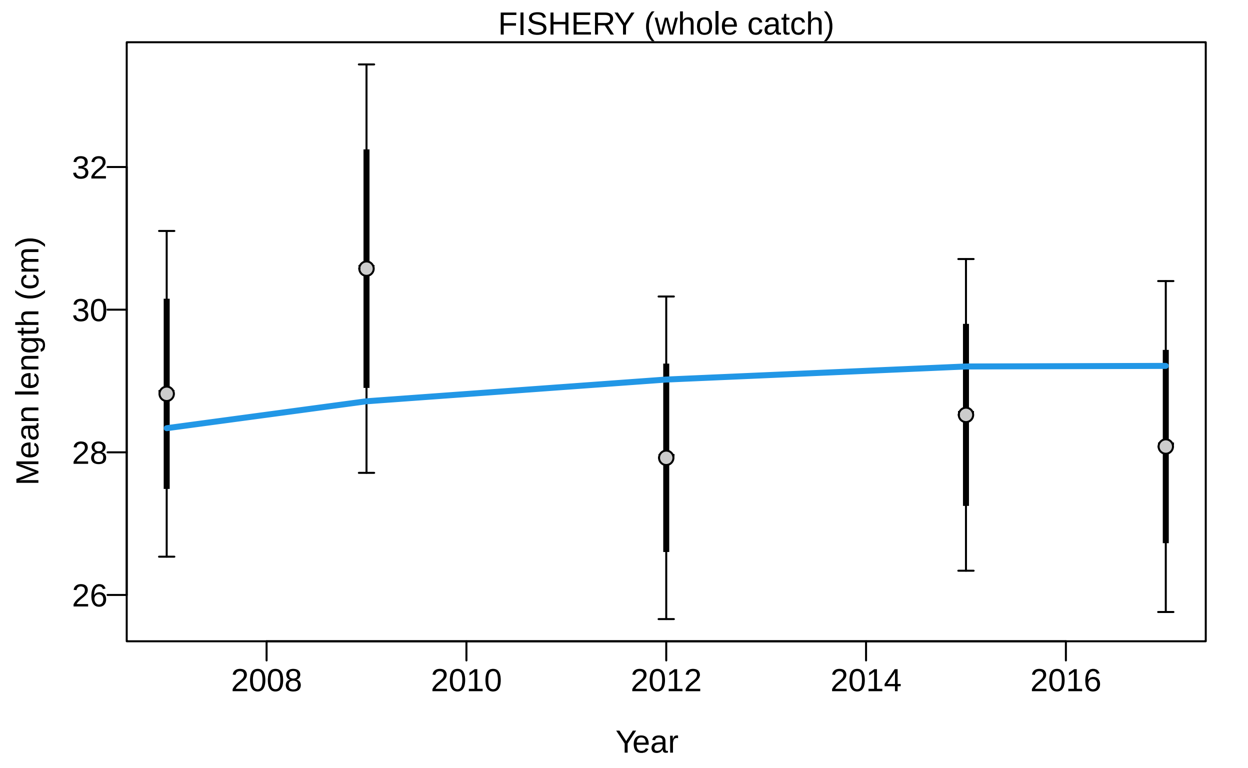


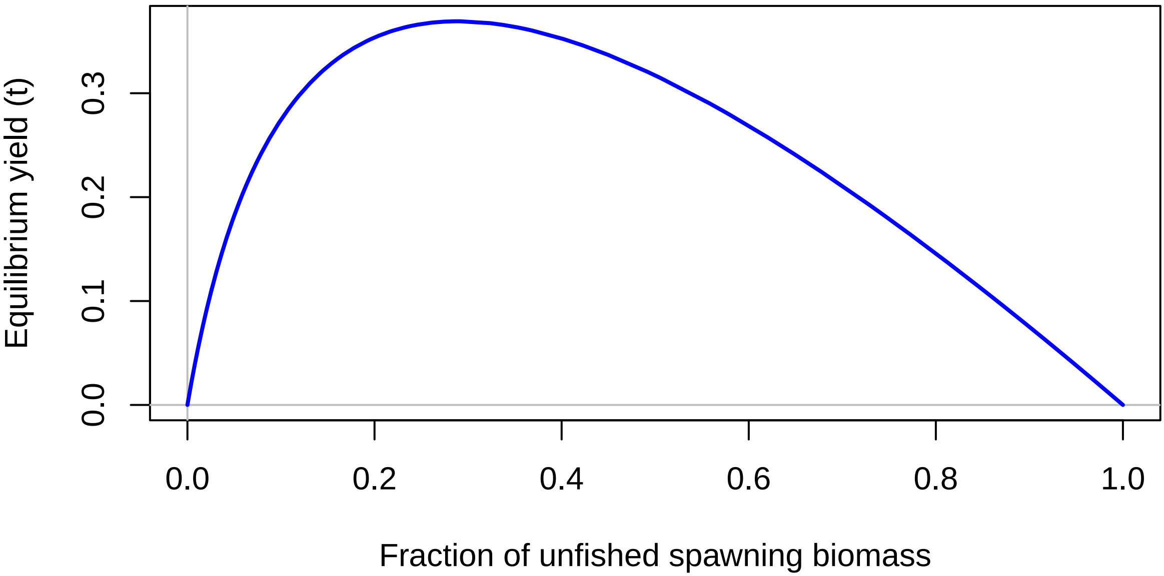
Proportion
2007
N input=57
N adj.=51.9
2009
N input=38
N adj.=34.7
2012
N input=62
N adj.=56.5
2015
N input=68
N adj.=61.9
2017
N input=61
N adj.=55.6
0.00
0.10
0.20
15
20
25
30
35
40
Length (cm)



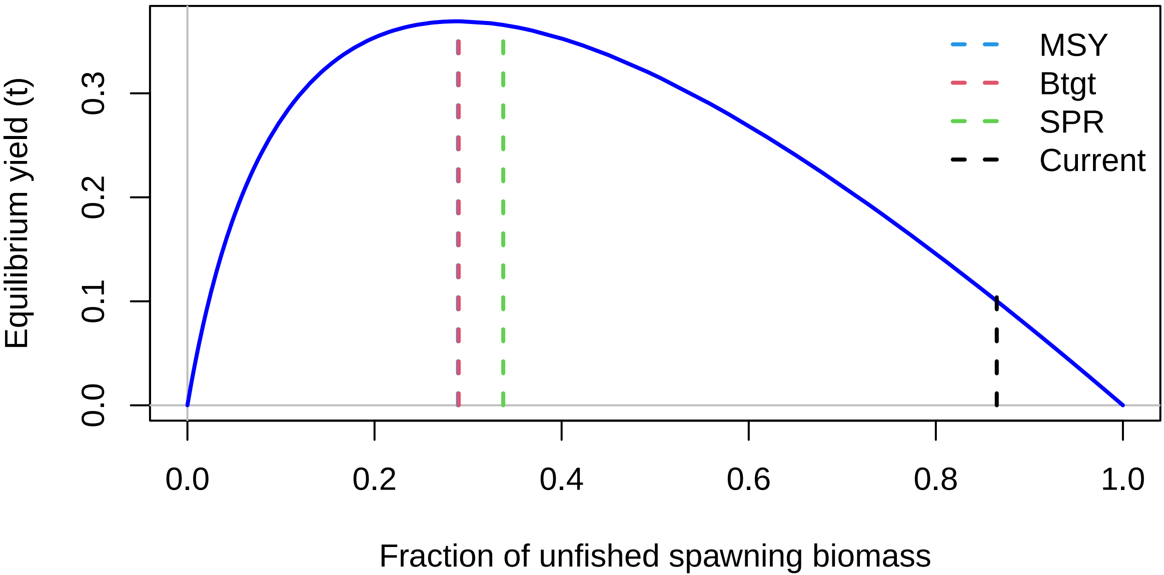
−2
0.1
2
40
36
32
28
24
20
16
Length (cm)
2007
2009
2012
2015
2017
Year


FISHERY (whole catch)
Mean length (cm)
32
30
28
26
2008
2010
2012
2014
2016
Year

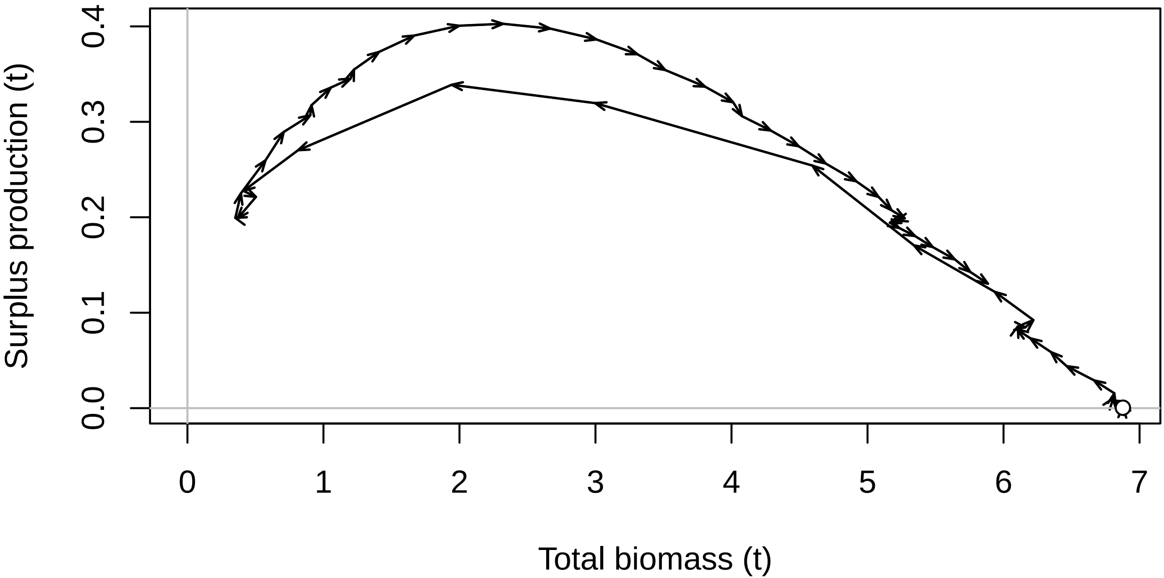

FISHERY (whole catch)
Mean length (cm)
26
28
30
32
2008
2010
2012
2014
2016
Year



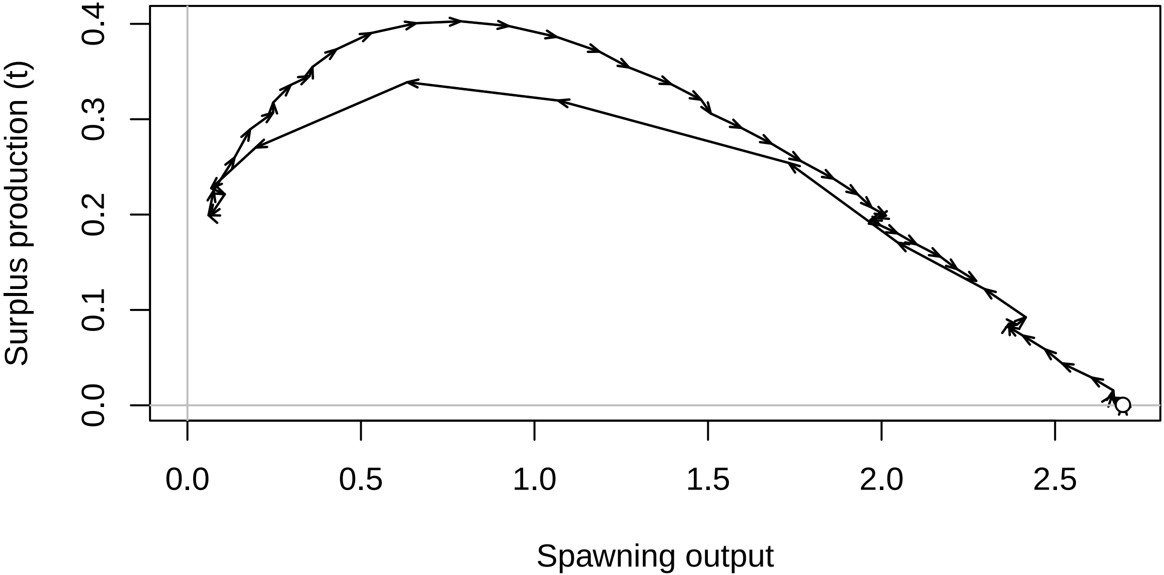
Equilibrium yield (t)
0.0
0.1
0.2
0.3
0.0
0.2
0.4
0.6
0.8
1.0
Fraction of unfished spawning biomass

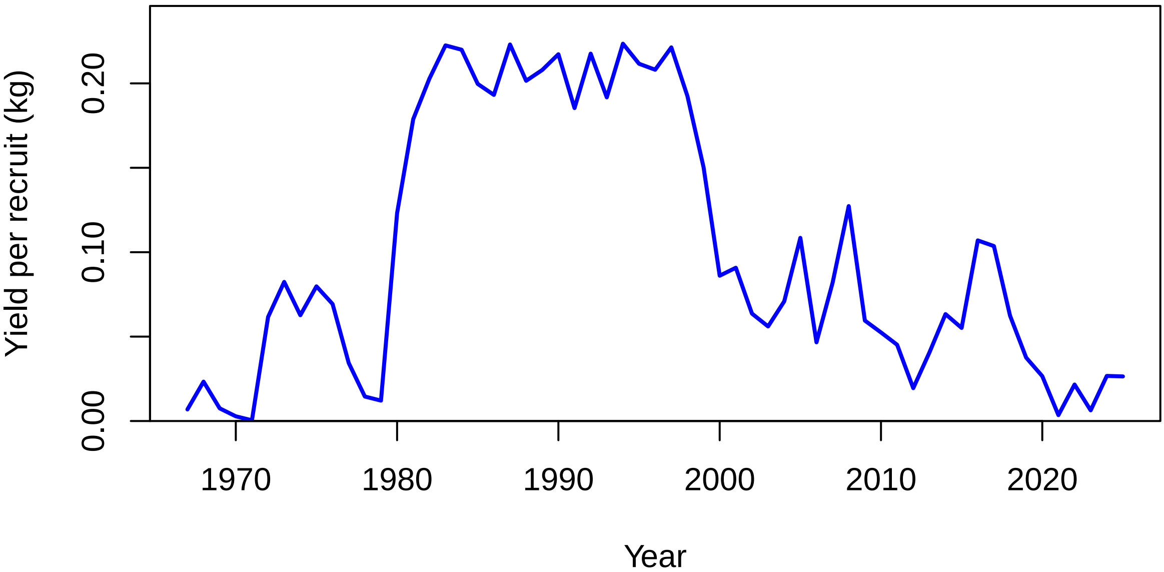


MSY
Btgt
SPR
Current
Equilibrium yield (t)
0.0
0.1
0.2
0.3
0.0
0.2
0.4
0.6
0.8
1.0
Fraction of unfished spawning biomass

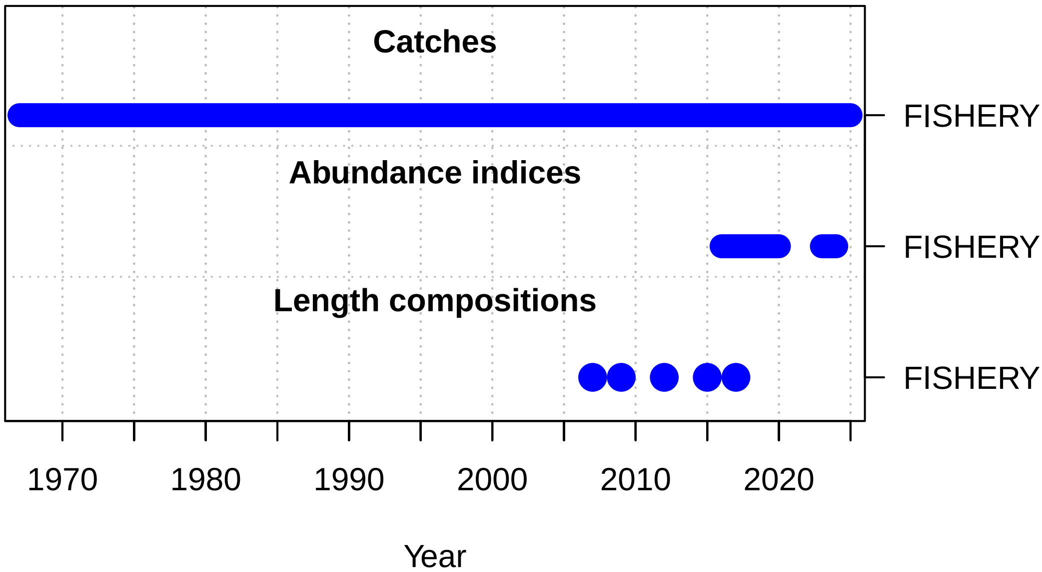


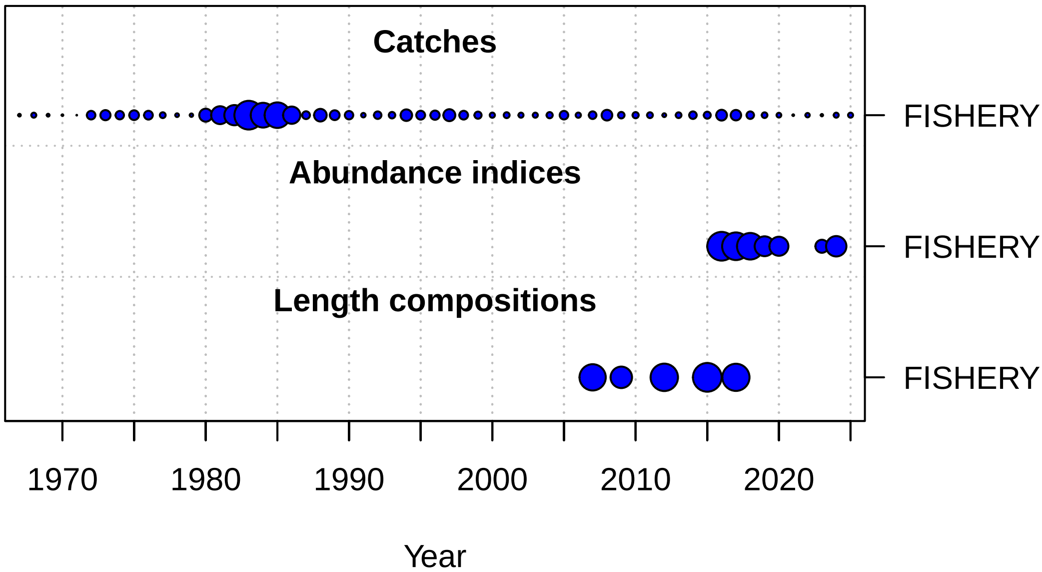
Surplus production (t)
0.0
0.1
0.2
0.3
0.4
0
1
2
3
4
5
6
7
Total biomass (t)

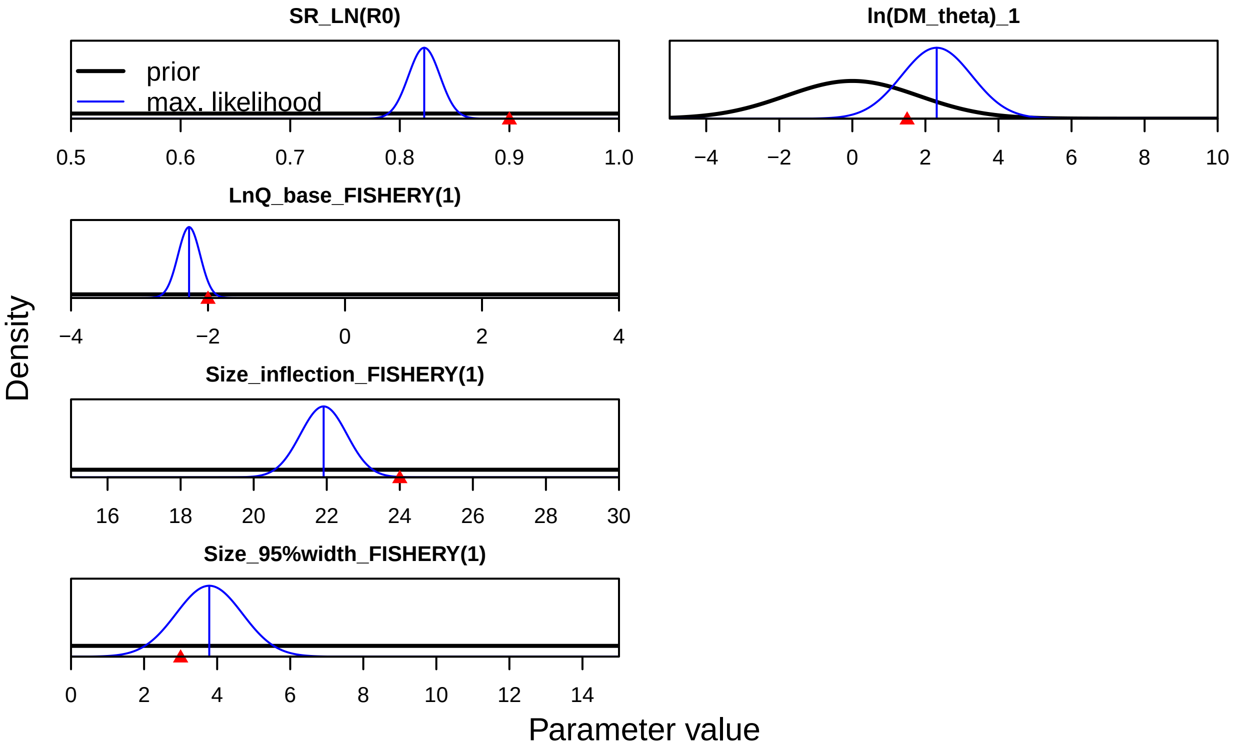

Surplus production (t)
0.0
0.1
0.2
0.3
0.4
0.0
0.5
1.0
1.5
2.0
2.5
Spawning output



Yield per recruit (kg)
0.00
0.10
0.20
1970
1980
1990
2000
2010
2020
Year


Catches
FISHERY
Abundance indices
FISHERY
Length compositions
FISHERY
1970
1980
1990
2000
2010
2020
Year


Catches
FISHERY
Abundance indices
FISHERY
Length compositions
FISHERY
1970
1980
1990
2000
2010
2020
Year


SR_LN(R0)
prior
max. likelihood
0.5
0.6
0.7
0.8
0.9
1.0
ln(DM_theta)_1
−4
−2
0
2
4
6
8
10
LnQ_base_FISHERY(1)
−4
−2
0
2
4
Size_inflection_FISHERY(1)
16
18
20
22
24
26
28
30
Size_95%width_FISHERY(1)
0
2
4
6
8
10
12
14
Density
Parameter value